



A Prefeasibility Analysis of a PV Mini Grid with Ice Plant on Buvu Island in Lake Victoria

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	TRITA-ITM-EX 2019:661 A Prefeasibility Analysis of a PV Mini Grid with Ice Plant on Buvu Island in Lake Victoria	
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Abstract

Improving energy access through the provision of sustainable electrification is fundamental to reducing poverty and facilitating development in many areas of rural Sub Saharan Africa. The region currently suffers from severe energy inequality, with 81% of people relying on traditional forms of biomass for direct energy provision. A major reason for this is that centralised grid infrastructure in many countries remains under maintained and grid extension to remote rural areas poses as too much of a financial burden to governments. Solar PV mini grids are gaining increasing attention from policy makers, researchers, and developers in the region as they pose as a means of leapfrogging the necessity for centralised fossil-fuel based infrastructures. Despite this, a high number of mini grid projects in SSA remain in their pilot stage. A major reasoning for this is the difficulty associated with accurately predicting the future electricity demand of currently un-electrified communities, and following on from that, it is also difficult to incentivize local economies once electricity is made available. The aim of this work was to assess the feasibility of Buvu Island on Ugandan Lake Victoria housing a PV mini grid and ice plant so as to provide power to its three un-electrified communities, and flaked ice to the local fishing industry. Structured interviews were first held with developers in the region so as to assess current weaknesses in mini grid design methodologies and following from that the literature was used to develop a suitable methodology. Data gathered in the field was used to predict the electricity demand of the island, and the HOMER optimization tool was used to size an appropriate system. The modelled system includes an AC-coupled PV array, a lead acid battery bank, a back-up generator, and a converter. Within Uganda, it is the role of the grid operator to provide capital investment for the distribution network. Accordingly, within this study, such a system has been proposed and costed, however this does not impact upon mini grid feasibility. A full financial analysis was used to as to assess the feasibility of the mini grid and ice plant project over a twenty year lifetime. A reference scenario returned promising results, with an IRR of 23.5%, DSCR of 529%, and NPV of \$114,651 assuming that daily electricity demand is 695 kWh / day, and 2.5 tonnes of flaked ice are sold per day. The business model is highly dependent on the sale of ice and if average sales drop below under 1.6 tonnes per day over the project lifetime the system becomes unviable. To overcome the problems associated with unpredictability of demand and encouraged new uses of electricity, the mini grid in this work is scaled to match current demand on the island – thus mitigating such risks. Additionally, a sensitivity study is included so as to fully characterise the range of mini grid operative models that may need be considered, and indeed the likely financial outcomes.

Sammanfattning

Att förbättra energitillgången genom tillhandahållande av hållbar elektrifiering är grundläggande för att minska fattigdomen och underlätta utvecklingen i många områden i Afrika söder om Sahara. Regionen lider för närvarande av allvarlig energifattigdom, då hela 81% av befolkningen förlitar sig på traditionella former av biomassa för energiförsörjning. En viktig orsak till detta är att den centraliserade nätinfrastrukturen i många länder förblir dåligt underhållen och att nätutvidgning till avlägsna landsbygdsområden utgör en för stor kostnad. PV-mini-nät får ökad uppmärksamhet från beslutsfattare, forskare och utvecklare i regionen då de utgör ett sätt att minska beroendet av centraliserad, fossilbränslebaserad infrastruktur. Trots detta kvarstår ett stort antal mini-nätprojekt i Afrika fortfarande i uppstartsfasen. En anledning till detta är svårigheten i att förutsäga det framtida elbehovet för de för närvarande oelektrifierade samhällena. Syftet med detta arbete var att bedöma lönsamheten av ett PV mini-nät lokaliserat på Buvu Island i Victoriasjön, Uganda. Strukturerade intervjuer hölls först med utvecklare i regionen för att utvärdera de nuvarande svagheter i designmetoderna för mini-nät. Därefter användes litteratur för att utveckla en lämplig metodik. Data som samlats in genom fältarbete användes för att förutsäga öns efterfrågan på el, och optimeringsverktyget HOMER användes för att anpassa ett lämpligt system. Det modellerade systemet inkluderar en AC-kopplad solpanel, en batteribank, en reservgenerator och en omvandlare. I Uganda är det nätoperatorens ansvar att tillhandahålla kapitalinvesteringar för distributionsnätet. Följaktligen, inom denna studie, har sådana system föreslagits och kostnadsförts. Detta påverkar dock inte lönsamheten av mini-nät på ön. En fullständig ekonomisk analys användes för att bedöma lönsamheten under en tjugofemårsperiod. Ett referensscenario gav lovande resultat, med IRR på 23,5%, DSCR på 529% och NPV på \$ 114 651 förutsatt att den dagliga efterfrågan på el är 695 kWh, och 2,5 ton is säljs per dag. Affärsmodellen är mycket beroende av försäljningen av is och om den genomsnittliga försäljningen sjunker under 1,6 ton per dag under projektets livslängd förlorar systemet sin lönsamhet. För att komma över problemen förknippade med oförutsägbarhet av efterfrågan och uppmuntra till ny användning av elektricitet, anpassas mini-nätet i detta arbete för att matcha den nuvarande efterfrågan på ön. Dessutom inkluderas en känslighetsanalys för att fullständigt karakterisera utbudet av de olika modeller av mini-nät som kan behöva övervägas.

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List of Equation Terms

A – Cross-sectional area of conductor (m^2)

A_m = amortization (USD)

$C_{ann,tot}$ – Total annualised cost of mini grid (USD)

CC = Customer connection costs

CPF – Capital recovery factor

Db - total debt (USD)

D = Distribution system costs

Dep = depreciation (USD)

E - shareholder equity (USD)

$E_{supplied}$ – The sum of the energy served to the load (J)

F_{PV} – The derating factor of the solar array

G_{STC} – Solar irradiation on PV array at standard test conditions (kW/m^2)

G_T – Solar irradiation on the PV array (kW/m^2)

H = Household installation costs

I – Current (A)

i – Real interest rate

I = interest (USD)

I_{sc} – Short circuit current of solar cell (A)

K – Conductor coefficient

K_d - cost of debt (USD)

K_e - cost of equity ((USD)

LCOE - Levelized cost of electricity ($\$/\text{kWh}$)

L_T – Project lifetime (years)

MPP – Maximum power point of solar cell

NI = Net income (USD)

NPC – Net present cost (USD)

NPV – Net present value (USD)

O = Operation costs (USD)

P = Power generating asset costs

p.f. – Power factor – 0.8

$P_{D,max}$ – Maximum discharge power of the battery (kW)

$P_{D,max,ksm}$ - Maximum discharge power of the battery calculated from kinetic model (kW)

P_{PV} – Power Output of PV array (kW)

$P_{PV,STC}$ – Rated capacity of PV array (kW)

T= total expenditure (USD)

T_{STC} = Array temperature at standard test conditions ($^{\circ}\text{C}$)

T_T – Temperature of the PV array ($^{\circ}\text{C}$)

Tx = taxes (USD)

U – Voltage (V)

V_{OC} – Open circuit voltage of solar cell (V)

WACC – Weighted average cost of capital

α_T – Temperature coefficient of power ($\%/^{\circ}\text{C}$)

η_D – Discharge efficiency of battery (%)

List of Abbreviations

ASCR- Aluminium-conductor steel-reinforced cable

ATP – Ability to pay

CAPEX – Capital expenditure

CC – Cycle Charging

DSCR – Debt service coverage ratio

EPC – Engineering, Procurement, Construction

HH – Household

HHH – High Income Household

IPP – Independent Power Producer

IRR – Internal rate of return

LCOE – Levelized cost of electricity

LDA – Load demand assessment

LF – Load Following

LHH – Low Income Household

MEI – Micro Energy International

MHH – Medium Income Household

NEP - The National Energy Policy (2002)

O&M – Operation and maintenance

OPEX – Operational expenditure

REP - Renewable Energy Policy (REP), 2007

RESP - Rural Electrification Strategy and Plan (2001)

SSA – Sub Saharan Africa

WTP – Willingness to pay

1 Introduction

1.1 Energy Generation in the Sub-Sahara

In light of ever increasing global emissions, and diminishing resources, the need for the integration of renewable energy into our electricity generation mix is of growing importance. Tackling this issue in Sub-Saharan Africa (SSA) poses as a particular challenge as it is an area that houses 13% of the world's population, yet only accounts for 4% of its energy consumption [1].

The region is particularly hard hit by energy inequality with only 215 out of 915 million people having access to grid electricity [2] and with 81% of people still using traditional, inefficient forms of energy that have long been proven to be damaging to health [3]. The level of electricity access varies greatly both between urban and rural communities and across countries. It is estimated that as of 2016 75.7% of urban populations in the region had access to electricity, compared to only 24.8% of the rural populations [2]. A major reasoning for this disparity is that centralised power networks are largely underdeveloped and the costs of grid extensions to remote rural communities are too high compared to the relatively low revenue that could be generated. Efforts to improve these conditions are gaining pace however are still outperformed by population growth [4] [3] [5].

In terms of resources, the SSA has an abundance of untapped renewable resources [6]. Despite this, centralised power supply largely comes from under maintained fossil based infrastructure, which suffers from regular power losses and high costs [7]. Although energy issues are country specific, recurring issues across the region are lack of government regulations, low investments, low corporate activity, and inappropriate and/or inactive policy making [7].

It is evident that sustainable development within the region is fundamental to improving energy access whilst and decentralisation of energy poses as one of the most promising means doing so. Indeed, it is now widely forecast by policymakers, developers, and researchers that large rural areas of the region will undergo a leapfrog process, whereby communities go from having no electricity to adopting decentralised renewable systems [8] [9] [10] [11].

The IEA now predict that in order for the region to meet its SDG goals by 2040, approximately 140 million people will need to be given access to electricity via decentralised energy systems that rely on renewable supply [3]. To do so, it is predicted that an additional 26 TWh of generating capacity will need to be added to the region and this could be provided by between 100,000 and 200,000 hybrid mini grids. Successfully rolling out such a high number of mini grids within this space of time will require much industry development. Currently however, the industry faces multiple barriers that prevent project development and one of the major ones is the lack of planning capabilities within the region [3].

Uganda is one of the countries in this region that is likely to soon undergo a period of rapid off-grid energy development and this is particularly true for island communities on Lake Victoria [12]. This is largely due to the high costs associated with grid extensions to the mostly un-electrified islands [12]. The Ugandan government are evidently aware of the supposed benefits of off-grid energy systems and have recently instigated energy policy that incentives independent power producers (IPPs) to develop renewables solutions in off-grid areas [13] – this is discussed in more detail in Section 1.3.

Considering the above, Buvu Island on Lake Victoria has been highlighted as a key island for mini grid development by the Ugandan mini grid developer GRS Commodities. As yet, no appropriate study has been conducted which assesses the local energy needs, potential for productive uses of energy, or socioeconomic conditions that would allow for an appropriately sized energy system to be proposed.

Once such a study has been completed it is intended that GRS Commodities will apply for a license to develop a mini grid on the island.

1.2 Project Aims

The aim of this study is to conduct a prefeasibility assessment of the electrification options for Buvu Island and suggest pathways by which electrification can increase living conditions on the island through increased supply for electricity. The results of the study will support GRS Commodities in their application for a detailed feasibility study and necessary licenses and approvals.

Primary data collected at the community level will be used to investigate the technical and financial feasibility of a PV mini grid being installed on the currently un-electrified island. In accordance with Ugandan Electrification policy, the mini grid will have to set a tariff at approximately the same cost as mains supply and the grid operator will cover the cost and oversee the installation of grid infrastructure. Considering this, the specific objectives of this work are as follows:

1. Identify weaknesses in previous methodologies used in mini grid design
2. Estimate hourly electricity demand for all communities on Buvu Island
3. Assess the suitability of introducing an ice plant to the local fishing industry
4. Assess the suitability for the island to house a mini grid based on the following considerations:
 - a. Size of community
 - b. Building dispersion
 - c. Electricity demand
 - d. Potential to improve local industry via the sale of ice
 - e. Financial feasibility
5. Size a PV mini grid using the HOMER optimisation tool
6. Propose a distribution system configuration with a quantified voltage loss
7. Use a financial tool to assess the economic feasibility of the HOMER model based on different scenarios. These scenarios differ in:
 - a. Load demand and pattern
 - b. Capital and operational costs
 - c. Connection rates and fees
 - d. Ice sales
 - e. Grants awarded for assets
 - f. Debt from investment
 - g. The impact that paying for grid infrastructure would incur
8. Conduct a risk assessment of the project with risk mitigation strategy

Finally, as PV mini grids are forecast to play an increasingly important role in island communities, this study also aims to contribute to the literature regarding island electrification within the context of developing countries.

1.3 Uganda Energy Profile

Similar to other countries in Sub-Saharan Africa, Uganda has abundant natural resources; however encounters widespread energy poverty. As of 2017 Uganda had a population of 42.86 million, 77% of whom live in rural areas (32.92 million) [14]. As of 2016 approximately 22-26.7% of the populace had electricity access of some sort (55-57.5% urban and 10-18% rural) and the main energy source remained to be basic biomass, with 90% of the population getting their primary energy supply from sources such as firewood (78.6%), charcoal (5.6%), and crop residues (4.7%) [15]. Uganda has 21 power plants connected to the grid with a total installed generation capacity of 947 MW, 68% of which is Hydro [16].

As of 2018, there remained to be approximately 6.9 million households without power and the government has a target of making a farther 1.5 million connections by 2020 [16]. Uganda has seen consistent GDP growth of approximately 6% in the last 20 years and consequently electricity consumption has increased at around 10% annually. Despite this, the country still has one of the lowest electricity consumption rates in the world - 215 kWh per capita per annum (SSA average is 552 kWh/year and the world average is 2,975 kWh / year) [16].

The country has an abundance of natural resources (hydro, biomass, solar, geothermal, peat, fossil fuels) that are relatively well distributed across the country. The energy resource potential of the country includes 2,000 MW from Hydro, 450 MW from geothermal, and 1,650 from biomass cogeneration. Additionally there is an average solar irradiance of 5.1 kWh/m², recoverable petroleum reserves of around 1.4 billion barrels, and renewable potential of 5,300 MW [16].

The overall power market structure in Uganda is conveyed in Figure 1:

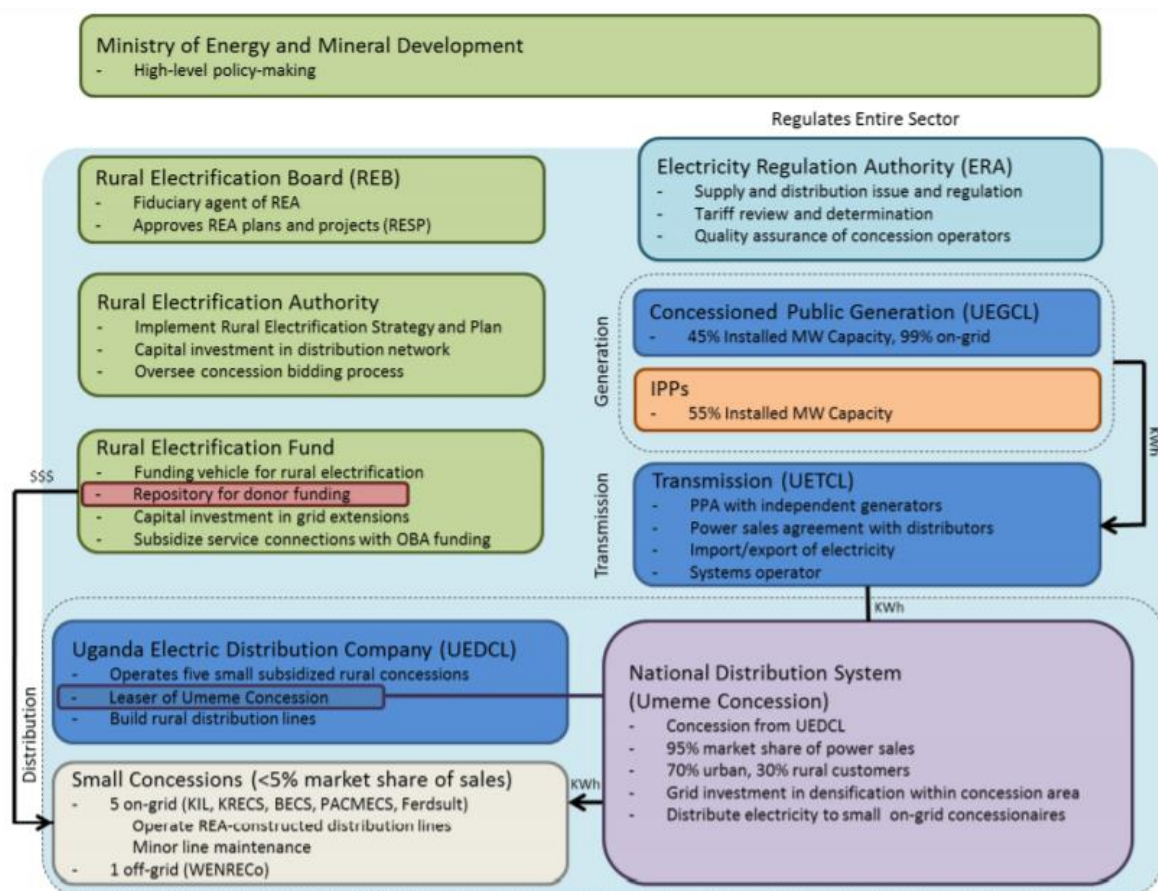


Figure 1 Ugandan Power Market Structure [17]

As shown in Figure 1, the power sector is organised via three main institutions, The Ministry of Energy and Mineral Development (MEMD), The Rural Electrification Agency (REA), and the Electricity Regulation Authority (ERA). A review of Ugandan energy policy instruments and the means by which they support off-grid development is included in the appendices as a supplementary study.

1.3.1 Ugandan Energy Policy.

As highlighted in the supplementary study, Uganda has a strong political motive to improve electricity access for communities that are unlikely to receive grid connection. Despite this, for much of the country there is little access to micro-finance services, and potential investors in off-grid energy

systems have to rely on moneylenders, suppliers, and short term seasonal loans. It is reported that pre-existent micro-finance institutes usually have thin credit product lines, little experience in rural markets, and limited access to best practice information [18].

GIZ are currently facilitating the instalments of various mini grids around the country with the support of the MEMD, REA, and ERA. These mini grids are being facilitated via bundled tenders as opposed to unsolicited single village projects that are currently operational [18].

In 2018, USAID set up its Power Africa Uganda Electricity Supply Accelerator. This accelerator aims to facilitate the installation of 1,000 MW of generating assets whilst also making 1,000,000 new connections by 2020 [12]. As of May 2018, only 10 mini grids were registered as operational in Uganda, all are operated by communities and NGOs, and five have leveraged private investment and sustainable operational models [18].

It was found that mini grids in Uganda are mainly driven by the public sector but managed by private companies and communities. Typically, the REA will identify suitable sites for mini grids and tender to developers. If it is a private sector initiated project then the company needs to apply for a license from the regulator with a letter of support from REA [12]. In accordance with the REP, Government led mini projects have certain advantages within Uganda, summarised as follows:

- Clearer planning across multiple territories takes into account economies of scale and reduced operational expenses
- Project has less risk as there is known predictability as to when the grid is likely to be extended
- Makes utility cheaper for end-users through subsidies on distribution and connection

Figure 2 maps the key mini grid developments across Uganda:

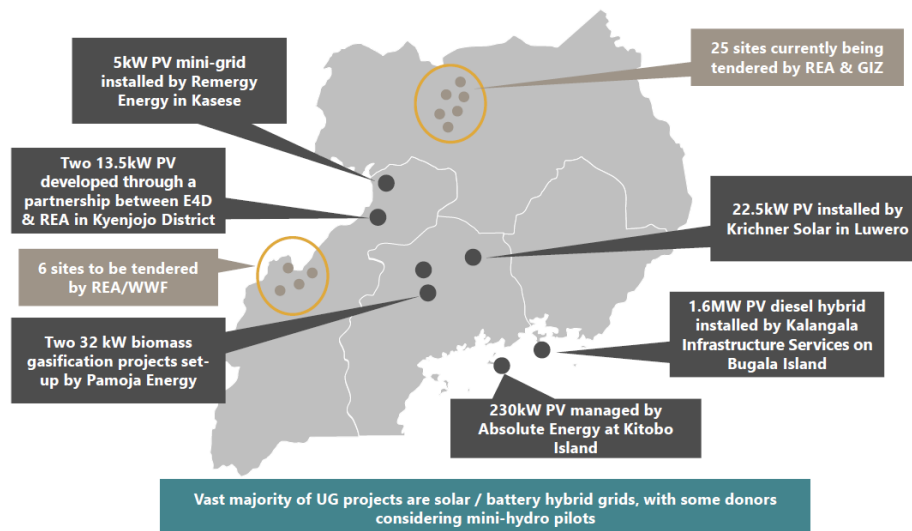


Figure 2 Map of Current and Developing Mini Grid Sites in Uganda [12]

As shown in Figure 2, currently on Lake Victoria there are only two large scale mini grids, one of 230 kW on Kitobo Island, and one of 1.6 MW installed on Buggala Island. The Buggala island mini grid was designed by the mini grid developer GRS Commodities, who now they have plans to develop on Buvu Island.

1.4 Buvu Island

Buvu Island is an island located in North East Lake Victoria. The island houses three communities, Kachanga, Luwungulu, and Kamese. The major industry is fishing and each of the three communities has its own landing site for fishing and transport vehicles. These landing sites are areas where the boats dock. The largest landing site on Buvu is at Kachanga and it includes a small jetty and an area where the local fishermen trade, treat products, and socialise. A satellite image of the island is shown in Figure 3:



Figure 3 Lake Victoria and Buvu Island

A detailed summary of Buvu Island is included in the appendices as a supplementary report – see section 8.1.2. This study provides a summary of Buvu Island’s community composition, its industries, topographies, as well as information on the local people’s relationship with energy. Particular attention is paid to the means by which people utilise energy, and the amount they pay for it. All information has been gathered by the GRS Commodities / MEI field worker who visited the island in May 2019.

1.5 Background to Mini Grids

The following sub-section summarises the means by which mini grids are typically proposed, designed, and implemented. Findings have been taken from literature, research, MEI internal documentation, and structured interviews with mini grid developers in SSA. Findings of this background study feed directly into the methodology employed for the Buvu Island study.

1.5.1 Mini Grids

A mini grid is an alternative to a centralised energy network in that it typically comprises of one or more electricity generators, a storage system, and a distribution network that supplies electricity to a limited number of end users. Mini grids can operate in complete isolation, not connected to centralised infrastructure, or can indeed be grid connected. Mini grids can serve different purposes depending on the size and technical specification of the power generation [19] and this is characterised in Table 1:

Table 1 Mini Grid Categories [20]

Mini Grids	Power Supply	Size of Powerhouse	Description
	AC	300 kW – 2 MW	Large towns usually far from central grids
	AC	1-300 kW	Typically used to supply single or multiple villages located far from grid
	AC	0.2 – 15 kW	Small village and/or an anchor load
	DC	0.2 – 5 kW	Single village far from grid (up to 100 HH)

Mini grids have the capacity to produce electricity at the same quality as the national grid. However, within rural electrification projects the level of electricity required is typically lower (4-8 hours per day).

The most commonly found configuration of a mini grid comprises of: an AC coupled Solar PV system that supplies primary power, a diesel genset for use as backup power, and a battery bank for energy storage and supplying demand in the evening. This option is typically the most flexible / least cost option for rural communities within SSA [19]. A higher use of the PV panels in supplying electricity results in a lower reliance on the genset and therefore the local diesel price has less impact on the levelized cost of electricity (LCOE). Other power source options for a mini grid developer include micro-hydro, biogas generators, and micro-wind [20] [19].

1.5.2 Solar Photovoltaic Systems

A solar PV array is a commonly used power generating technology within mini grids due to its low O&M costs, scalability, and usefulness in regions that experience high solar irradiance yet have low energy access [20].

Solar PV systems are now a well-established energy provider in both on-grid and off-grid applications internationally. Recent years have seen an increase in global investment in the technology and this has led to rapidly reducing costs in production with increased levels of efficiency. China is currently a market leader in solar panel production, and indeed, is SSA's largest supplier of solar panel technology [19]. Off-grid solar solutions have played an important role in rural electrification and the number of people benefiting from them has grown twelve-fold since 2010, exceeding 120 million by 2016. Currently, however, despite growing investment in mini grids, the most commonly found solar products in SSA remain to be those used for SHS [21].

As the primary purpose of this study is to assess the financial feasibility of the mini grid, the major consideration for the PV array is the cost per kilowatt, and thus the technical specification of the PV panel is not considered. One particular aspect of a specific PV panel that could impact upon results is its unit area as this effects shipping costs and the land area occupied by the system – which impacts on land rental fees. Accordingly, a summary of the solar photovoltaic effect and how to assess the ground area of a PV array is attached in the appendices as a supplementary study, found in Section 8.1.3

1.5.3 HOMER

HOMER (Hybrid Optimisation Model for Electric Renewables), developed by the National Renewable Energy Laboratory (NREL) is a widely used industry tool for deciding the configuration of decentralized systems. HOMER uses an hourly simulation approach to mini grid modelling and considers a broad range of technologies (including PV, wind, hydro, fuel cells, generators and boilers) as well as various load types (AC/DC, thermal, and hydrogen). For studies in developing countries a large number of studies exist that use HOMER to assess mini grid prefeasibility [20] [22] [23]. The methodology

employed by HOMER to propose an optimal mini grid system has been included in the appendices as a supplementary study to this work, see section 8.1.5.

1.5.4 Willingness and Ability to Pay

A key consideration in mini grid planning is the ability and willingness of potential customers to pay for its use and this plays a key part of the load demand assessment (LDA). A supplementary study in which these concepts are discussed is found in the appendices in section 8.1.4

1.5.5 Operative Models

There are four commonly used operational modes for mini grids in rural areas and the optimal one to use is highly project specific. Factors to be considered in choosing an operational mode are environment, geography, and socio-economic variables [24]. These operational modes largely differ by the way in which the mini grid system is designed, organised, operated, and managed. Mini grids can be composed of two operational bodies, the small power producer, and the small power distributor. The four main operational modes are: utility operated; private sector operated; community operated; or a combination of the previous three [24]. A summary of the specifics of each of the models is attached in the appendices as a supplementary study, see section 8.1.6

Within this work, GRS Commodities intend to maintain ownership of the mini grid on Buvu Island. However, due to Ugandan energy policy, the local grid operator will be in charge of installing the grid infrastructure and will provide a subsidy for metering costs. Accordingly, a hybrid operative model will likely be used.

1.5.6 Business Models

It is often difficult for private developers to make mini grids profitable business ventures and typically innovative business models have to be adopted [24]. Mini grid business models have developed greatly in recent years so as to be more accommodating to the low revenue nature of rural electrification. The most commonly found private business models are:

- The Franchise Approach
- Anchor, business, and Consumers “ABC” Approach
- The Clustering Approach

A summary of these business models is attached in the appendices as a supplementary study to this work, see Section 8.1.7.

Within this work, as Buvu consists of three communities, it is evident that a clustered approach will need to be used, but with an anchor load so as to make the system economically viable – that being the ice plant.

1.5.7 Distribution System

Major considerations in the design of mini grid distribution systems include: location of the powerhouse; line configuration; and placement of poles. The line configuration and pole locations are largely determined by the configuration of the village, the nature of the loads to be served, the type of conductor to be used by the grid operator, the available pole options, and the Ugandan grid codes which determine variables such as system clearance [25].

A background study on mini grid distribution infrastructure has been conducted and is included in the appendices as a supplementary study, see section 8.1.8.

The specific methodology employed in this work to design the distribution system is detailed in section 3.4

1.5.8 Quality Assurance

Although the overall governing electrical engineering principles of mini grids are well established, the specifics of construction and operation vary drastically between projects and systems can experience very different variations in power quality and reliability [26].

In order for mini grid business models to be effective at expanding energy access, tariffs need to be affordable to end users and the electrical supply needs to meet their needs. Accordingly, a mini grid's capacity to offer a meaningful service at a reasonable cost to an end user is detrimental to a project's overall success. As previously detailed, this is of course highly difficult due to the high capital investment requirement, the fragmented approach to design commonly seen, the severe lack of governing standards or data, and the lack of enabling environments [26].

It is evident therefore that wherever possible, mini grid designers should seek to follow quality assurance framework which addresses the sectors two root problems, affordability, and quality [26]. NREL has produced such framework and this has been summarised in the appendices, found in section 8.1.9

The methodology used in this work to assess grid quality is detailed in Section 3.4.4.

1.5.9 Background Research Summary

It is evident from the background study on mini grids that from a technical perspective, they are well understood systems and constructing them in the correct environment is a relatively straightforward engineering task. However, a lack of industry standardisation often results in systems that do not deliver upon the predetermined service quality.

2 Literature Review

Following on from the background research study the literature was reviewed so as to assess best methods for conducting a prefeasibility assessment of a mini grid and identifying likely errors that may arise in the process. Accordingly, major themes of the literature review were:

- Best methods for conducting an LDA of a mostly un-electrified rural community using:
 - Energy surveys
 - Energy measurements
- Best methods for increasing energy demand through:
 - Introducing new productive uses of energy
 - Electrifying pre-existent industry

2.1 Load Demand Assessment

The accurate prediction of the electricity demand of a community is a governing factor within mini grid design and is paramount to achieving perceived financial outcomes [20]. Accurately predicting future electricity feeds directly into the design of the mini grid system as it is the main decider in sizing the components and thus the price of the energy produced. Lack of understanding of electricity demand can, according to research, result in the following outcomes:

- Oversized – Oversizing a mini grid can result in too high initial investment, longer payback times, higher O&M costs, and low efficiency [20].
- Undersized – Under sizing a mini grid leads to unreliable supply, reduced service hours, and blackouts. This in turn has a negative impact on the end-users satisfaction. The components of the system can also become damaged due to overloading thus increasing O&M costs. If the service quality is low, the developer may also suffer additional problems when collecting payments from end-users [20].

Both cases highlighted above lead to a system operating incorrectly, having adverse effects on costs, however it is widely reported in research, case studies, and directly by developers, that mini grids are most often oversized [20] [27] [28].

Having an accurate LDA therefore leads to an accurately sized system and this increases a project's chances of success. A major barrier to doing so is that oftentimes the developer will have little to no experience in ascertaining accurate prediction of future electricity use from a currently un-electrified potential customer.

In the majority of cases mini grid developers will be looking to begin projects in new markets [20]. The most commonly used approaches to LDA therefore are through the developer conducting housing surveys/interviews supplemented with other community engagement tasks.

LDA conducted through surveys rely on getting accurate data on the hours of use of specific appliances, as well as their rated power. This data is then aggregated into hourly load profiles that reflect the interviewee's daily electricity demand. GIZ prepared a handbook that specifies recommended procedures for conducting load demand assessments via surveys [20]. Likewise, The Heinrich Boll Foundation produced a methodology for conducting load demand assessments with surveys for community owned projects [29]. The Green Mini Grids helpdesk facility is an online platform that has the overall aim of incentivizing mini grid developers and policymakers from SSA through offering open source information. The platform offers a methodology for conducting load

demand assessments through a survey approach [30]. Developers in Nigeria and Haiti have used this; however results have yet to be transferred into a scaled mini grid.

Multiple studies have used a survey based approach for conducting the demand assessment for mini grids [22] [27] [31]. *Blodgett et al* [27] report that energy-use surveys deployed in the pre-feasibility stage are widely known for being unreliable and their accuracy is yet to be proven. The research compares survey-predicted energy use to actual measured consumption at eight mini grids in rural Kenya. The errors associated with assessing past electricity consumption on a customer-by-customer basis were too high and an average consumption was used for the comparison. The system analysed was scaled for 426 Wh per day per customer and the recorded use after one year was on average 113 Wh per day per customer. The result is that the system has higher capital and operational costs and less revenue to support it then was first expected. Again, it is reported that until more data sources are made available from operational mini grids, energy-use surveying is likely to remain a dominant methodology.

Bahaj et al [23] investigated the overall impact that a solar PV mini grid had on a rural community in Kenya through reviewing the design and construction process, the preconstruction interaction with villagers, and subsequently the impact that the mini grid has had since its conception in 2013 and the time of the study (2017). 1,069 households were surveyed during the load demand assessment that used a survey approach and a comparison of that to the electricity use five years later are conveyed in Figure 4:

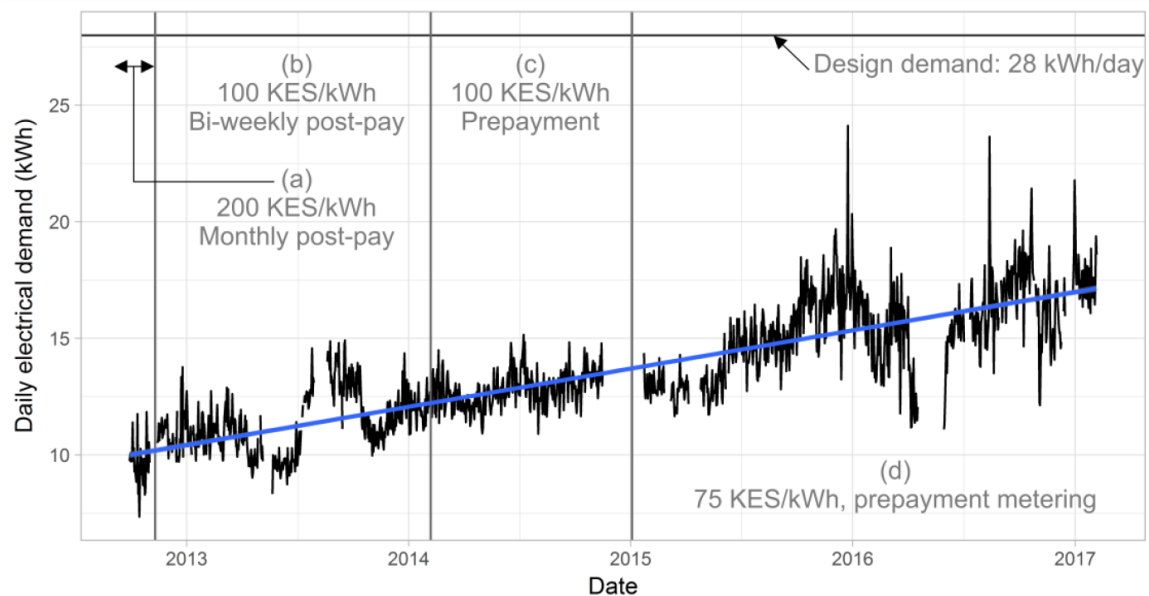


Figure 4 Electricity demand compared to load demand assessment [23]

As shown above, the system was designed to produce 28 kWh/day based on the electricity demand assessment, however, in 2013 the average demand was 10 kWh rising to 17 kWh in 2017. This implies that the system was initially oversized by three times having adverse effects on project costs. Additionally, it was observed that the main driver behind the increased load demand was the reduction that was made on the tariff. It is noted in the work that the final tariff of 75 KES / kWh was not financially feasible.

Similarly *Hartvigsson et al* [32] claim in their work that there is currently a lack of measured load profiles from mini grids and the most common method for estimating electricity usage is through

appliance data collected via interviews. The paper compares and analyses the differences between measured daily load profiles and daily load profiles created from appliance data collected through interviews. This feeds into an analysis of how findings effect mini grid sizing. It is found that the interview method does not provide accurate estimations and that repercussions on developers and investors can be significant. Figure 5 conveys the results of using an interview and appliance data to estimate demand and this is compared to manual measurements. Results are for an entire community, a house, and an SME respectively:

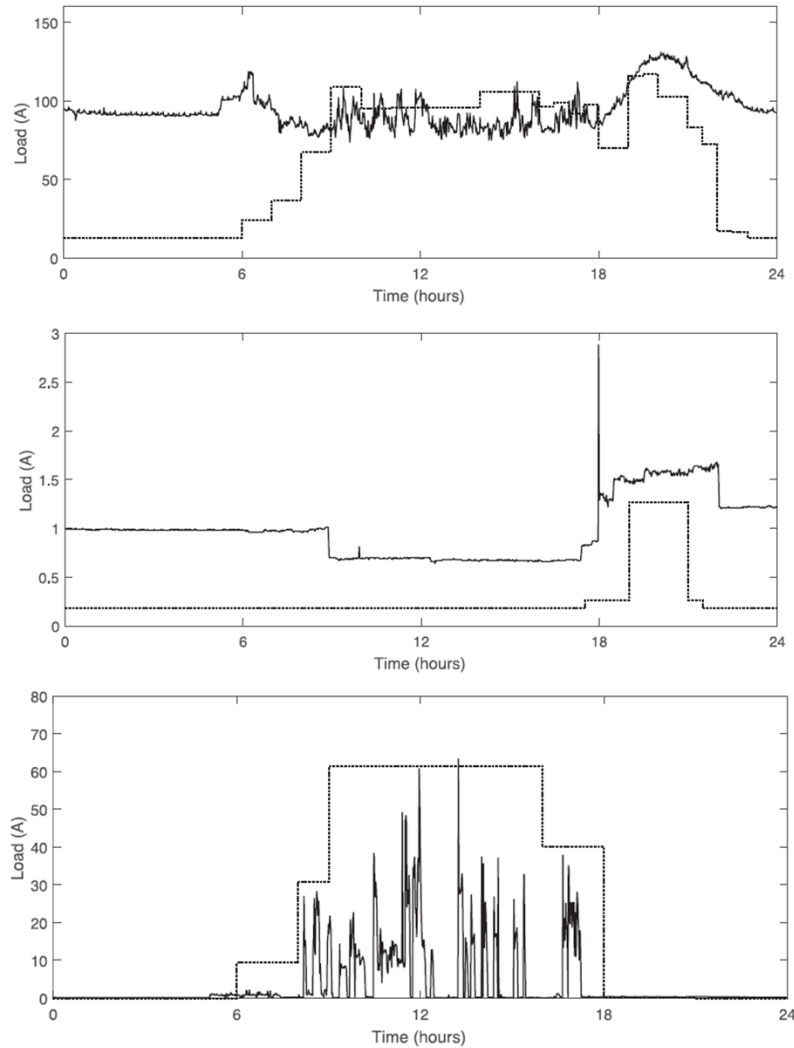


Figure 5 Interviewed based Load Profile formation (dotted line) and measured load profile (solid line) for: mini grid (top) household (middle) and SME (bottom) [32]

As shown in Figure 5 there is disparity between the two methods used. An issue is that when constructing load profiles based on appliance's rated powers and hours of use, the model only considers hourly resolution and thus ignores the impacts of rapid changes, such as peak demand arising from switching appliances on and off [32]. Improvements that could be made to the interview process are suggested and it was found that if the interviewer probes for specific usage patterns, it is possible that these will be evident in the results, regardless of whether they actually exist. The following suggestions are made for improving the interview method whilst doing field work [32]:

- Considering the risk of not obtaining necessary data within a limited time in the field (morning and evening peak) – questions need to be formulated so as to identify appliances and use patterns.
- The interview should not probe for specific usage patterns as the results are likely to reflect this probing even if not true. The interviewer therefore needs to be open to all unknown outcomes and thus the survey should include both specific and general questions
- Questions need to be specific enough to identify multiple usage times of appliances (such as morning and evening) yet general enough to marginalise bias. Questions should thus avoid the following structure ‘are there any lights left on during the night?’
- A suggested questions include:
 - ‘What are your household members doing in the evening?’
 - Followed by, ‘do any of these activities involve electricity?’
 - These questions then lead onto detailed questions on appliance usage
- It is fundamental that interviewees view the interview team as neutral throughout all procedures

Overall, *Hartvigsson et al* [32] suggest that the interview approach has several advantages over manual measurements as long as above mentioned suggestions are included. Manual measurements are time consuming and require special skills and equipment that may not be readily available. This will in turn reduce access to data, which as previously discussed, is a major barrier to the mini grid sector. A final benefit to using interviews is that data can be gathered on specific appliances and times of use and this information is valuable to utilities who aim to introduce load management strategies [32].

It is often suggested that instead of using surveys and manual measurements, developers should rely on data from previous projects and analyse how electrification impacted upon the demand of a community. Unsurprisingly however, within these often extremely remote environments, access to data from previous projects, or relevant open source datasets, is almost always non-existent [33][38][34]. It is widely reported in literature that one of the major barriers to conducting accurate demand assessments is the lack of up-to-date publically available data relating to demand and costs of mini grids on a location specific basis [28]. This lack of data was reported as an issue by *Cross and Guant* [33] who developed a residential load model for South Africa. Likewise, *Wijaya and Tezuka* [35] reported that lack of electricity use data was the greatest challenge to creating accurate models and implementing appropriate energy policy, and *Moner-Girona et al* [28] report that there is a lack of transparency and consistency within literature regarding mini grid costs – this being a main barrier to wide scale mini grid uptake. Additionally, The World Bank reported that gathering accurate load data for regions of SSA is fundamental to incentivizing renewable energy investment in the region [34].

Peters et al [36] reviewed major challenges facing the uptake of rural mini grid and again highlighted that over optimistic demand projections were among the key challenges. It is stated that if business models were based on realistic demand forecasts and took into account high transaction costs in rural areas, the commercial viability of mini grids would improve. Overestimated uptake of productive uses of energy is also cited as a major reasoning for overestimated demand. *Taneja* [37] goes further by stating that “planning should depend on reasonable electricity demand growth projections that are produced by parties without incentives to overstate growth”. *Peters et al* state that productive uses of energy (PUEs) are fundamental to making mini grids a viable option however demand projections need to be well substantiated and reasonable. It is also suggested that developers should partner with microfinance institutes to make such productive uses more readily available to end users.

2.2 Productive Uses of Energy (PUE)

PUEs are commonly reported in industry to be the main driver behind making mini grids more financially feasible. Multiple intergovernmental bodies have produced frameworks for introducing productive uses into previously un-electrified communities including Energy4Impact [38] and GIZ [39]. It is widely reported in these aforementioned frameworks that the successful introduction of productive use loads into rural electrification projects will bring benefit to both the livelihoods of the customers but also the revenues of the operator.

Indeed, it is a truism that electrification brings with it the opportunity for communities to adopt new modern industrial practices through the purchasing of new equipment. For this reason there are numerous studies that promote strategies through which new PUEs can be introduced to currently un-electrified rural communities [40] [20] [39] [41] [42]. However, there is little evidence to suggest that such strategies have been successfully applied and much research supports this claim [43] [38] [23] [44] [40] [36]. What is evident, *Peters et al* argue, is that electricity is only one of many variables that effect the uptake of PUEs within rural communities [36] and that the issue expands further into the complexity of local commercial markets. Likewise, *Terrapon-Pfaff et al* [43] analysed the results of an impact evaluation of 30 electrification projects so as to better understand how sustainable energy provision supports productive end uses and whether these activities supported overall goals. It was found that access to energy does not automatically lead to increased electrical consumption. It is suggested that training, equipment provision, and dedicated research need be integrated into all stages of the energy project. Indeed, one study completed by MEI on behalf of the mini grid developer JUMEME aimed to assess PUE potential for a Tanzanian community and suggestions made in the report remain unacted upon two years after project completion [45].

2.3 Energy Planning Tools for LDA

It is evident from the literature surrounding demand assessment methodologies that lack of electricity use data is a major barrier to wide scale uptake of mini grids, and that data driven planning tools should to be developed [27].

Numerous electrification planning tools are under development that use spatially explicit GIS-based models and are applied to countries across Africa. These tools identify a community's distance to existing grids, local topographies, population density and growth, poverty indices, solar irradiance, cost of diesel, energy demand, and feasible connection rates, amongst others. Using these indicators the least cost electricity option for investment can be estimated. These quantitative based models pose as a useful alternative to speculative based electricity projections, however, as highlighted by *Trotter et al* [46] these tools still need much improvement prior to being useful to developers. To date such tools have been primarily used by policy makers and investment bodies as they present results in an easily understandable format. Before such tools become useful for mini grid developers in the load demand assessment phase there needs to be more readily available data sets that characterise the livelihoods of people within a given geographical area [46].

2.4 Literature Review Summary

Overall, it can be concluded from the literature review, and from consultations with industry partners, that anticipating the uptake of new industrial practices is not a useful strategy for making mini grids financially feasible. There is however evidence to suggest that pre-existent industries can be improved upon through electrification from a mini grid (as noted by GRS Commodities staff who currently operate an ice plant on Buggala island). It is also clear that the best strategy to do so is to model the technical solution for current demand and not predict load growth. As the main productive activity on

Buvu Island is fishing, and currently the local community suffers from having inadequate access to ice, it was decided to analyse the effect that a flake ice plant would have on the feasibility of the mini grid.

3 Methodology

This chapter reviews the methods used to answer the research questions and project aims. The study relies on data gathered in the field over the course of one month during the field visit as well as data gathered previously from GRS Commodities and data found in literature, which is all cross-referenced with MEI.

3.1 Interviews with Developers

Structured interviews were conducted with multiple mini grid developers across Sub-Saharan Africa. The aim of which was to ascertain information regarding the following:

- Methodologies employed for the load demand assessment
 - Accuracy of said load demand assessment
 - Mechanisms for correcting inaccuracies
 - Mechanisms for assessing impact of poorly sized systems
 - Actions that need be taken to correct wrongly sized systems
- Electricity production data from a mini grid following an installation
- Methodologies for planning:
 - Business models
 - Tariff structures
 - Increasing connection rates
 - Risk mitigation

Overall, it was found that mini grid developers are very protective of data gathered internally as it gives them their competitive advantage regarding scaling accurate systems. It was also found that developers were highly interested in research that analysed mini grid output data. However, they were not as willing to share their own gathered data.

Notably, the Ugandan developer GRS commodities reported that if mini grid data were made readily available then systems would likely become better designed and at lower cost to the end user. The company currently operate one mini grid and report that it was designed for 32 kWh / per person / per day, however reported end use was 8 kWh / per person / per day – this overestimation agrees with findings from the literature review whereby LDAs are often four times over estimated. End usage data from this mini grid is not available and it was further reported that this data is very rarely recorded. Similarly the Tanzanian mini grid operator JUMEME who operate one mini grid report that their designed system was over scaled.

The summary of findings of thesis discussions is found in the appendices, summarised in the Table 37. As is shown in Table 37, the majority of developers interviewed are highly interested in understanding how best to introduce productive end uses into their target communities, however, are reserved to share electricity production data which in turn would allow for a robust study to be conducted.

3.2 Project Information from Field Trip

The information and data that needed to be collected during the field trip are listed as follows:

- Island Information
 - GPS coordinates
 - Number of inhabitants

- Local diesel price
- Existing power generating systems
- Available land area
- Local industrial practices
- Contact details for local leaders and stakeholders
- Customer Based Information
 - GPS coordinates of customers
 - Household occupation levels
 - Economic activities
 - Ability to pay and willingness to pay
 - Existing power generating systems
 - Customer appliance list and hours of use

All information was ascertained during the field visit in May/June 2019. Information was gathered through interviews, focus group discussions, and structured transect walks.

3.2.1 GIS Analysis

A GIS analysis was conducted of Buvu Island so as to better understand the local topography, location of households, businesses, productive centres, and land ownership. During the field trip the field worker logged GPS coordinates of the following places so as to conduct an analysis:

- Agricultural lands
- Businesses
- Individual productive use loads
- Community institutions
- Current market centres
- Potential development areas
- Trading centres
- Ports and commuting routes

A map was created as highlighted in Figure 6:

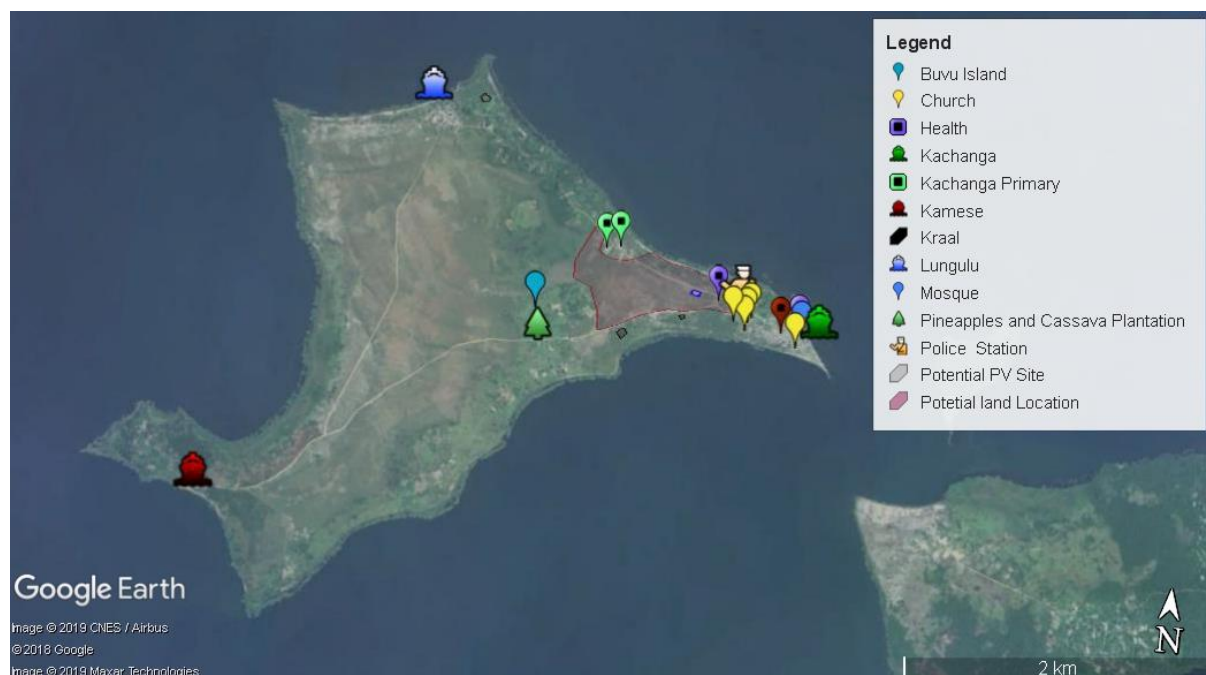


Figure 6 GIS Map of Buvu Island and its key features

As shown in Figure 6, the majority of the industry and key locations are centred on the Kachanga community located in the East of the island. More detailed views of each community are shown in *Appendix D – Photos of Landing Sites*.

Specific GPS coordinates were not taken for the majority of customers and thus assumptions had to be made of their location. For large buildings, location assumptions have been made through inspection of the GIS map for key indicators such as the size and shape of buildings.

3.2.2 Electricity Surveys

Based on the findings of the literature review and an assessment of available skills and data on Buvu Island, a survey based approach was chosen for the load demand assessment. A copy of the electricity survey used is attached in *Appendix E – Electricity Survey*.

Following completion of these surveys, results were aggregated and load profiles created for each building type on the island as well as each community. This method employed is detailed in *Section 3.2.3*.

3.2.3 Customer Segments

In accordance with field data and literature [30], households on the island have been split into three categories – high, medium, and low income. High income households (HHH) represent households that currently have a larger SHS and use it to power indoor/outdoor lighting, a TV, and a phone charger. Medium income households (MHH) are those households that have a smaller SHS and use it to power indoor/outdoor light and a phone charger, and low income households (LHH) are those households that are currently un-electrified but with a mini grid will be connected so as to operate one light and a phone charger.

A full list of customers on the island is shown in Table 38, *Appendix F – Customer List*. As shown, 73.9% of customers are located in the Kachanga community.

3.2.3.1 Appliances

The key information that needed to be ascertained from electrical appliances were the rated power, the hours of use, and where possible the measured power consumption. In most instances the rated power was stated on the appliance and in cases where this was not possible and data was not ascertained, the field worker and GRS Commodities were consulted so as to better understand the likely power demand. Consultations were cross referenced with likely rated powers referenced in previous project work and literature [47]. A full appliance list is shown in *Appendix G – Appliance List*.

An assumption that had to be made was the assessment of the fridges' energy demand as the rated power does not represent hourly energy consumption. From the field notes, fridges observed ranged from 100 to 120 litre capacity and on average used 1/8 horse power compressors. Through consultation with GRS Commodities an hourly demand of 93.125 W was assumed.

3.2.3.2 Energy Demand of Each Customer

In accordance with methods detailed in literature [27], the daily energy consumption of each customer segment has been taken from the appliances used and the hours of use. For each appliance the daily energy demand is assessed using Equation 1:

Equation 1 Appliance Energy Consumption

$$E = No * P * Hr$$

Where;

E = Daily energy demand (kWh/day)

P = Rated power of the appliances (kW)

No = Quantity of the appliance

Hr = hours of use per day

For each customer segment the daily energy demands of each appliance are summed and a customer energy demand is defined. A sample inventory list from the secondary school is conveyed in Table 2.

Table 2 Secondary School Energy Assessment

Secondary School					
Types of Appliances per customer	Quantity	Power (W)	Hours/day	Total Power of Appliances (W)	Daily Energy Demand (kWh/day)
Indoor Lighting points	8	5	1	40	0.04
Outdoor lighting points	4	5	1	20	0.02
Phone Charging	4	5	1	20	0.02
Lighting points in school	7	5	1	35	0.035
Television + decoder	1	36	1	36	0.036
Total				151	0.151

3.2.4 Ice Plant

An ice plant was modelled based on the following considerations:

- Daily flake ice demand on Buvu
- Cost of selling flake ice
- Capital and shipping cost
- O&M Cost

There is an estimated daily flake ice demand of 2.5 tonnes on Buvu Island. It was therefore necessary to model an ice plant facility that could produce and store such quantities on a daily basis, ideally within sunlight hours so as to marginalise additional expenses on storage or backup supply.

It was confirmed by GRS Commodities that the company would seek to retain ownership of the ice plant throughout the project lifetime and they would ideally sell flaked ice to the community at a rate of 300 UGX per kg which is sold in 83 kg bags for 25,000 UGX. It was also specified that they would look to invest in a plant that could produce up to 5 tonnes of flaked ice per day.

Considering the above, the technical specifications of multiple Ice Plants were reviewed. Based on consultation with GRS Commodities, the Focussun Ice Machine FIM-50K was decided upon (see Ice Plant Data Sheet for full data sheet). Table 3 summarises a quote for the system that has a load demand of 17.7kW and a production rate of 208 kg/hr (5,000 kg/day).

Table 3 Ice Machine Data [48]

Component	Cost (USD)
Ice Machine	26,000

Ice Storage facility	3,600
Steel structure	1,200
Inverter	1,750
Shipping/handling/insurance	N/A - Covered in component costs
Total CAPEX	32,550
OPEX	700 / year

Based on the current daily demand of 2.5 tonnes the ice plant will have to operate 12 hours per day. The hours at which the ice plant operates have impact on the sizing of the system and a sensitivity study was conducted on the impact that operating hours have on system sizing, see Section 4.1.3.4.

12 hours of daily operation results in a 213 kWh daily energy demand for the ice plant. This value has been included in the load demand assessment which is subsequently input into the HOMER sizing study. In the reference scenario a constant ice demand of 2.5 tonnes per days is presumed for the entire project lifetime. Additional scenarios considered and results are seen in Section 4.5.4.

3.2.5 Load Profile Creation

Having determined the daily energy consumption of each customer segment on the island the dispersion of demand over the course of the day had to be estimated. The literature has been used to determine a most likely electricity demand pattern. A study completed by USAID assessed the electricity use pattern for multiple operational mini grids and created different scenarios for demand dispersion [49] [50]. The 'Business Heavy Profile', as highlighted in Figure 7, has been used as a reference for the overall electricity demand on Buvu as the commercial demand is higher than that of the domestic demand.

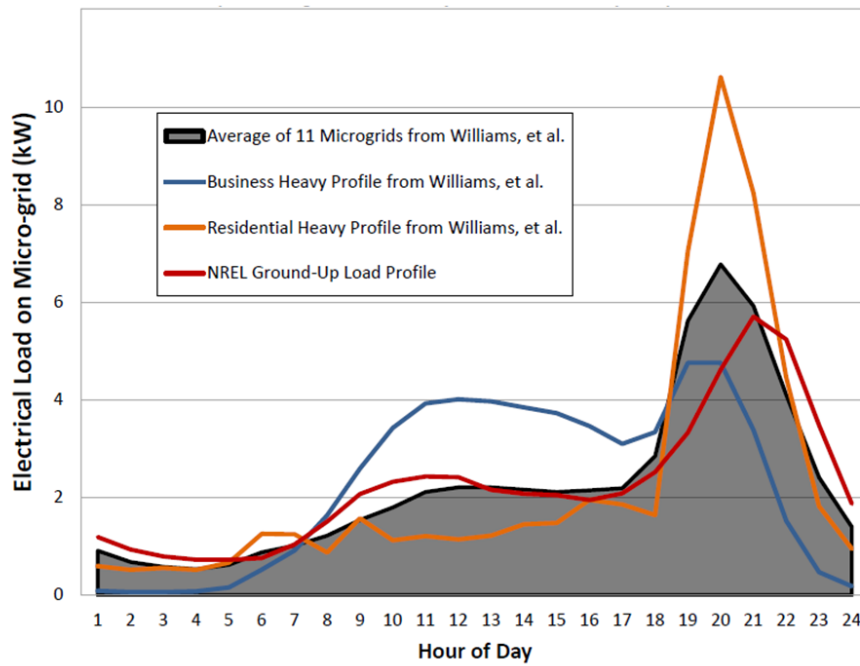


Figure 7 Average load distribution for mini grids [49] [50]

Figure 8 displays the load profile of the reference case scenario of Buvu Island which has a daily energy demand of 694.56 kWh. In this scenario the productive use load demand is solely coming from the 17.7 kW ice plant that operates 12 hours a day so as to produce 2.5 tonnes of flaked ice.

a

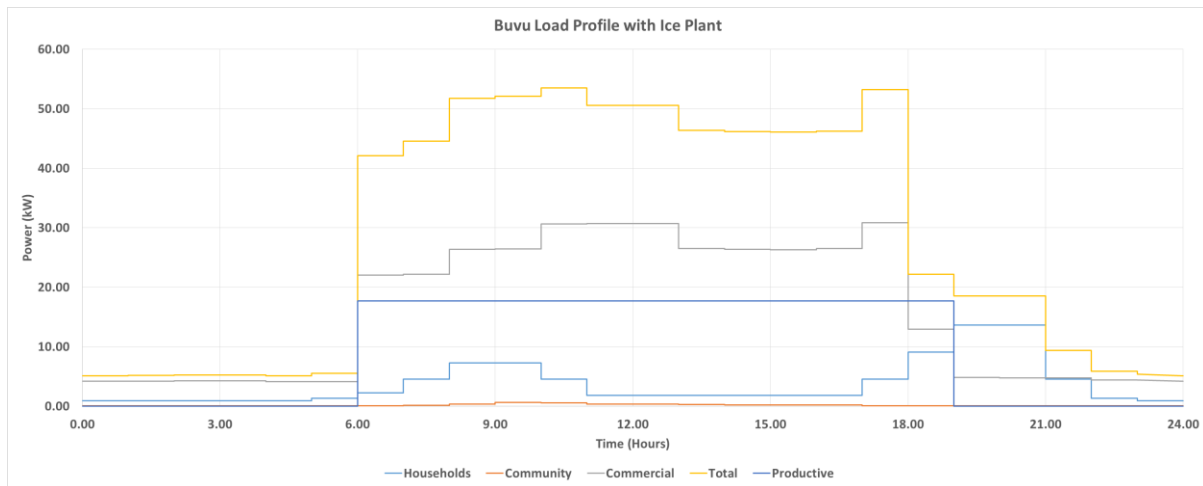


Figure 8 Buvu Island Load Profile

As shown in Figure 8, the community has a peak demand of 53.49 kW daily. Following on from discussions with GRS Commodities, this daily electricity demand is assumed to be constant throughout the year. This hourly load demand information is input directly into the HOMER sizing tool.

Figure 9 displays the load profile of the island without the ice plant with a total daily energy demand of 482 kWh:

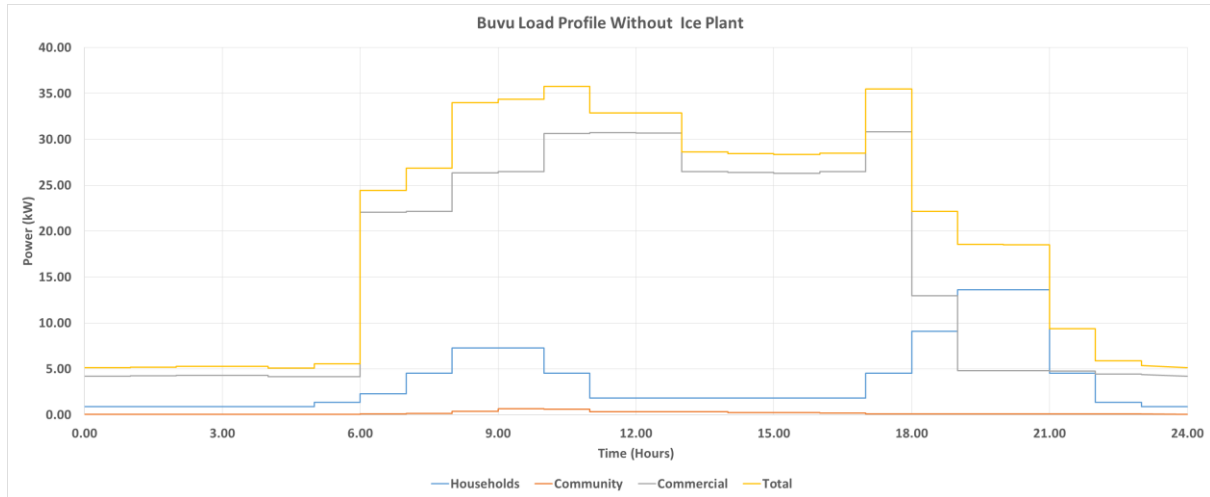


Figure 9 Buvu Island Load Profile without Ice Plant

As shown in Figure 9, the peak demand reduced to 36 kW.

Within the HOMER analysis a daily load fluctuation is applied to the daily load demand and as recommended by MEI, a 5% daily fluctuation was used in the analysis. The result of this is that peak loads that are up to 5% above those predicted occur at infrequent intervals throughout the year and accordingly, the system is sized to account for these.

3.3 HOMER Analysis

The HOMER tool is used to size the optimal configuration of the system. The variable data input into the software includes:

- Hourly solar irradiance [51]
- Hourly load demand
- Technology capital costs
- Technology performance characteristics
- Constraints for the battery bank size

The component installation costs have been estimated based on projects conducted previously by GRS Commodities and INENSUS (a Berlin based mini grid developer), cross referenced with literature [49] [20] [52]. These costs are listed in Table 4:

Table 4 Costs for Power Assets

Component	Life (years)	Low (USD)	Medium (USD)	High (USD)
Installed PV	20 [20]	1,400/kW [49]	1,800/kW [49]	2,200/kW [49]
Lead-acid battery	7 [20]	300/kWh [49]	400/kWh [49]	500/kWh [49]
Inverter/ Charge Controller	10 / 20 [20]	600/kW [49]	900/kW [49]	1,200/kW [49]
Generator	N/A	400/kW [49]		
O&M Assets	2% of capital investment in power generating assets [52]			
O&M Distribution	2% of cost of distribution system [52]			
Discount rate	10% [52]			

As shown in Table 4, three costs are quoted for mini grid power assets. The medium cost range is used as the Reference Scenario and the impact that the Low and High costs ranges have on results is investigated in the sensitivity analysis, see Section 4.5.1.

The key outputs assessed from the HOMER study are:

- Renewable penetration (%)
- Solar PV array size (kW)
- Diesel generator size (kW)
- Battery bank size (kWh)
- Converter size (kW)
- Net Present Cost (\$) (excluding revenues from ice sales)
- LCOE (USD / kWh) (excluding revenues from ice sales)

Overall, within this study, little attention was paid towards reviewing or modelling specific technical components beside the battery bank and converter. Reasons for this are three fold. Firstly, the capital cost of installing, ordering, and installing specific components is still highly project specific and it is advisable to use the average costs reported in literature [49] and from previous project experience. Secondly, unlike the battery and converter that have configurative constraints, the PV Array only impacts on results in terms of \$/kW as there is no need to model specific configurations within the HOMER tool. It should be noted however that the specific PV panel chosen does impact on the land area required for the installation due to differences in efficiency. This could potentially impact on land use decisions in certain projects, however due to the abundance of available land on Buvu, the PV array area has not been specifically calculated. A methodology to do so however is included in the appendices as a supplementary study, found in Section 8.1.3. Likewise, diesel generators are widely available and used across the SSA, and thus the costs predicted for installation are accurate, and indeed, the technology is highly scalable to specific conditions – this removes the need to consider specific models at this early project stage.

3.3.1 Solar Irradiance

The size and technical configuration of the solar PV array is directly related to the specific solar irradiance of the site and thus it is fundamental to use accurate solar irradiance data within the techno-economic assessment. Within this study the solar irradiance data for Buvu Island has been taken from the NREL database. The hourly irradiance data is input directly in HOMER and is characterised in Figure 10:

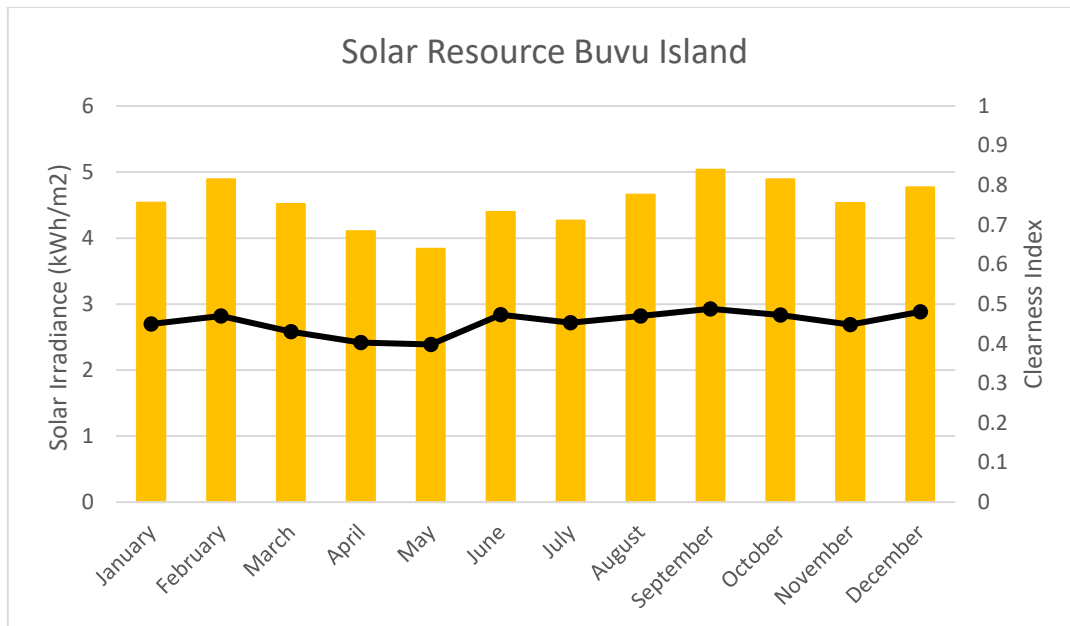


Figure 10 Solar Irradiance for Buvu Island

As shown in Figure 10 the average annual solar irradiance observed on Buvu Island is 4.54 kWh/m²/day.

3.3.2 Diesel Generator

Within this study a diesel generator is sized in the HOMER software to a level that ensures that it can provide sufficient back-up support in a cost effective manner. The main variables that are considered within the diesel generator analysis are: the local cost of diesel; the expected fuel cost inflation rate; and the operation and maintenance costs of the selected diesel generator. These costs have been taken from literature and cross referenced with the field worker at GRS commodities [49].

3.3.3 Batteries

Two batteries are considered within this study, a generic lead acid battery of 1kWh capacity and the second is the BAE 26 BVS of 6.72 kWh capacity. Their BAE 26 PVS model was assessed in a homer study [53], and its datasheet is found in Battery Data sheet.

3.3.4 Converter

A generic converter has been modelled in HOMER and costs have been assigned based on the values specified by USAID, [49] (low scenario - \$600 / kW, medium scenario - \$900 / kW, high scenario - \$1200 / kW). It is an industry standard that the total converter size should be scaled to be 10% greater than the peak demand. Additionally, as there is a requirement for three phase supply for the ice plant, the number of converters needs to be multiple of 3.

The SMA Sunny Island series are a range of converters that are suitable for rural electrification projects and based upon recommendation from MEI and INENSUS, this range of converters has been selected within this work. The Sunny Island 6.0H and 8.0H have rated powers of 4600 W and 6000 W respectively and these two are considered within the analysis in this study. The peak estimated demand on Buvu can be accommodated by 12 Sunny Island 8.0H units which have a combined capacity of 72 kW. The datasheet of this model is attached in the Appendix N.

3.3.5 Sizing Sensitivity

As the load demand upon which the system is sized is subject to much variation and inaccuracy, a sensitivity analysis had to be conducted specifically within the HOMER tool. The following variables were adapted so as to highlight potential variations:

- Investment Costs
- Renewable Penetration
- Diesel Cost
- Ice Plant Operating Hours
- Ice demand
- Demand from customers

The results of the sizing sensitivity study are shown in Section 4.1.3.

It should be noted that this sensitivity study served the purpose of identifying the range of system sizes that would be optimal, based on the input variables. A separate sensitivity study is conducted of the financial model which serves the purpose of highlighting the range of financial incomes that are observable once a system size is finally decided upon.

3.4 System Layout

In accordance with Ugandan energy policy, if a license is granted for the mini grid, the grid operator will finance the design and installation of the mini grid. Accordingly, a study was conducted in which a grid layout was proposed so as to act as advisory information to the grid operator, as advised by GRS Commodities. A major consideration for the grid layout is that it complies with local grid codes and ensures that each customer is supplied with electrical power at the quality paid for. A voltage loss assessment has therefore been conducted to ensure that the total voltage loss is below 10%, as highlighted in the NREL Quality Assurance Framework [26]. A copy of the grid conditions specified in this framework is included in *Appendix A - Quality Assurance Framework*, Table 35.

The following sub-section highlights the methodology employed for proposing a grid layout that effectively serves all customers referenced in the load demand assessment.

3.4.1 Powerhouse Location

The potential location of the powerhouse in this study was determined from the field study. Through focus group discussions with community leaders in Kachanga, a potential land area was mapped, as shown in Figure 11, marked by the shaded red area:



Figure 11 Allowable Area for PV System

A specific location for the PV system was decided upon based on the available land and the results of the voltage loss assessment. The final location is marked by the rectangular blue area in Figure 12.

3.4.2 Grid Layout

As the dispersion of buildings in remote rural communities often vary and there is little integration of infrastructure, it is advisable for potential grid configurations to be proposed and voltage losses assessed at a prefeasibility stage [25]. Doing so removes the necessity of an additional cost intensive site visit where all specific customers are located.

The proposed grid layout is made for use by the grid operator and will be reviewed alongside other mini grid technical documents during the process of deciding upon whether to support the project or not. In this high level work, the lines are made so as to follow road networks and confirm that a skeletal configuration, that has a minimal reliance on anchoring to counteract loads, is possible.

Two grid designs are considered within the analysis - one design uses only one three phase line (as recommended by MEI), and the other has only three phase lines (as has previously been used by the Ugandan grid operator (information provided by GRS Commodities)).

The layouts of the Kachanga, Luwungulu, and Kamese grid are shown in the following three figures, where red lines denote three phase, yellow lines denote single phase lines, and green lines denotes service drop cables (each green line represents a maximum of 30 potential customers). The Ice Plant is located at Pole 47, and Pole 1 connects the three phase line to the PV plant.

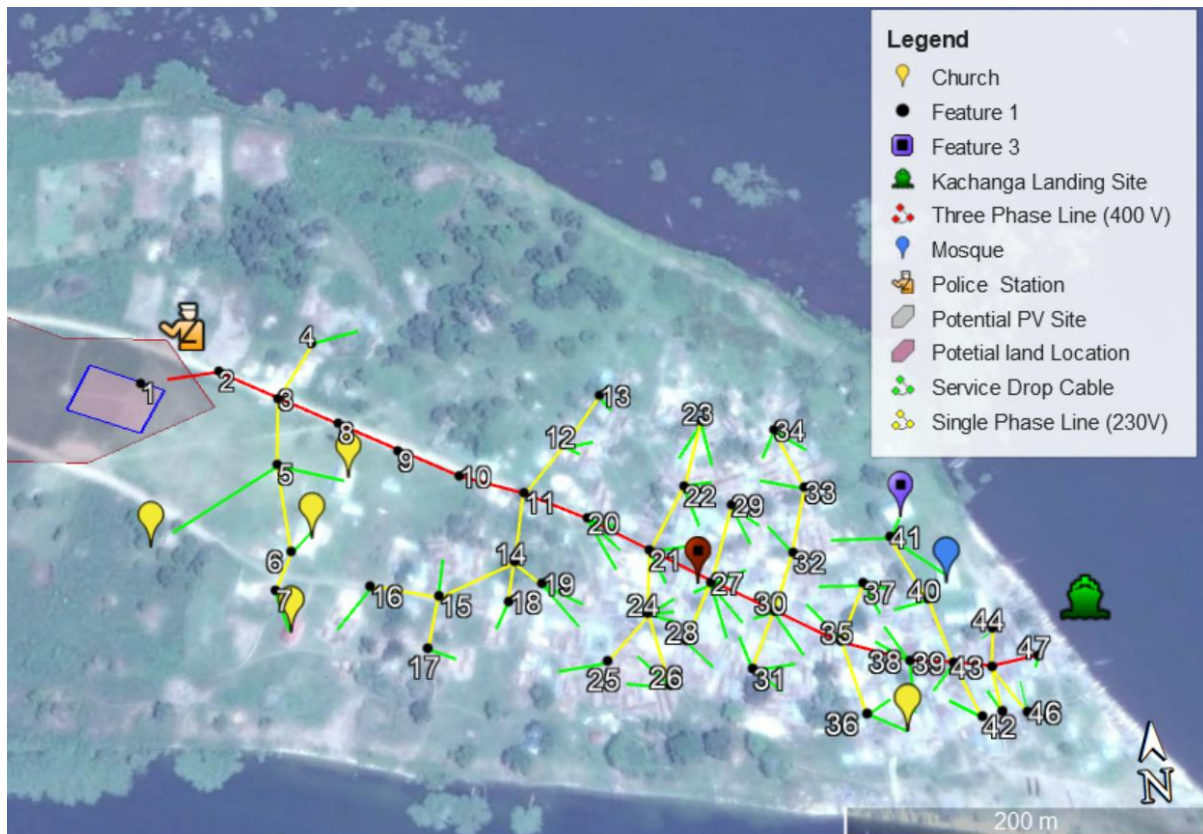


Figure 12 Kachanga Proposed Grid

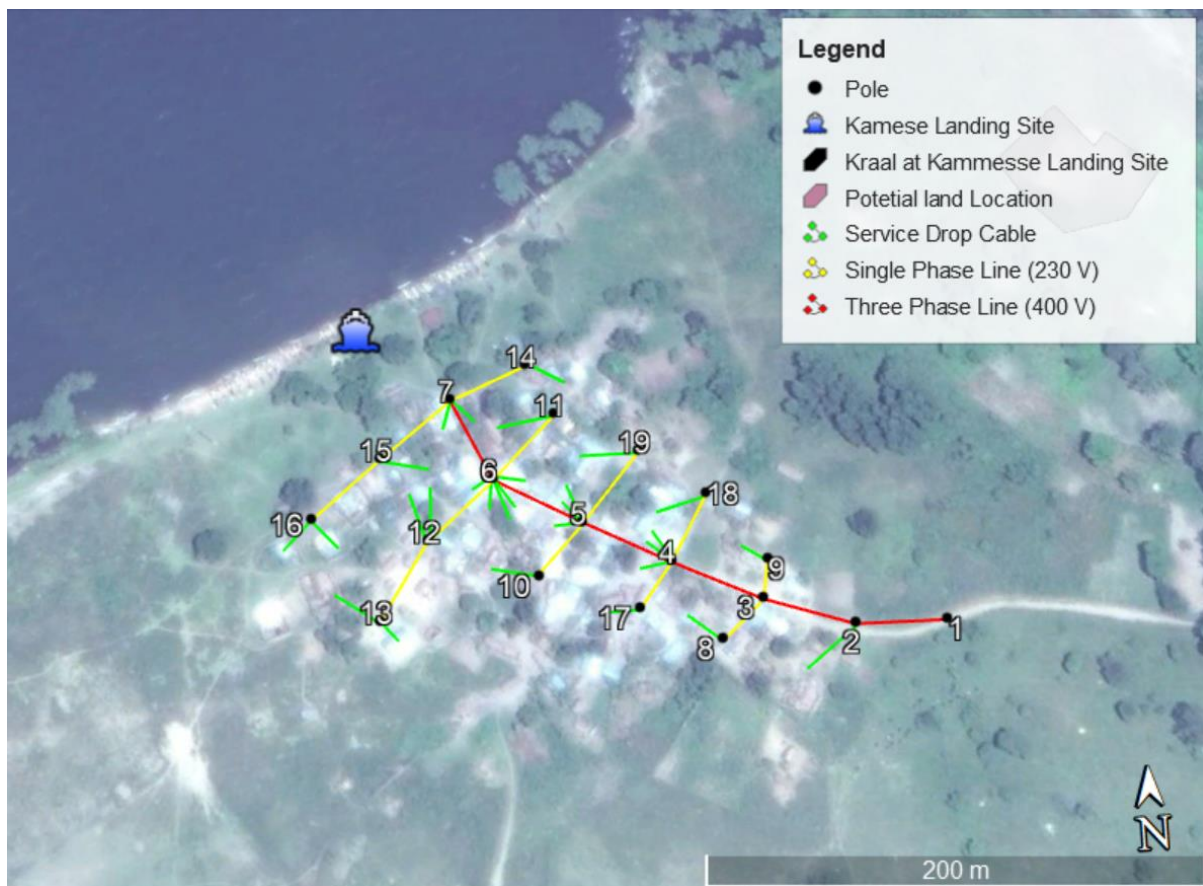


Figure 13 Kamese Proposed Grid

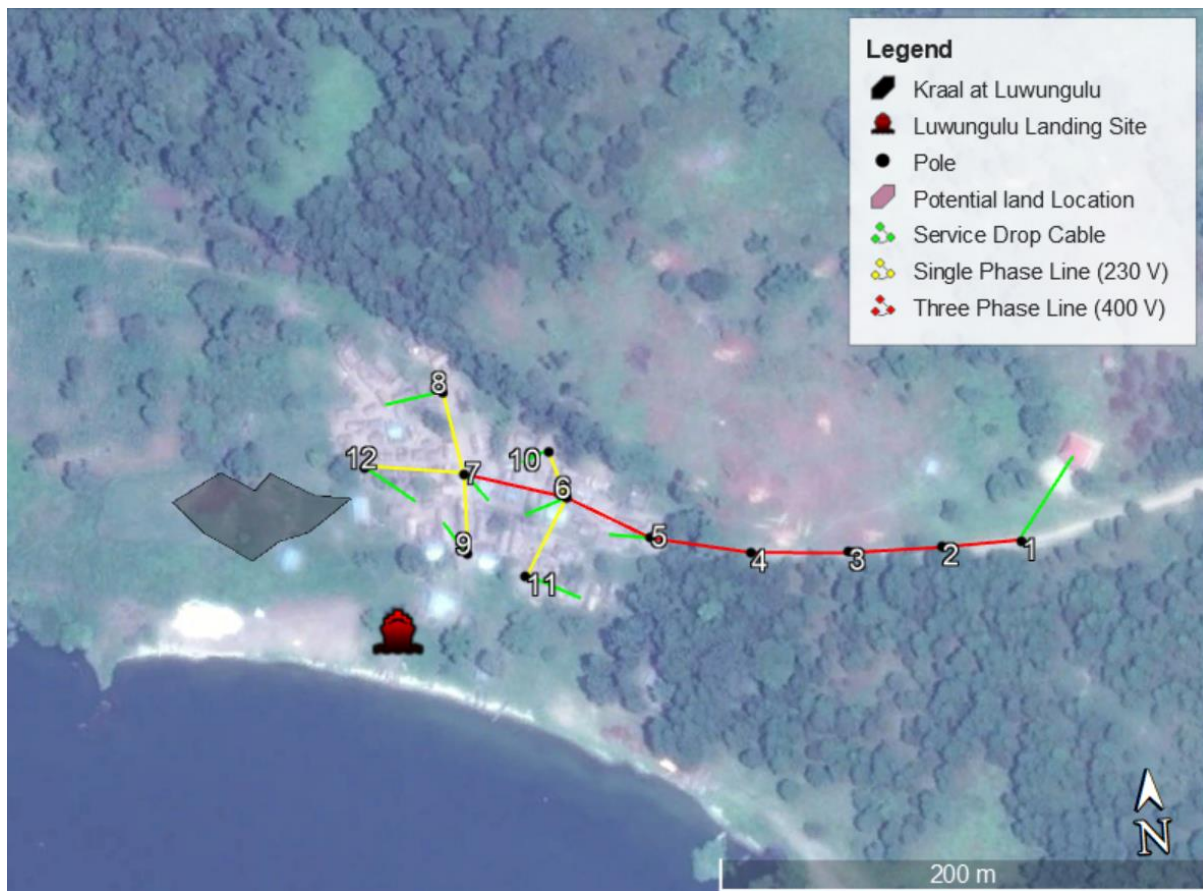


Figure 14 Luwungulu Proposed Grid

As shown in Figure 12 the allocated area for site development is within close proximity of the Kachanga community in the East of the island. To connect the PV system to the remaining communities whilst minimizing losses a Medium Voltage system distribution system is proposed. This has been considered upon the request of GRS Commodities who specify that the grid operator would facilitate such a system once a license has been ascertained.

As shown in Figure 15, Medium Voltage cables (33 kV) have been plotted that connect the PV site to the Kamese and Luwungulu land sites via the island's road network.

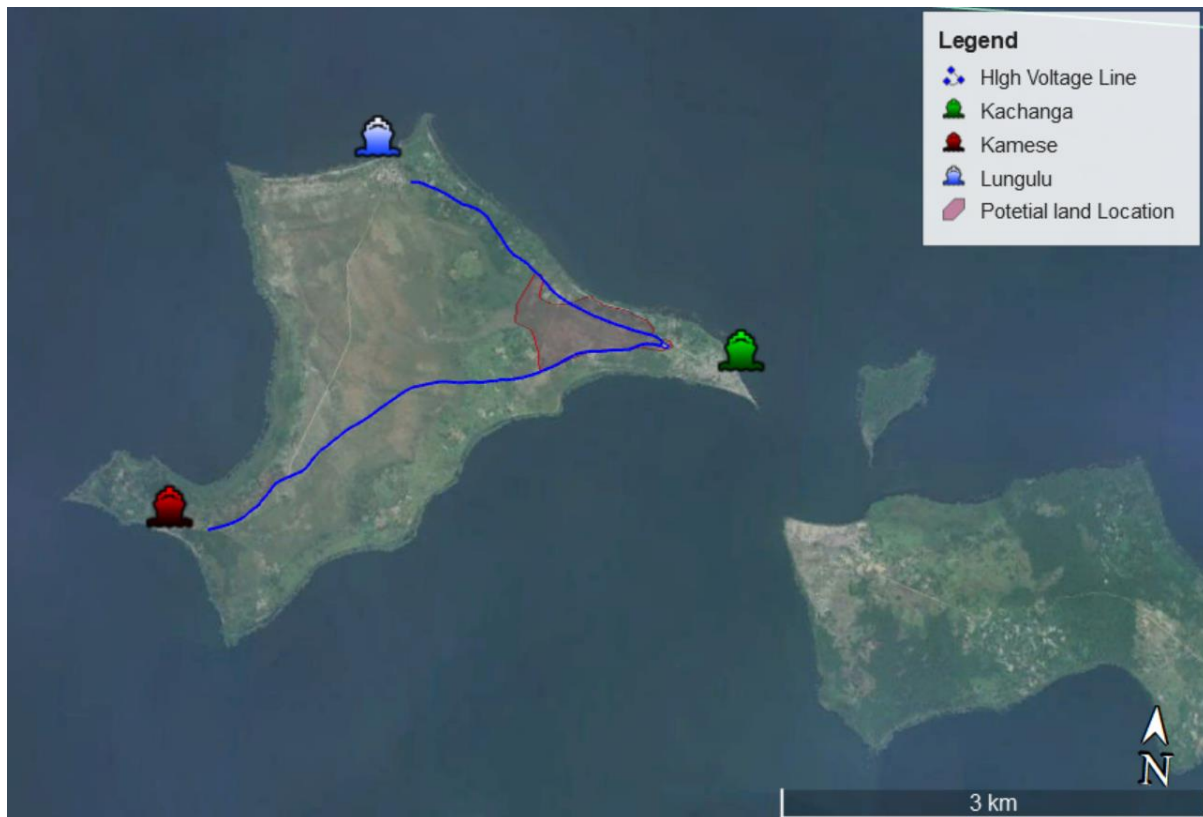


Figure 15 Medium Voltage Lines

Each Medium Voltage line is supported by poles spaced between 30 and 40 metres. Transformers are proposed at the end of each line to step down the voltage to 400 V (three phase).

3.4.3 Conductors

It is typical for the mini grid developer to select the conductor material and size based on the load demand and the project budget. However, within Uganda, as the distribution system is installed by the grid operator, the conductors used are standardised.

Within this work two scenarios are considered for conductor selection. The first scenario relies on MEI prior project experience and the second scenario replicates what has previously been used by the Ugandan grid operator (information supplied by GRS Commodities). The characteristics of each as well as the conductor characteristics of those found in the HV lines are detailed in Table 5:

Table 5 Technical Specifications for Cabling [54]

Component	MEI	Ugandan Grid Operator
Single Phase (230)		
Material	Copper	N/A
Conductance (k)	56	N/A
Cross Sectional Area (mm ²)	50	N/A
Three Phase (400V)		
Material	Copper	Aluminium
Conductance (k)	56	38
Cross Sectional Area (mm ²)	120	100
Medium Voltage (33kV)		
Material	ASCR	ACSR
Conductance (k)	38	38
Cross Sectional Area (mm ²)	100	100

3.4.4 Voltage Loss Analysis

The aim of the voltage drop analysis is to assess the maximum possible voltage drop that could occur within the system. This is assessed by determining the power demand and subsequent voltage drop that would arise from all electrical loads being turned on simultaneously on the island.

To assess total voltage drop, the voltage drop is calculated for each line individually and these values are summed. The grid therefore is first split into individual lines which are categorised as 'End', 'Intermediate', or 'Source' segments. A sample line is characterised in Figure 16:

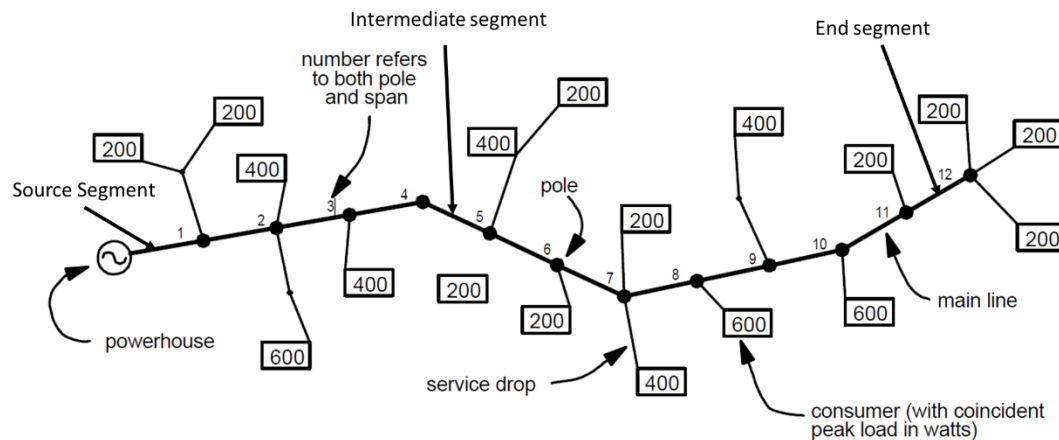


Figure 16 Sample Line [25]

The maximum power demand at each node is assessed and this is done by summing the rated powers of all appliances for all connected customers at that node. The maximum customer current demand is then assessed using Equation 2:

Equation 2 Line Current

$$I = \frac{P}{V}$$

Where;

I = current (A)

P = power (W)

V = voltage (V)

At each node the maximum current of each connected customer and connected lines are summed and the line voltage drop can be assessed. Equation 3 and Equation 4 are used for calculating the voltage drop in single and three phase lines respectively:

Equation 3 Single Phase Voltage Drop

$$VD\% = \frac{200 \times I \times L \times p.f.}{K \times A \times V}$$

Equation 4 Three Phase Voltage Drop

$$VD\% = \frac{\sqrt{3} \times 100 \times I \times L \times p.f.}{K \times A \times V}$$

Where;

VD% = voltage drop (%)

I = current (A)

L = line length (m)

p.f = power factor

K = coefficient of conductance

A = cross sectional area (m²)

V = voltage (V)

As shown in Figure 16, the voltage analysis begins in Line_{11, 12} and works towards the source which each subsequent current being added to the successor. Line voltage drops for the whole network are summed and a total voltage drop for the system is ascertained. A sample table from the voltage loss analysis is characterised in Table 6.

Table 6 Line 42-43 Voltage Drop Table

End Segment		Phase	1		Type of Conductor	Copper	Size of Conductor	50	mm2	
Line Segment 43-42	Customer Segment	Pole	Demand (KW)		Voltage (V)	Current (A)		Length (m)	Voltage drop	
Connection			Spurs	Total		Spurs	Total		(V)	%
1	MHH	44	0.015	0.015	230.000	0.082	0.082	43	0.022	0.01
2	MHH		0.015	0.030	230.000	0.082	0.163			
3	MHH		0.015	0.045	230.000	0.082	0.245			
4	MHH		0.015	0.060	230.000	0.082	0.326			
5	MHH		0.015	0.075	230.000	0.082	0.408			
6	MHH		0.015	0.090	230.000	0.082	0.489			
7	MHH		0.015	0.105	230.000	0.082	0.571			
8	MHH		0.015	0.120	230.000	0.082	0.652			
9	MHH		0.015	0.135	230.000	0.082	0.734			
10	MHH		0.015	0.150	230.000	0.082	0.815			
11	CHU		0.015	0.165	230.000	0.082	0.897			

3.5 Financial Modelling

Following on from sizing a system using HOMER and designing an appropriate grid, the system is financially modelled in order to assess what financial support is required in order to satisfy the proposed tariff structure.

The methodology employed for conducting the financial feasibility assessment has been adapted from MEI in-house standard practice. The following section of the report details the methodology employed for financially assessing the mini grid and determining a financial structure that can support the proposed tariff. The methodology takes into account: assets and sales; investments; profits and losses; cash flow, outgoings, and key performance indicators. A reference case scenario was modelled for the initial analysis. The main variable parameters of this reference case model are listed in Table 7:

Table 7 Financial Analysis Reference Case Conditions

Variable Parameter	With Ice Plant	Without Ice Plant
Ice Sales volume	2500 kg per day every year	N/A
Ice Cost	300 UGX / kg (\$0.08 / kg)	N/A
Diesel Cost	1.07 (USD/kg)	
Technical Configuration	See Table 42	
Asset investment costs	Medium cost scenario (see Table 4)	
Distribution Network Costs	Covered by grid operator	
Project Lifetime	20 years	
Connections year 0	455	
Connections year 20	910	
Connection fee to customer	185 (USD)	
Load growth	Linear	
Grants for Assets	50% - Specified by GRS Commodities	
Grants for Development Costs	30% - Specified by GRS Commodities	
Debt from investment costs (minus grants)	10% - Specified by GRS Commodities	
Loan in foreign currency	100% (USD)	
Interest rate of foreign current loan	8.5% - Specified by GRS Commodities	

The following sub-sections detail the financial model employed for the reference case.

3.5.1 Overheads

3.5.1.1 Capital Investment.

The major capital investment costs taken into account are highlighted in section 3.3.

3.5.1.2 Operation and Maintenance Costs

Table 8 details the major operation and maintenance costs taken into account as specified by GRS Commodities.

Table 8 O&M Considerations within Financial Modelling

O&M Costs	
O&M Assets [USD/year]	242
Local operational management of assets [USD/year]	3,221
Insurance [USD/year]	256
O&M of distribution system [USD/year]	2,788
O&M Ice plant [USD/year]	700
Mobile payment fee [%]	2.00

3.5.1.3 Project Development and Overhead Costs

Table 9 details the major project development and overhead costs taken into account. The figures listed below are taken from GRS Commodities.

Table 9 Project Development and Overhead Cost Considerations for Financial Modelling

Project development cost		Overhead cost	
Component	USD	Component	USD
Feasibility study	805	Salary Managing Director	2,148
Environmental Impact Assessment	134	Management and technical staff	1,879
License acquisition	134	Accounting	242
Acquisition of capital incl. due diligence	134	Travel and vehicle cost	268
Company foundation and establishment	0	Office costs	27
Acquisition of land usage rights	1,208	Consultancy cost and fees	81
Community engagement	268	Company insurance	107
		Miscellaneous	81
TOTAL	2,685	TOTAL	4,832

3.5.1.4 Total Expenditure

The capital investments, O&M costs, and project development and overhead costs are summed to assess the total annual capital expenditure each year taking into account inflation, as shown in Equation 5:

Equation 5 Total Annual Expenditure

$$TEX = O + P + D + CC + H$$

Where;

T= total expenditure (USD)

O = Operation costs (USD)

P = Power generating asset costs

D = Distribution system costs

CC = Customer connection costs

H = Household installation costs

3.5.1.5 Reinvestment Costs

The lifetime of all assets considered within the financial model and thus the reinvestment costs are also considered at the end of each assets lifetime. The reinvestment costs for technical assets have been taken from discussions with GRS Commodities as has the amount of money that is salvageable from the sale of end-of-life assets.

3.5.1.6 Grants

The amount of investment that comes from grants is taken into account whereby a percentage of total required investment can be allocated to 'grants for assets' or 'grants for development costs'. Both of these are considered variables within the analysis and optimum values are found based on the tariff applied for and the amount of funding available locally. It was specified in the project documents that 50% of costs for assets and 80% of costs for development will come from grants. The grants modelled in the reference scenario are listed in Table 10:

Table 10 Investments from Grants

	USD
TOTAL investment	514,154
Grant for assets	255,643
Grant for development	2,148

3.5.1.7 Debt from Investment

It is specified in the project documents that 10% of the private investment will be from debt (not including grants). The required private investment and the size of the foreign currency loan are listed in the following table:

Table 11 Debt from Investment

	USD
Total Private investment	256,363.77
Debt from investment costs (minus grant)	25,636.38
Equity from private investment costs	230,727.39

The specific conditions of the debt loan required are highlighted in Table 12:

Table 12 Conditions for Debt Loan

Debt	Loan in foreign currency [USD]	Loan in local currency [UGX]
Share for each loan	100.0%	0.0%
Amount of the Loan	25,636.38	0
Starting Year of the Loan	2019	2019
Load Tenor [years]	10.0	5.0
Interest rate for Loan	8.5%	21.8%
Debt Grace Period [years]	1.0	1.0

As shown in Table 12, the entirety of the loan is taken in USD to take advantage of the preferential interest rates.

3.5.1.8 Weighted Average Cost of Capital (WACC)

The weighted average cost of capital (WACC) is the average rate at which a company should pay to all security holders to finance assets. The WACC can be considered as the cost of capital to a business, and represents the minimum return that a company must earn on an already existent asset to satisfy relevant stakeholders. The WACC is assessed using Equation 6,

Equation 6 Weighted average cost of capital

$$WACC = \frac{Db}{Db + E} K_d + \frac{E}{Db + E} K_e$$

Where;

Db = total debt (USD)

E = shareholder equity (USD)

K_d = cost of debt (USD)

K_e = cost of equity ((USD)

3.5.1.9 Tax

The local tax conditions in Uganda are taken into account within the model as an input. The taxes considered are detailed in Table 13 which in turn enables the WACC to be assessed.

Table 13 Tax Considerations within Financial Model

Income tax	30.0%
Regulator Tax	0.0%
Other Tax	0.0%
Communal Tax	0.0%
Interest Rate Equity	25.0%
WACC (discount factor)	14.2%

3.5.1.10 Reinvestments

Depreciation and reinvestment for all assets within the system is taken into account in the model. Table 14 lists the asset lifetimes and total costs for reinvestment. The reinvestment costs are adapted to inflation within the model.

Table 14 Reinvestments for Assets

Asset	Lifetime (years)	Reinvestment Cost (USD)
Solar PV	21	318,600
Battery	7	65,200
Inverter	10	64,800
Ice plant, storage, freight, insurance	21	32,550
Ice Plant Inverter	7	1,735
Connection cost 1 phase users	12	83,794
Connection cost 3 phase users	12	184
Total Reinvestment Cost		561,823

3.5.1.11 Cost and Revenue Indexation

The financial viability of the system is intrinsically linked with the local diesel cost, cost indexation, indexation for reinvestments, and the local exchange rate to international currencies. Table 15 summarises the major cost and revenue indexation parameters used in the model. All values have been taken from discussions with GRS Commodities as well as research:

Table 15 Cost and Revenue Indexation

	Indexation	Year 1
Nominal Discount Rate	10%	N/A
Diesel fuel [USD/L]	3.5%	1.07
General cost indexation [%]	2.0%	1.00
Indexation for reinvestments	2.0%	1.00
Exchange rate USD - UGX	5.0%	3,911

3.5.2 Profit and Loss Statement

The main indicators of the project's income statement are: the estimated earnings before interest, tax, depreciation, and amortization (EBITDA), estimated earnings before interest and taxes (EBIT), the estimated earnings before tax (EBT), and the annual net income.

The EBITDA is used as an alternative to income or net earnings and is calculated using Equation 7:

Equation 7 EBITDA

$$EBITDA = NI + I + Tx + Dep + Am$$

Where;

NI = Net income (USD)

I = interest (USD)

Tx = taxes (USD)

Dep = depreciation (USD)

Am = amortization (USD)

The EBIT is an indicator of a company's profitability and is assessed from the revenue minus expenses, excluding tax and interest, as highlighted in Equation 8

Equation 8 EBIT

$$EBIT = TR - TEX$$

The EBT is a measure of a company's operating and non-operating profits and is assessed from revenue minus expenses, not including taxes. All of the above measures are calculated annually within the financial model, taking into account inflation, reinvestments, loan conditions, and transient revenues.

3.5.3 Key Performance Indicators (KPIs)

The KPIs used to assess the feasibility of the system are the NPV, project IRR, equity IRR, lower DSCR, revenue, average operating profit margin, and lower operating profit margin.

The NPV is the present value of future cash flows discounted at a certain discount rate and determines the sum of cash flows for a long term investment. It is characterised in Equation 9.

Equation 9 Net Present Value

$$NPV = \sum_{t=0}^n \frac{C_t}{(1+r)^t}$$

Where;

N = project lifetime,

t = time of the cash flow,

r = discount rate, and

C_t = net cash flow at time t.

The Internal Rate of Return is the discount rate which, when applied to a stream of cash flows, generates NPV of zero. It is a widely held industry standard that for an isolated mini grid, the IRR should average above 15% - this value has been confirmed by MEI, INSENSUS, and GRS Commodities [55].

Calculation of the internal rate of return considering the cash flows net of financing gives the equity IRR. If the project is fully funded by equity, the project IRR and Equity IRR will be the same. If the project is fully funded by the debt, equity IRR does not exist.

The Debt Service Coverage Ratio (DSCR) is the ratio of cash available for debt servicing (e.g. interest payments) and this is a measure of the cash flow available to pay current debt obligations. The ratio considers net operating income as a multiple of debt payment obligations due within the next 12 months. This relationship is characterised in Equation 10

Equation 10 Debt Service Coverage Ratio

$$DSCR = \frac{\text{Net Operating Income}}{\text{Total Debt Service}}$$

Within the financial model it is calculated using Equation 11 on an annual basis:

Equation 11 DSCR Calculation

$$DSCR = \frac{EBIT + \text{Sum Profit/loss}}{\frac{\text{Loan redemption} - \text{interest expenses}}{(1 - \text{corporation tax})} + \text{interest expenses}}$$

Within mini grid financing an industry norm is to aim for a DSCR of over 110% [56]. This value has been further confirmed through discussions with MEI, and GRS Commodities.

The levelized cost of electricity (LCOE) is a measure of the average cost of electricity over a project's lifetime. It is assessed by taking all expected lifetime costs and dividing it by the expected power output of the system. It is the NPV of the cost of one unit of electricity over the lifetime of the project, and is thus the average minimum price at which electricity should be sold in order to break even. It is assessed using Equation 12,

Equation 12 Levelized Cost of Electricity

$$LCOE = \frac{\text{sum of costs over lifetime}}{\text{sum of electrical energy produced over lifetime}} = \frac{\sum_{t=1}^n \frac{I_t + M_t + F_t}{(1+r)^t}}{\sum_{t=1}^n \frac{E_t}{(1+r)^t}}$$

Where;

I_t = capital cost,

M_t = O&M cost,

F_t = fuel cost,

E_t = electrical energy generated,

r = discount rate, and

n = expected lifetime:

The LCOE is highly dependent on the lifetime of the project considered. Typically mini grids are assessed for a project lifetime of 15-25 years and this project has a lifetime of 20 years.

By assessing the IRR, equity IRR, NPV, and LCOE, a decision can be made on the financial feasibility of the system. A sensitivity analysis is employed for various factors in the model so as to assess the impact that they play on the feasibility of the system. The overall goal is to achieve an LCOE that is lower than the proposed tariff and an average project IRR of over 110% over the course of the project lifetime.

3.5.4 Risk Analysis and Financial Analysis Sensitivity

The literature findings clearly indicated that a major problem facing the effective design of mini grids is lack of data accuracy regarding both the load demand and the cost of technology and services. Accordingly, it was fundamental to conduct a risk assessment that quantified these perceived risks. A SWOT analysis was first conducted, followed by a risk identification analysis.

The sensitivity analysis assumes that the system size is that of the optimised HOMER model. The following parameters were varied in financial model:

- Investment cost
- Connection fee
- Connection rate
- Load demand per capita
- Ice sales
- Investment from grants
- Debt from investment
- Inclusion of grid infrastructure costs

The results of the above studies are shown in Section 4.3.

Two additional financial studies were completed so as to verify the impact that Ugandan energy Policy and PUEs have on mini grid feasibility. The first additional study includes the cost of grid infrastructure, and the second additional study excludes the use of the ice plant respectively. The results of these studies are detailed in the Appendices K and L respectively.

3.5.5 Revenues

3.5.5.1 Income from Connection Fees

The system sizing study is conducted for 100% connection rate on the island. Within the reference scenario of the financial analysis it is assumed that a full connection rate is achieved in year 20 of the project. In the reference scenario it is assumed that connections are increased at a linear rate from year 0 to year 20. The pattern by which the connection rate increases annually has impact on the financial feasibility of the system and is thus subject to a sensitivity analysis, see Section 4.5.2.

A connection fee is charged for all single and three phase users and thus a revenue is generated by annual increases in connections. Through discussion with GRS Commodities, the proposed connection fee in year 0 is \$185 and this increases with inflation. This fee covers the costs associated with connecting the customer to the system and thus the developer breaks even.

3.5.5.2 Income from Electricity Sales

A revenue is generated from the sale of electricity at the set tariff rate. As discussed in Section 1.3, within Uganda the mini grid operator applies for a tariff cost that is roughly that of local grid costs. Accordingly, an initial tariff is set as well as an annual tariff growth rate which is fixed with inflation. The number of sales (N° of kWh) is then multiplied by the tariff (UGX/kWh) to assess the revenue generated. The tariff used in the reference model for single phase users is 927 UGX/kWh (0.25 USD/kWh) and this is fixed with inflation.

3.5.5.3 Income from the Ice Plant

Based on the business model employed, the Mini Grid will generate revenue from the ice plant through the sale of flake ice.

As reported in the field data, there is a daily ice demand of 2.5 tonnes. Currently on the island, ice is sold at 300 UGX per kg (\$0.14 / kg). This is modelled as a revenue for the system. A sensitivity analysis is employed that investigates the effect that ice sales have on overall economic performance, see Section 4.5.4.

3.5.5.4 Total Revenues

The total annual revenues can be assessed by summing all revenue streams, as characterised in Equation 13

Equation 13 Total Annual Revenue

$$TR = ES + CF + IC + IS$$

Where;

TR = Total revenues (USD)

ES = electricity sales (USD)

CF = connection fees (USD)

IC = income from charges (USD)

IS = ice sales (USD)

3.5.5.5 Gross Profit

The annual gross profit for the system is calculated by the Equation 14:

Equation 14 Annual Gross Profit

$$GP = TR - TEX$$

Where;

GP = annual gross profit

4 Results

The following section provides results summary of the following:

- The HOMER sizing study that suggested the best sized components for the mini grid
- The configuration of the proposed distribution network, including:
 - The estimated capital cost of the network to the grid operator
 - The estimated O&M cost for GRS Commodities
 - The estimated voltage drop of the network
- The financial analysis of the 20 year project
- The risk analysis of the project
 - Risk map
 - Risk mitigation strategy
- The sensitivity analysis which assessed the projects financial risks

4.1 HOMER Sizing Study

Two HOMER sizing studies were undertaken, one with optimised conditions whereby there are no limitations on the technical specification of the components, and one study where the realistic battery bank and converter conditions are applied.

4.1.1 Optimised Sizing Study

The configuration and size of the power generating assets was assessed using HOMER. Within this reference case, the following inputs are applied:

- The load profile highlighted in Figure 8
- The medium costs highlighted in Table 4
- The solar irradiance shown in Figure 30.

It should be noted that the NPC and LCOE results stated in this section are only preliminary results as they do not account for the revenues generated from the tariff or ice sales. Also, HOMER uses the net present cost (NPC) result, which is the inverse of the NPV, whereby costs are positive, and revenues are negative. The technical configuration proposed for the reference case is highlighted in Table 16. The values referenced are the optimised outputs based on the demand, costs, and solar resource inputs.

Table 16 HOMER Reference Scenario Technical Configuration

Renewable Penetration (%)	Solar PV (kW)	Diesel Generator (kW)	Battery (kWh)	N° Batteries	Converter (kW)	NPC (USD)	LCOE (USD/KWH)
76.98	227	71	412	412	66	1,299,030	0.40

As stated in Table 16 the optimal battery bank size is 412 kWh assuming that an idealised battery is used within the system. This configuration of the battery is not realistic as within rural mini grids of low power quality it is advisable to have only 4 strings of batteries and a maximum line voltage of at 48 V; the current idealised battery has a voltage of 12 V. It is also an industry standard that the total converter size should be scaled to be 10% greater than the peak demand. Twelve Sunny Island 8.0H converters have a capacity of 72 kW and would be a more realistic selection for the model.

Additionally, there is a 77% renewable penetration despite a relatively large generator having to be in place for times of low solar resource (71 kW) and this implies that a high initial investment is needed for an underused generator. This is highlighted in Figure 17 where the red line is the load demand the blue line is the generator output power, and the green line is the battery state of charge:

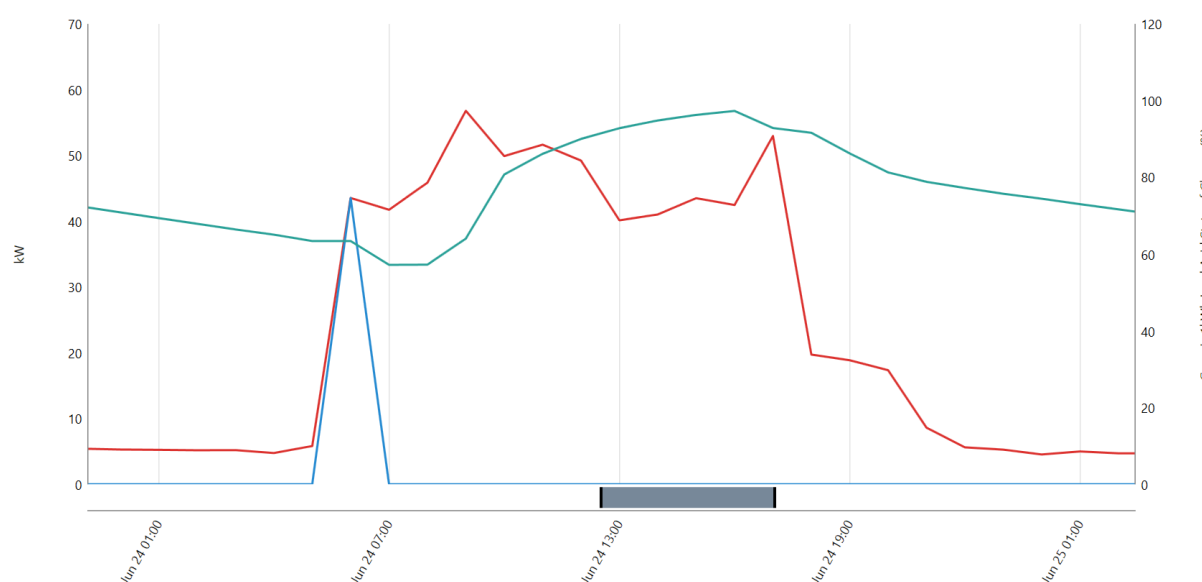


Figure 17 Battery state of charge (green), generator output power (blue), and power demand (red) for Optimised Homer Simulation

As shown in Figure 17, for a typical day in June the generator is only used for two hours and the battery state of charge never reduced below 60% despite having a minimum state of charge of 30%. Considering this, a subsequent study was conducted with realistic battery conditions set, as detailed in Section 4.1.2.

4.1.2 Sizing Study with Specific Components

The BAE 26 PVS model and Sunny Island 8.0H components were implemented into the model. The configuration outputs of the proposed system are conveyed in Table 17:

Table 17 Battery Study Results

Renewable Penetration (%)	Solar PV (kW)	Diesel Generator (kW)	Battery (kWh)	N° Batteries	Converter (kW)	NPV (USD)	LCOE (USD/KWH)
54.6	178	71	163	24	72	1,26,2823	0.40

The state of charge of the battery, and output power of the generator over the course of a typical day are conveyed in Figure 18:

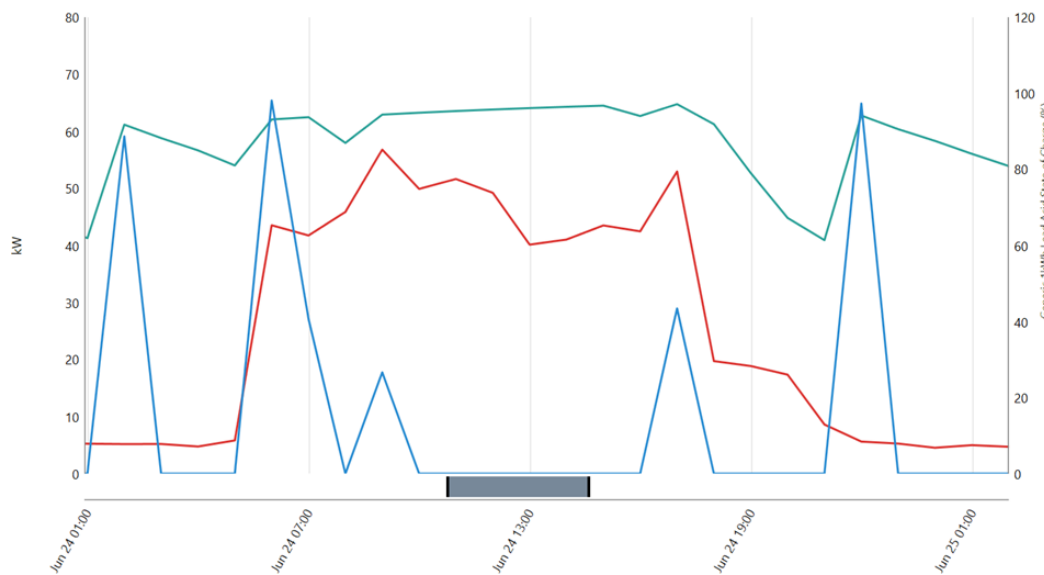


Figure 18 Battery state of charge (green), generator output power (blue), and power demand (red) for Battery Homer Simulation

As shown in Figure 18, the generator is used multiple times throughout the day and the battery has a minimum state of charge of 60% and this is a more economical use of equipment than that of the previous solution.

An additional sizing study was conducted investigating the impact of not modelling an ice plant. The results of this are seen Appendix J. The model without the ice plant has a larger LCOE which at this stage in the work indicates that the project may be less profitable - as the tariff is fixed to grid cost.

4.1.3 Sizing Sensitivity Study

Within the sizing study it was fundamental to conduct a sensitivity study so as to assess the implications that difference in costs and load demand have on the technical design.

4.1.3.1 Investment Cost Variation

The cost variations shown in Table 4 were used for the cost sensitivity study. The results of the study are conveyed in Table 18:

Table 18 Homer Sizing Study with Ice plant and variable renewable penetration

Cost Scenario	Renewable Penetration	Solar PV (kW)	Diesel Generator (kW) Diesel Generator (kW)	Battery (No)	Converter (kW)	NPV (USD)	LCOE (USD/KWH)
Medium (ref)	54.6	178	71	24	72	1,262,823	0.40
Low	62.29	219	71	48	72	1,098,625	0.34
High	52.61	158	71	24	72	1,393,023	0.43

As shown in Table 18, the low cost scenario results in a significantly larger PV system and battery bank being used. It also has the lowest LCOE and is therefore the more favourable scenario. It is imperative that GRS Commodities reassess the optimised technical design once more accurate capital cost data is obtained.

4.1.3.2 Renewable Penetration

The impact that a fixed renewable penetration had on performance was investigated. The daily energy demand in this study is 694.54 KWh with a load distribution as shown in Figure 7, with results shown in Table 19:

Table 19 Homer Sizing Study with Ice plant and variable renewable penetration

Minimum Renewable Penetration	Optimised Penetration	Solar PV (kW)	Diesel Generator (kW)	Battery (No)	Converter (kW)	NPV (USD)	LCOE (USD/kWh)
Optimised	54.69	177	71	24	72	1,262,823	0.39
60	60.21	232	71	24	72	1,276,032	0.40
70	70.08	330	71	48	72	1,392,379	0.43
80	80.01	293	71	72	72	1,458,339	0.45
90	90.06	371	71	96	72	1,565,516	0.49
100	100.00	2085	0	96	72	4,342,180	1.35

As highlighted in Table 19, the LCOE increases with the renewable penetration. A reason for this is that the ice plant, which is a significant load to the system, operates in early evening hours of low sunlight. The optimised system has a high dependence on the diesel generator and this increases project risks. This is discussed further in the risk analysis, see Section 4.4.

4.1.3.3 Diesel Cost

Alongside the differing levels in investment needed for the PV array, batteries, and converter, the fuel expenditure for the generator will have impact on the optimised size of the system. Additionally, seeing as diesel prices in SSA are subject to high variation, it was paramount to assess the effect it has on system sizing considerations. Table 20 lists the results of the diesel cost sensitivity study whereby the reference scenario is used (1.07 USD/l [57]) with the ice plant installed.

Table 20 Impact of Diesel cost on System Sizing

N°	Diesel Price (\$/L)	Renewable Penetration	Solar PV (kW)	Diesel Generator (kW)	Battery (No)	Converter (kW)	NPC (USD)	LCOE (USD/KWH)
1	1.07 (ref)	54.69	177	71	24	72	1,262,823	0.39

2	0.8	51.40	152	71	24	72	1,136,207	0.35
3	0.9	51.29	152	71	24	72	1,184,645	0.37
4	1	54.72	177	71	24	72	1,231,307	0.38
5	1.1	54.68	177	71	24	72	1,276,477	0.40
6	1.2	57.49	203	71	24	72	1,320,527	0.41
7	1.3	57.48	203	71	24	72	1,362,622	0.42

As highlighted in Table 20, as the diesel input variable price increases, the HOMER model calculates that it is more cost effective to have a larger PV system with less reliance on the diesel genset over the system lifetime. As shown, at \$1.07/l the optimised PV array size is 177 kW, however at \$1.3/l, the optimised PV size is 203 kW. The increasing fuel costs for operation make the increased investment cost for PV preferential. Again, an increasing, or indeed unpredictable, fuel price greatly heightens projects risks this is discussed further in the risk analysis, see Section 4.4.

4.1.3.4 Ice Plant Operating Hours

The hours at which the ice plant operates has significant impact on the configuration and size of the system due to the hours of sunlight and peak load variation. Different operating hours have been investigated for the ice plant. The plant shall operate for twelve hours a day so as to produce 2.5 tonnes of flaked ice. Table 21 lists the results of the assessment:

Table 21 Impact of Ice Plant Operating Hours on System Sizing

Nº	Operating Hours	Renewable Penetration	Solar PV (kW)	Diesel Generator (kW)	Battery (No)	Converter (kW)	NPC (USD)	LCOE (USD/kWh)
1	Ref (6am - 6pm)	54.69	177	71	24	72	1,262,823	0.39
2	5am-5pm	53.49	167	71	24	72	1,275,789	0.40
3	7am-7pm	53.49	167	71	24	72	1,275,789	0.40
4	8am-8pm	54.12	170	71	24	72	1,277,286	0.40
5	9am-9pm	55.10	174	71	24	72	1,249,741	0.39
6	6pm-6am	38.08	121	53	24	72	1,425,789	0.44

As highlighted in Table 21, it is highly preferential for the Ice Plant to operate during daylight hours. As shown in scenario six listed above, night time operation results in low renewable penetration and high fuel expenditure, vastly increasing project risks. As shown hours of 6am-6pm and 9am – 9pm propose the lowest LCOEs. However, operating after 7pm brings about addition risks due to peak domestic load being at this time and this could result in unwanted voltage surge and reduced service. For this reason it is preferential for the ice plant to remain operating from 6am – 6pm.

4.1.3.5 Load Scaling

As highlighted in the literature review it is highly likely that the energy demand from customers will be notably different from that predicted in the load demand assessment. The literature highlighted cases of loads being up to four times smaller than those predicted and this has been replicated in the sensitivity study [20] [27] [28].

Accordingly, two studies have been conducted that investigate the impact of difference in load scaling. The first study scales the entirety of the load, including households, community, commercial, and ice plant connections. The second study assumes that the ice demand remains constant and only scales the loads from households, community and commercial loads. Within this sensitivity study the battery

bank limitation are consistent with industrial reality, and the converter capacity is scaled proportional to 10% above the peak demand. Table 22 lists the results of full load scaling:

Table 22 Scaled Load

No	% Load	Daily Demand	Renewable Penetration	Solar PV (kW)	Diesel Generator (kW)	Battery (kWh)	Converter (kW)	NPC (USD)	LCOE (USD/KWH)
1	Ref	694.55	55.44	184	71	24	72	1,266,596	0.39
2	25%	174	65.33	48	18	24	18	420,426	0.52
3	50%	347	59.01	94	36	24	36	685,267	0.43
4	75%	521	56.20	136	53	24	54	961,506	0.40
5	125%	868	55.44	212	89	48	90	1,544,108	0.38
6	150%	1042	55.67	264	110	48	108	1,821,861	0.38

As shown in Table 22, as the load increases, as do the systems components and the cost of electricity provision reduces. It can be seen that if the system is sized for scenario 2, the cost of electricity provision is 33% higher than the reference case.

Table 23 lists the results of the scaling study where the ice plant demand remains constant:

Table 23 Scaled Load with Constant Ice Demand

No	% Scale	Daily Demand	Renewable Penetration	Solar PV (kW)	Diesel Generator (kW)	Battery (kWh)	Converter (kW)	NPC (USD)	LCOE (USD/KWH)
1	Ref	694.55	55.44	184	71	24	72	1,266,596	0.39
2	25%	333	62.32	87	36	24	36	639,836	0.41
3	50%	454	56.99	114	48	24	41.4	841,552	0.40
4	75%	574	56.03	147	59	24	54	1,033,373	0.39
5	125%	815	54.35	234	94	48	82.8	1,654,376	0.38
6	150%	936	54.35	234	94	48	82.8	1,654,376	0.38

Again, in this study, as the load reduces, the cost of electrical provision increases. It can be seen again that if the single phase customers only reach 25% of predicted demand (Scenario 2), then the cost of electricity provision is 5.2% higher than the reference case. It is therefore paramount that GRS Commodities conduct a more robust load demand assessment at a later project stage.

4.2 Distribution Network

Grid configurations were modelled and costed for each of the communities on the island as well as for the Medium Voltage distribution network needed to connect all three.

For the community grids two configurations have been considered. The first is built upon MEIs past experience and the second is modelled upon what the Ugandan grid operator may install. The main difference is that in the second case only three phase cables are used for the entire grid network whereas in the first instance single cables are used for the breakaway branches. The results of the voltage drop analysis are listed in Table 24:

Table 24 Voltage Drop Analysis

Grid	Voltage Drop (%)	
	Three Phase and Single Phase Copper	Three phase aluminium

Kachanga	9.808	9.879
Kamese	0.395	0.487
Luwungulu	0.085	0.113
Medium Voltage Lines		
PV→ Kam	9.37968E-05	
PV → Luw	1.03921E-05	

As shown in Table 24, the combined voltage loss is 10.38% this makes the grid of a suitable standard in accordance with grid quality assurance standards highlighted in NREL framework [26].

A cost analysis was conducted for the grid configuration. The costs used for this study have been taken from World Bank reports that estimate the installed costs of infrastructure in Uganda and this has been confirmed by GRS Commodities [54]. The results are listed in Table 25.

Table 25 Grid Infrastructure Cost

Component	Cost (USD)	Quantity	Total (USD)
33 kV 3 Ph (100 mmsq ACSR)	13,085/km	5.493	71,876
400 LV Line (100mmsq Al)	11,165/km	2.831	31,608
50 kVA Transformer	5,439/unit	3	16,317
Pole	121.99/unit	220	26,838
Total Investment			146,639
Annual O&M Cost			2,788

As shown in Table 25, the proposed grid infrastructure has a total estimated cost of \$146,639 that shall be covered by the operator, and the annual O&M costs are \$2,788 that shall be covered by GRS Commodities.

4.3 Financial Analysis

The following subsection provides a summary of the financial analysis, which assesses the projects performance over a 20 year lifetime. In each study the mini grid has two revenue streams: the sale of single phase electricity, and the sale of ice. Included are the results of the sensitivity analysis. Each scenario is labelled with a numerical indicator for the comparative analysis detailed in Section 4.5.8.

The variable conditions for the reference scenario are summarised in Table 26. The KPIs for the reference case scenario are highlighted in Table 26:

Table 26 Key Performance Indicators for Reference Case

KPIs	
NPV (USD)	114,651
Project IRR	23.5%
Equity IRR	24.3%
Lowest DSCR	529.0%
LCOE in [USD/kWh]	0.15
Revenues in [USD/kWh]	0.16
Avg. Operating Profit Margin	0.35
Lowest Operating Profit Margin	0.26

As highlighted in Table 26 the IRR is above the WACC (15.3%) and the DSCR is above 110% which are the minimum requirements for the model being financially viable. The LCOE is stated at 0.16 USD/kWh which is below the proposed tariff of \$0.26 / kWh (927 UGX). The income statement for the mini grid over a 20 year lifetime is summarised in Figure 19:

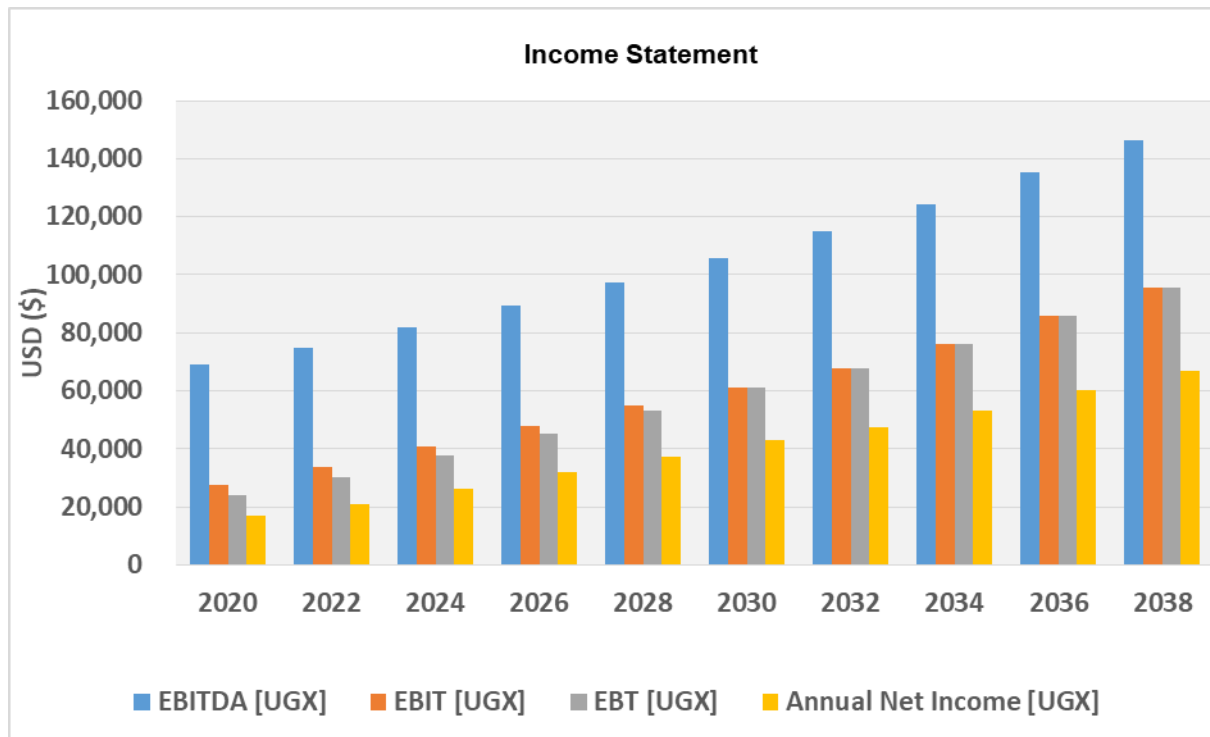


Figure 19 Income Statement for Reference Case Scenario

As highlighted in Figure 19, the project returns a profit from 2020 (year 1). A full summary of the project costs are highlighted in Figure 20:

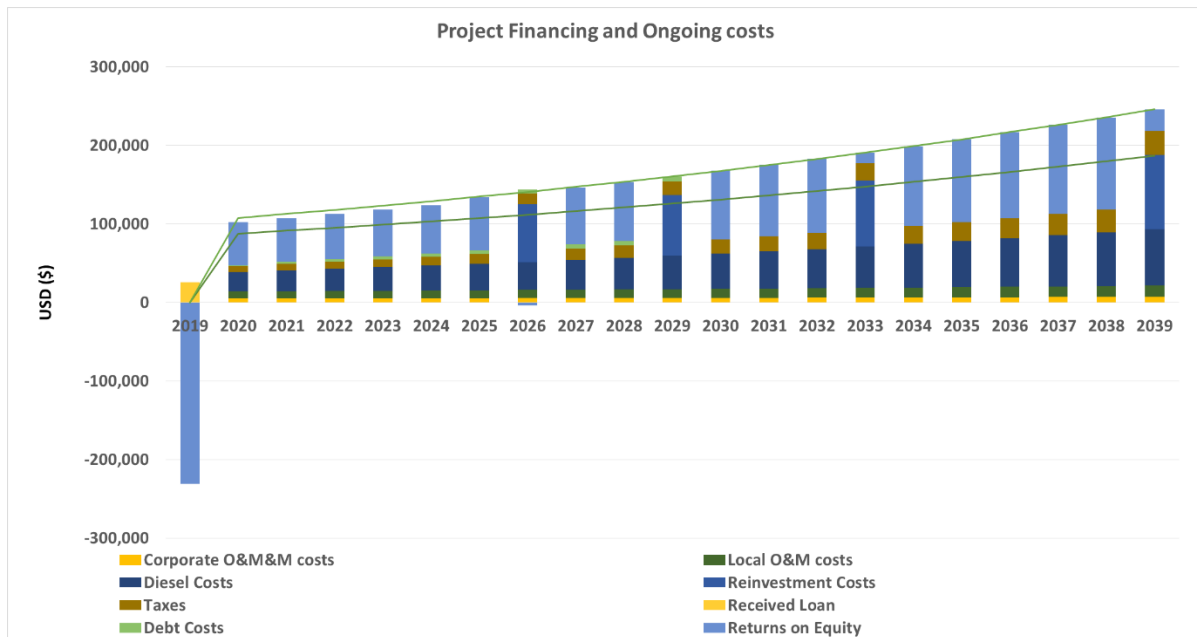


Figure 20 Reference Case Project Financing and Outgoing Costs

As shown in Figure 20, following on from an initial investment in 2019, the project returns a profit. Reinvestment costs for assets can be seen in years 2026, 2029, 2033, and 2039 where batteries, converters, and customer connections need to be replaced.

There are three key reasons for the preferential performance of the mini grid, as follow:

1. Ice sales represent a high portion of total revenues
2. The project is part funded by assumed grants (50% for assets)
3. The company investment only requires a 10% debt loan
4. The CAPEX for grid infrastructure is not included

All of the above points are highly uncertain and are assessed in the sensitivity study – see Section 4.5.

Although it is almost certain that the government will pay for the grid infrastructure, a study was assessed on the impact that paying grid CAPEX will have on operative performance, and this attached in Appendix K. As highlighted in the additional study, the inclusion of the grid still returns feasible results however the NPV is \$100,000 lower and the project is far more susceptible to financial risk.

Likewise, a study was completed of the financial performance of the mini grid if no ice plant were planned. The results are seen in Appendix L. As shown, the project is unfeasible and returns an NPV of -\$100,481.

4.4 Risk Analysis

The following section highlights the results of the Risk Assessment in the form of a risk map and a list of strategies so as to mitigate the risks highlighted in the financial risk analysis. The precursors to this study were a SWOT analysis and a Risk Impact Assessment. These studies are included in Appendix O.

4.4.1 Risk Map

The results of the Risk Impact Assessment were input into a risk map, adapted from the Alliance for Rural Electrification Framework [58]. This is conveyed in Figure 21. Risks have been categorised based on the action that needs to be taken to mitigate them, summarised as follows:

- Risk Tolerance – These risks have low probability and low consequence. Stakeholders should be aware of them but little action is required [58]
- Risk Transfer – These risks have low probability but high consequence. They can cause severe harm to the project and can typically be mitigated through outsourcing the issue to a third party [58]
- Risk reduction – These risks have high probability and low severity. Developers should implement resource buffers or utilise emergency measures to prevent occurrence [58].
- Risk avoidance – These risks have high likelihood and severity. Direct action should therefore be taken from an early project stage to remove them. Failure to do so could result in an early project exit [58].

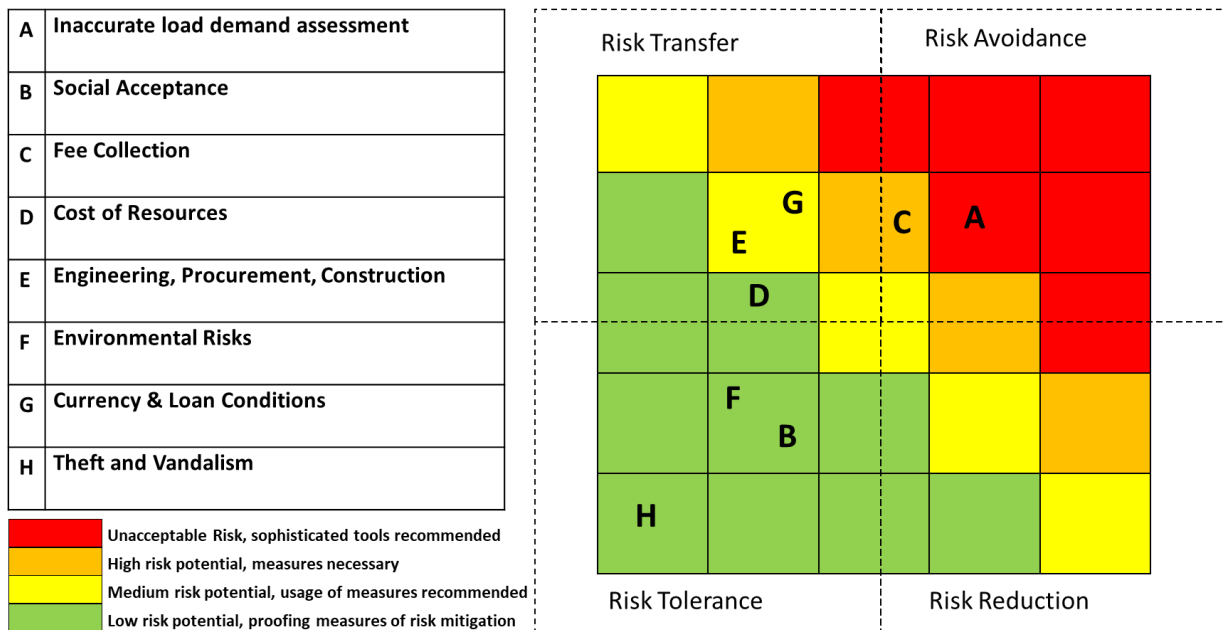


Figure 21 Risk Map (author's own diagram)

As shown in Figure 21, it is recommended that Risks 'A' and 'C' have risk avoidance measures applied to them. It is recommended that Risks 'D', 'E' and 'G' have risk transfer measures applied and Risks 'B', 'F' and 'H' need to be tolerated. For the project to be a sensible investment therefore, Risks 'A' and 'C' need to be mitigated as much as possible, Risks 'D', 'E' and 'G' need to be partially mitigated through outsourcing to a third party, and Risks 'B', 'F' and 'H' need to be made of aware of, however, action is not needed. These results are based on the likely outcomes highlighted in the risk assessment framework [58] and through consultation with MEI based on their project experience.

4.4.2 Risk Mitigation Strategy

The following sub-section highlights measures that should be taken so as to high the risks highlighted in the Risk Map.

4.4.2.1 Risk Avoidance Measures

The following actions are proposed as means of removing the risks associated with inaccurate load demand data:

- A secondary load demand assessment should be conducted that manually measures daily load demand from specific customers on the island. This should be conducted at a different time of year to that of the first study

- The surge voltage of multiple appliances being turned on at once should be measured through structured testing
- The load demand data from the mini grid on neighbouring island should be ascertained and analysed for load growth projections
- A more detailed survey be completed of the ice demand on the island
- The ice demand from the local fishing community should be strictly monitored for a set period of time and preferably at multiple times across the year to check for seasonality.

The following actions are proposed as a means of removing the risks associated with fee collection:

- The project team should conduct a robust programme of workshops and structured events with the community so as to showcase the benefits of electrification
- The developer should engage local business throughout the project development phase so as to encourage them to purchase new electrical equipment that can in turn improve their own business ventures
- Monitor the seasonality of the local industry to ensure that the ice demand is consistent throughout the year
- Employ a community engagement officer to work on the island. They should be involved in initial community engagement tasks as well as in overseeing fee collection
- Conduct a robust study of the best fee paying mechanism for the island. The study should consider multiple options and choose one that is suitable to the local context
- The developer should consider partnership with local microfinance institutions so as to give customers the option of taking out loans to cover the upfront connection fee.
- For mobile payment customers, it must be ensured that a reputable be used who have a proven ability of facilitating payments for energy use via meters

4.4.2.2 Risk Transfer Measures

The following actions are proposed as means of removing the risks associated with currency and loan conditions:

- The risk associated with fluctuating currency exchange rates can be mitigated through using foreign currency hedging which has been shown to reduce this risk in mini grid projects [58]
- Project developers should allocate reserve funds so as to deal with this relatively unlikely risk if it arises

The following actions are proposed as means of removing the risks associated with the costs of resources:

- Ensure that prior to project development there is thorough understanding of the local topography and land ownership.
- Conduct a robust and structured tendering process for acquiring technology providers and third party companies for overseeing logistics
- Hire and train community members to support the construction and operative stages
- Ensure that robust insurance policies are implemented for all assets
- Partner with a known and experienced logistical company for transporting assets

The following actions are proposed as means of removing the risks associated with resource costs fluctuations:

- Partner with multiple fuel suppliers to ensure that the risk of having no diesel is marginalised.

- Monitor the mini grid performance and wherever possible try to minimize reliance on the diesel generator
- Implement demand side management strategy if high night time loads are observed

4.4.2.3 Risk Tolerance Measures

The following actions are proposed as means of removing the risks associated with social acceptance:

- The developer should highly prioritise a community engagement phase in the project plan. An experienced project team should be formed who make early contact with the community
- Community wide workshops should be conducted focused on:
 - Payment methods
 - Productive uses
 - System operation
- Facilitate the creation of a board of community members who have say in specific project decisions

The following actions are proposed as means of removing the risks associated with theft and vandalism:

- Create an environment of ownership within the community through employing local people and involving them in project decisions
- Ensure cooperative bonds are formed throughout the community engagement phase
- Design security protocol and procedures for onsite staff

The following actions are proposed as means of removing the risks associated with environmental risks:

- Ensure technical assets have appropriate protective equipment i.e. housing structures
- Ensure insurance for assets covers the costs of technical components
- Ensure that procedures are in place for events where construction or operative works need to be stopped due to adverse weather conditions

The financial implication of the risks highlighted are analysed in the sensitivity study, see Section 4.5.

4.5 Sensitivity Study – Financial Risk Assessment

The financial implications of the risks highlighted above have been quantified and are discussed in detail in the following sensitivity study. The feasibility of scenarios is primarily assessed from the IRR (>15%), DSCR (>110%), and NPV (positive). The analysed risks include:

- Capital investment
- Connection Rate
- Connection Fee
- Ice Sales
- Grants for assets
- Debt from investment
- Load demand per capita

4.5.1 Capital Investment Cost Sensitivity

A sensitivity study was conducted for the varying levels of capital investment highlighted in Table 4. Table 27 summarises the KPIs for the financial modelling of the reference, high, and low cost scenarios:

Table 27 Financial Results of Cost Comparison Study

KPIs			
	Ref	1.1 High	1.2 Low
NPV (USD)	114,651	65,837	163,464
Project IRR	23.5%	19.4%	29.3%
Equity IRR	24.3%	19.8%	30.7%
Lowest DSCR	529.0%	315.0%	858.3%
LCOE in [USD/kWh]	0.15	0.17	0.13
Revenues in [USD/kWh]	0.16	0.16	0.16
Avg. Operating Profit Margin	0.35	0.30	0.41
Lowest Operating Profit Margin	0.26	0.18	0.33

As shown in Table 27, the high cost scenario does not return favourable results. If high cost scenario is realised then the NPV specifies a \$48,814 loss over the projects lifetime. This has serious implications to third party investors who would expect certain return on investment. If the low cost scenario is realized then the LCOE is 10% lower and generates \$49,000 more revenue over the project lifetime in real terms.

4.5.2 Connection Rate Sensitivity

The reference case assumes that in Year 0, 50% of all total connections are made and this increases linearly to full connections in year 20. This is highly uncertain and thus different variations of connection rate need to be assessed. The considered scenarios are as follows:

- 2.1 Increasing linearly from 0% in year 0 to 100% in year 20
- 2.2 Increasing linearly from 25% in year 0 to 100% in year 20
- 2.3 Increasing linearly from 75% in year 0 to 100% in year 20
- 2.4 Increasing linearly from 50% in year 0 to 75% in year 20
- 2.5 Constant at 50%

The results of the study are highlighted in Table 28

Table 28 Connection Rate Sensitivity Results

KPI	Connection Rate Scenario					
	2.1	2.2	2.3	2.4	2.5	Ref
NPV	46,881	80,645	148,093	64,621	45,581	114,651
Project IRR	18.4%	20.9%	26.2%	19.9%	18.7%	23.5%
Equity IRR	18.7%	21.4%	27.4%	20.4%	19.1%	24.3%
Lowest DSCR	368.9%	447.9%	602.1%	589.0%	489.4%	529.0%
LCOE [USD/kWh]	0.15	0.15	0.15	0.15	0.15	0.15
Revenues [USD/kWh]	0.14	0.15	0.16	0.16	0.15	0.16
Avg. Operating Profit Margin	0.38	0.37	0.34	0.39	0.36	0.35
Lowest Operating Profit Margin	0.23	0.25	0.26	0.32	0.30	0.26

As shown in Table 28, all scenarios return favourable results. If the connection rate does not increase above 50% however the project makes a net loss of \$69,070 over its lifetime in real terms.

4.5.3 Connection Fee Sensitivity

The reference case uses a connection fee of \$185 (686,008 UGX) which is charged to the end user as a connection fee. This variable was altered to assess the impact that it has on the financial viability of the project. The scenarios modelled are as follow:

- 3.1 \$0 connection fee
- 3.2 \$50 connection fee
- 3.3 \$100 connection fee
- 3.4 \$150 connection fee
- 3.5 \$200 connection fee

Table 29 lists the results of the study:

Table 29 Connection fee sensitivity results

KPI	Connection Fee Scenario					
	3.1	3.2	3.3	3.4	3.5	Ref
NPV	25,700	49,850	74,000	98,150	122,300	114,651
Project IRR	16.7%	18.2%	20.0%	21.9%	24.3%	23.5%
Equity IRR	16.9%	18.6%	20.5%	22.6%	25.2%	24.3%
Lowest DSCR	334.3%	377.0%	426.3%	483.8%	551.9%	529.0%
LCOE in [USD/kWh]	0.16	0.16	0.15	0.15	0.15	0.15
Revenues in [USD/kWh]	0.15	0.15	0.15	0.16	0.16	0.16
Avg. Operating Profit Margin	0.33	0.34	0.34	0.35	0.35	0.35
Lowest Operating Profit Margin	0.23	0.24	0.24	0.25	0.26	0.26

As shown in Table 29, all scenarios return feasible results implying that under reference conditions, the connection fee is not necessary for making the project profitable, however it does make the project a significantly less promising investment due to IRR being 30%.

If there are financial concerns due to lack of WTP or ATP from the community it may be difficult to encourage the local community to connect to the grid and purchase single phase electricity. A strategy to mitigate this risk is by temporarily removing the connection fee. It can be seen that by removing the connection fee the projects generates \$88,951 less over its lifetime however this is a lesser risk than not connecting customers and selling electricity to them.

4.5.4 Ice Sale Sensitivity

Different scenarios were modelled for different volumes of ice sales. The scenarios considered are listed as follows

- 4.1. 0 tonnes per day
- 4.2. 0.5 tonnes per day
- 4.3. 1 tonne per day
- 4.4. 2 tonnes per day
- 4.5. Linearly increasing from 1.25 tonnes in year 0 to 2.5 tonnes in year 20

The KPIs for each of the ice sales studies are listed in Table 30:

Table 30 KPIs for Ice Plant Sales Sensitivity Study

KPI	Ice Sale Scenario					
	4.1	4.2	4.3	4.4	4.5	Ref
NPV	-230,137	-147,628	-70,172	56,620	10,334	114,651
Project IRR	-11.2%	2.8%	9.7%	19.4%	15.9%	23.5%
Equity IRR	-13.0%	2.1%	9.4%	19.9%	16.1%	24.3%
Lowest DSCR	-1706.0%	-1134.1%	-562.1%	287.0%	-276.1%	529.0%
LCOE in [USD/kWh]	1.30	0.51	0.31	0.18	0.22	0.15
Revenues in [USD/kWh]	0.60	0.30	0.22	0.17	0.19	0.16
Avg. Operating Profit Margin	-0.41	-0.14	0.05	0.28	0.23	0.35
Lowest Operating Profit Margin	-0.92	-0.45	-0.17	0.15	-0.07	0.26

As highlighted in Table 30, only scenario 4.4 returns feasible results. For the project to be feasible it needs to sell on average 1.6 tonnes of ice per day over the lifetime of the project. In this instance the NPV is \$15,999 implying that the project losses \$99,000 over its lifetime in revenue. This study emphasises the necessity that the ice plant has in the project's long term success.

4.5.5 Grants for Assets Sensitivity

The reference scenario assumes that 50% of costs for assets from grants. This condition is not a certainty and a sensitivity analysis had to be conducted of the mini grid's financial feasibility. The scenarios considered are listed as follows and results reported in Table 31:

- 5.1. 0% investment for assets
- 5.2. 20% investment for assets
- 5.3. 40% investment for assets
- 5.4. 60% investment for assets
- 5.5. 80% investment for assets

Table 31 Investment for Assets Sensitivity

KPI	Investment for Assets					
	5.1	5.2	5.3	5.4	5.5	Reference
NPV	-103,184	-16,050	71,084	158,217	245,351	114,651
Project IRR	11.2%	14.5%	19.6%	29.2%	57.2%	23.5%
Equity IRR	11.0%	14.5%	20.1%	30.6%	61.8%	24.3%
Lowest DSCR	252.8%	322.0%	437.1%	666.9%	1353.0%	529.0%
LCOE in [USD/kWh]	0.15	0.15	0.15	0.15	0.15	0.15
Revenues in [USD/kWh]	0.16	0.16	0.16	0.16	0.16	0.16
Avg. Operating Profit Margin	0.35	0.35	0.35	0.35	0.35	0.35
Lowest Operating Profit Margin	0.26	0.26	0.26	0.26	0.26	0.26

As shown in Table 31, scenarios 5.1-5.2 are unfeasible due to net losses and inability to pay back debt. For the project to become feasible there needs to be a minimum initial investment of 47%.

4.5.6 Debt from Investment Sensitivity

The reference scenario assumes 10% debt on private investment and this would be entirely in foreign currency (estimated interest rate of 8.5%). This condition is not a certainty and a sensitivity analysis had to be conducted on financial feasibility with varying levels of debt from investment. The scenarios considered are:

- 6.1. 0% debt from investment.
- 6.2. 20% debt from investment.
- 6.3. 40% debt from investment.
- 6.4. 60% debt from investment.
- 6.5. 80% debt from investment.

The results of the study are listed in Table 32:

Table 32 Debt from Investment Sensitivity

KPI	Debt from Investment					
	6.1	6.2	6.3	6.4	6.5	Ref
NPV	95,625	135,290	182,106	237,681	303,844	114,651
Project IRR	23.2%	23.8%	24.5%	25.2%	25.9%	23.5%
Equity IRR	23.2%	25.8%	30.0%	38.8%	70.3%	24.3%
Lowest DSCR	0.0%	252.4%	114.1%	68.0%	45.0%	529.0%
LCOE in [USD/kWh]	0.15	0.15	0.14	0.13	0.13	0.15
Revenues in [USD/kWh]	0.16	0.16	0.16	0.16	0.16	0.16
Avg. Operating Profit Margin	0.35	0.35	0.35	0.35	0.35	0.35
Lowest Operating Profit Margin	0.26	0.26	0.26	0.26	0.26	0.26

As shown in Table 32, only scenario 6.2 is feasible, 6.1 has a net loss over its lifetime and scenarios 6.4 & 6.5 have DSCRs too low to be able to generate revenues to repay debt obligations. The project becomes economically unviable when debt for investment goes above 28%.

4.5.7 Load Demand Per Capita

Finally, from the literature review it became evident that it is probable that the predicted per capita energy demand be up to four times lower than that predicted in the load demand assessment. Accordingly, the load demand per capita has been varied and the effects on the fully sized system analysed. It should be noted that in this scenario the ice sales remain at 2.5 tonnes per day. The modelled scenarios have the following conditions:

- 7.1. 10% demand from single phase loads
- 7.2. 25% demand from single phase loads
- 7.3. 50% demand from single phase loads
- 7.4. 75% demand from single phase loads

Table 33 lists the results of the study:

Table 33 Load Demand Per Capita Sensitivity

KPI	Scenario				
	7.1	7.2	7.3	7.4	10 (Ref)
NPV	98,854	104,240	113,217	122,194	114,651
Project IRR	22.6%	23.0%	23.6%	24.2%	23.5%
Equity IRR	23.4%	23.8%	24.4%	25.1%	24.3%
Lowest DSCR	506.6%	526.0%	558.5%	590.9%	529.0%
LCOE in [USD/kWh]	0.13	0.14	0.14	0.14	0.15
Revenues in [USD/kWh]	0.14	0.14	0.15	0.15	0.16
Avg. Operating Profit Margin	0.40	0.40	0.39	0.38	0.35
Lowest Operating Profit Margin	0.30	0.30	0.30	0.31	0.26

As shown in Table 33, all scenarios considered are economically viable. This implies that as long as all other references conditions are met, notably the 2.5 tonne daily sale of ice, then the project remains feasible even if the per capita energy demand is 10% of that predicted in the LDA. This finding further necessitates the need for the ice plant as an over estimated load demand assessment is the most likely risk to occur, and with the highest consequences.

It is of course highly likely that the load demand from single phase users will be four times below what is estimated and if this happens the project generates \$10,411 less revenue over the course of the project. This value is low due to the ice plant that generates the majority of the revenues.

4.5.8 Results Summary

All scenarios modelled in the sensitivity analysis have been numbered and their KPIs compared. A comparison of each scenario's IRRs and DSCRs is characterised in Figure 22 where the dotted lined highlights the scenarios that produced favourable results. A table of these results that highlights scenario criteria is shown in Appendix M.

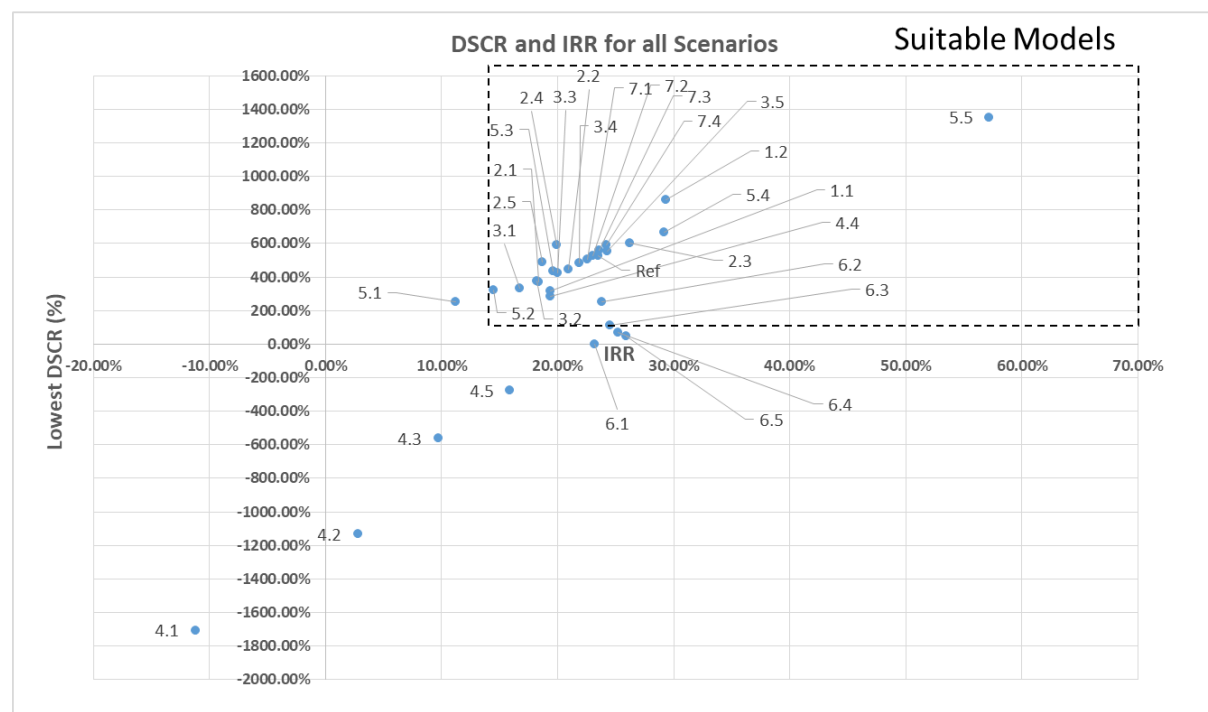


Figure 22 Comparison of IRR and DSCR for Sensitivity Results

As shown above, the IRR and lowest DSCR vary dramatically across the considered scenarios. This range characterises the broad range of outcomes that may arise in this project. Likewise, the NPV and operating profit margin have been plotted for all scenarios, as shown in Figure 23

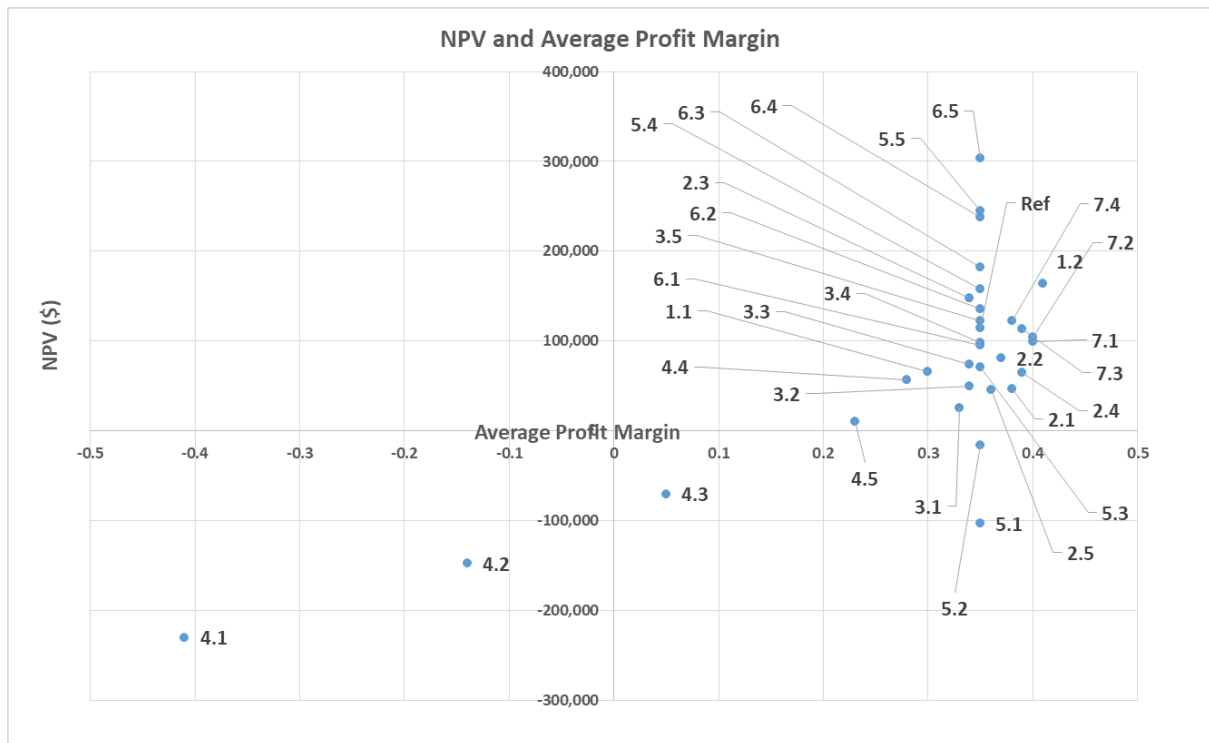


Figure 23 NPV and Average Operating Profit Margin for All Scenarios

As highlighted above, there is generally direct correlation between the NPV and operating profit margin. Scenarios that have positive NPV, and suitable IRR and DSCR are plotted on Figure 24:

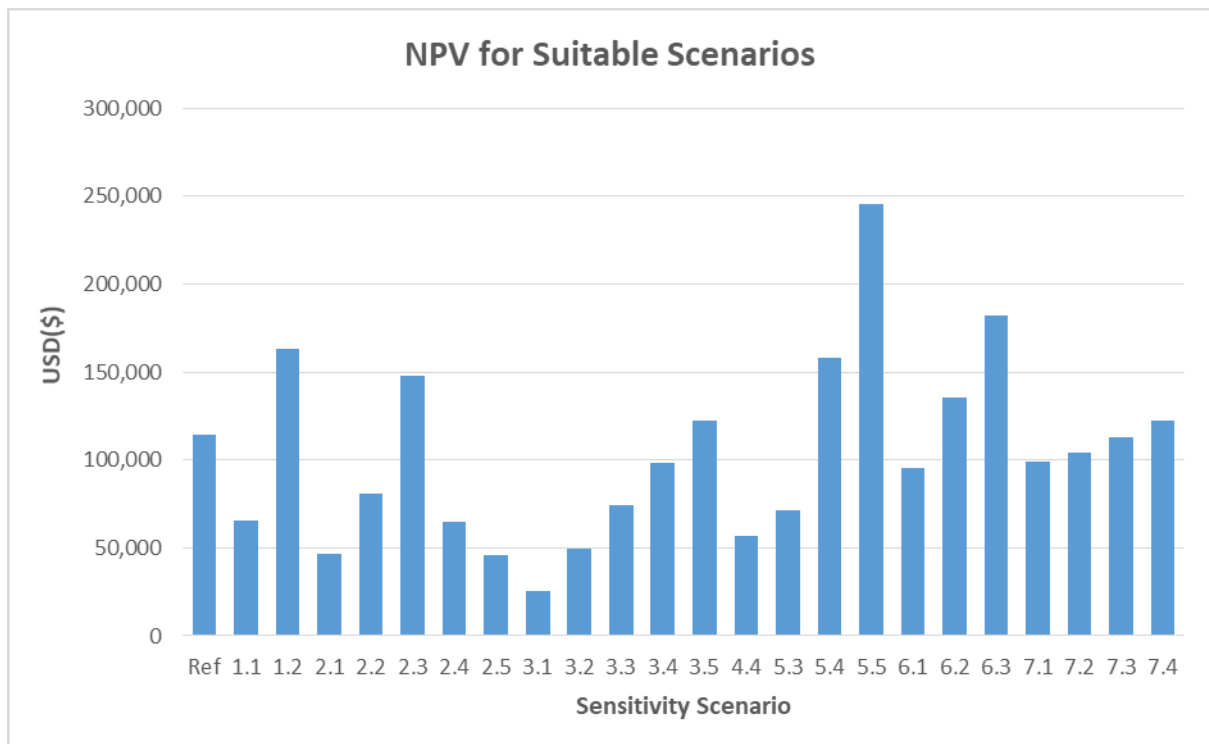


Figure 24 NPV for Scenarios of Suitable Conditions

As shown above, Scenario 3.1 has the lowest NPV as no connection fee is applied. The highest NPVs are from Scenario 5.5 in which the investment for assets is 80%, and 6.3, in which 40% of the investment is funded by debt (this scenario consequently has a low DSCR of 114%).

5 Discussion

The aim of this work was to size and financially assess a PV mini grid for the currently un-electrified Buvu Island in Uganda. To do this work effectively the following objectives were set:

1. Identify weaknesses in previous methodologies used in mini grid design
2. Use field data to assess demand on Buvu Island and identify uncertainties based on research findings
3. Assess the suitability of introducing an ice plant
4. Assess the suitability for the island to house a mini grid
5. Size a PV mini grid using the HOMER optimisation tool
6. Use a financial tool to assess the economic feasibility of the HOMER model based on different scenarios.
7. Conduct a risk assessment and from that assess the sensitivity of results presented

The following subsection discuss to what extent the above six objective were completed and what future work should do to expand on findings and remove result uncertainty.

5.1 Buvu Energy Assessment

It is evident from the literature review that data scarcity is currently a major barrier facing the mini grid sector across SSA. Developers who are keen to begin new projects currently suffer from not being able to rely on previous experience so as to size and manage contextually specific mini grid systems. Additionally, there are very few examples of projects where the mini grid has successfully developed local industry, and load growth is typically only seen when tariffs are reduced to unfeasible levels [23] [36] [38] [40] [44].

Due to the project risk that arises from proposing PUEs as a way of making a mini grid a feasible venture, this work proposed a mini grid that is scaled to what is currently observable on Buvu Island – a strategy supported by MEI and GRS Commodities. The predicted load demand from single phase users therefore is based on what domestic, commercial, and community customers currently use and the ice plant proposed is scaled so as to support the current ice demand of the fishing industry and not an unlikely future value.

Unfortunately, due to lack of skills on the ground, the LDA did have to be conducted via a survey approach whereby a select number of customers were queried about their electricity demand needs. However, in agreement with the literature surrounding load growth, the LDA did not presume that new, currently unfamiliar, electrical items would be used following the installation of the mini grid. The LDA instead presumes that HHHs and MHHs will maintain the same demand, and currently un-electrified LHH will purchase a bulb and a phone charger. The LDA predicted that the island daily energy demand is 695 kWh, inclusive of the ice plant.

It was evident from the literature that domestic energy demand is often up to four times overestimated and accordingly such studies have been built into the sensitivity analysis. Additionally, to factor in uncertainty in daily load dispersion, a weighted average has been used, as suggested by multiple sources.

5.2 Mini grid System Sizing

Within this study HOMER has been used to decide upon the optimal component size and operative strategy for the mini grid. The HOMER study takes into account an AC-coupled PV array, a converter, a battery bank and a diesel generator.

The optimised HOMER model proposed a system that includes a 178 kW PV array and has a renewable fraction of 54.6%. The reason for the relatively low renewable penetration is the need for the Ice Plant to operate twelve hours a day so as to meet the ice demand. An additional reason for such a low renewable penetration is the limited flexibility within the battery bank configuration. It is an industry rule of thumb that within rural mini grids, the number of battery strings should not exceed four unless additional control systems are purchased [19]. Additionally, the line voltage observed by the batteries needs to match that of the converter which is scaled to 10% above the daily peak demand [19]. In this case, the peak demand is 55 kW (three phase) and therefore twelve Sunny Island 8.0H converters are modelled each with a 48V voltage - this implies that each battery string can be 24 in length (2V each). These conditions limit the size of the battery bank and results in the generator having to meet demand more frequently throughout the day.

The sensitivity study of the HOMER sizing study revealed that the initial capital investment for assets greatly effects the optimal configuration. In the high cost scenario the proposed PV array is 10% smaller and the suggested LCOE is 8% higher. This is because in the high case scenario the long terms operative costs of the diesel generator are lower than the upfront capital required for the a PV that would supply the same power. It also became evident that the optimal operating times for the ice plant are 6am – 6pm and that if diesel prices were 30% higher, the PV array should be 15% larger, and with it, the LCOE would increase 10%. Finally, it was observed that if the ice plant remained at constant demand but the single phase demand was four times smaller than predicted (as is likely), then the optimal system would have a PV array 52% smaller and the LCOE would be 4% larger. The vast variability in system sizing options presented in the sizing results, see Section 4.1, reinforces the necessity for developers to ensure that load demand data and cost data are as accurate as possible. If the project becomes operational and the site is inaccurately sized then there will be adverse consequences.

5.3 Distribution System

Ideally, the results of the distribution system study will be passed onto the grid operator to act as advisory information on costs and configuration. One of the main barriers to mini grid scale up across SSA is the lack of quality assurance framework that standardises quality checking procedures and ensures that projects are, wherever possible, delivered with predetermined levels of quality [26]. The NREL have taken a positive step towards overcoming this barrier in their quality assurance framework and this has been used as a benchmark for assessing the grid quality in this work. The major variable assessed was the system voltage drop which is recommended to be below 10%. The result suggested an estimated total loss of 10.48%, however, this is based on an assumption that all loads are switched on simultaneously which is an extremely low risk event.

5.4 Financial Modelling

The aim of the financial modelling study was to determine the projects financial viability over a 20 year project lifetime. To achieve this an Excel based tool was adapted from previous MEI experience to accommodate the specific context of this work.

The model however is dependent on a number of variables that are highly project specific and at this early project stage multiple difficulties were faced in ascertaining data that reflected the Ugandan context. To overcome this a sensitivity analysis was conducted. The analysis would have been improved upon by the following conditions:

- Valid cost data – this data should be ascertained in later project stages
- More complex data analysis whereby multiple variable are altered simultaneously against reference conditions. Due to the complexity of the financial model used in this work it was not

possible to conduct a two variable What-If analysis using Excel. It is this recommended that the tool 'R' be used in later work to conduct this more thorough analysis.

The reference model produced very promising results for the mini grid with an NPV of \$114,651, an IRR of 23.5%, and a lowest DSCR of 529%. It was also found that if the project is sized to account for reference scenario costs, but the reality is that capital costs are of the high scenario, then the project loses \$48,814 over its lifetime with an LCOE 13% higher.

It is likely that the developer will not be able to convince 50% of the populace to connect to the system in year 0, and thus, the connection rate was also assessed. It was found that the project remains financially viable even if no connections are made in year 1 but this has to increase linearly to full connection rates in year 20. This case however results a \$67,770 reduction in revenue over the project lifetime in real terms.

A study of the ice sale price against reference conditions revealed that their needs to be on average 1.6 tonnes of daily ice sales for the project to be feasible. Additionally, from year 1 there need to be minimum daily sales of 1.6 tonnes so that debt repayments can be made in the first years of operation.

Finally, it was found that if the ice plant maintains fully operation but single phase demand is 10 times smaller than predicted, the project remains to be financially viable. However, in this scenario, if debt from investment increases to above 20%, or grants for assets drops below 30%, then it becomes unviable. This final point particularly highlights the benefits that a well-managed PUE can bring to an un-electrified community, however it is paramount that the developer is as sure as possible in the conditions in which it shall be operating.

From analysis of the sensitivity study results it can be concluded that ice sales, securing grants for assets, and ascertaining connection fees are the three most prominent risks in the project. The final key suggestions are thus made:

- Ensure ice sales are continually over 1.6 tonnes per day
- Ensure a minimum grant for assets of 47% is obtained
- Prioritise connecting new customers over incentivizing load growth from single phase customers
- Aim to connect a minimum of 455 single phase customers by year one of the project and increase linearly until full connections are made.
- Ensure enough staff is on hand at all times so as to ensure that the distribution system and ice plant are running at required standards as the project's feasibility is highly sensitive to this.
- Highlight prioritize community engagement tasks within the early project stages so as to maximise the chances of gaining customer trust, eliciting accurate information from them, and securely gathering usage payments in a timely manner

6 Conclusion

It is evident that mini grids have much potential to be leading clean energy provider for vast areas of rural SSA. Despite this, a large amount of mini grid projects still remain in their pilot phase. This is evidently due to a severe lack of industry experience and knowledge of the financial conditions that make mini grids economically viable.

The Ugandan government has made good progress towards creating a policy environment that encourages IPPs to develop mini grids in remote rural areas of the country – including island communities on Lake Victoria who mostly do not have access to electricity.

To date, there are two operational PV mini grids on Lake Victoria, one of which was designed and implemented by GRS Commodities. This local developer now wishes to develop a similar system on Buvu Island and this work aimed to act as a prefeasibility assessment of the site.

Information about the Buvu Island communities was gathered in the field in May/June 2019 and this formed the basis of the LDA. Predicting a likely future electricity demand enabled a solar PV system to be modelled using HOMER and a lifetime financial analysis to be conducted. It became evident from the analysis that in order to install a financially viable system, additional income would need to be generated, and the best way to do so would be through selling ice to the local fishing industry.

Overall, the reference financial model returned favourable results when the ice plant was incorporated into the design of the mini grid and it can thus be concluded that Buvu Island is well suited to housing a PV mini grid with diesel generator back-up. The methodology used for this work has proven to be a useful method of analysing mini grid prefeasibility. However, it does rely on a large amount of project specific data relating to site information and asset costs. It is therefore recommended that before moving to the EPC stage, a more robust study be conducted of the island's electricity demand, its ice demand, and the costs of installing a system. Once such work is completed, the methodology employed in this work can be used so as to generate suitable results upon which a finalised mini grid design can be made.

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8 Appendices

8.1 Appendix A - Supplementary Studies

8.1.1 Ugandan Energy Policy

The MEMD is the leading agency in the energy sector. This body oversees the development and implementation of policy framework and is also the chief legislating body for the country's energy sector. Uganda's energy policy remains to be centralised and there are no government bodies at regional level that represent the MEMD. The MEMD consists of four divisions: Energy Efficiency, Innovation and Renewable Energies, Electricity, and Provision with Oil Products.

The ERA was established with the 1999 Electricity Act and is the body tasked with issuing licenses for the generation, transmission, distribution, and sales of electricity. The ERA is also mandated with the power to design tariff structures, investigate tariff charges, and approves rates for charge. They give out licenses, reviews and approve tariffs, and enforce rules and regulations [13]. The key instruments they utilise are:

- The Electricity Act, 1999
- Rural Electrification Strategy and Plan (RESP) (2001)
- The National Energy Policy (NEP) (2002)
- Renewable Energy Policy (REP), 2007

The REA acts as the secretariat to the Rural Electrification Board who oversee the MEMD's overall plans as dictated in the Indicative Rural Electrification Master Plan. The REA control the funds for subsidizing rural electrification projects.

The Ugandan government first addressed the issue of renewable energy in the National Energy policy of 2002. The government proposed an overarching vision of renewable energy playing a major role in providing electricity through the following objectives: maintaining and improving responsiveness of legal and institutional framework to promote renewable energy investment; setting up a financial and fiscal policy framework for investments; increasing public awareness; promoting R&D as well as international cooperation in renewable energy technology; using biomass energy in an efficient and sustainable manner; and promoting the uptake of industrial waste to energy schemes [59].

Currently, the leading instrument for renewable energy promotion is the REP which establishes a Standardised Power Purchase Agreement (SPPA) and Feed in Tariffs (FiT) for renewable generation projects. The framework brings forward favourable financial and fiscal conditions for renewable project, listed as follows [13]:

- Tax exemptions and levy's
- Accelerated depreciation
- Provision of credit enhancement tools and risk mitigation procedures
- Credit mechanisms for end users

Targets for improving electricity access are highlighted in the RESP which specifies that the government has a target of connecting an additional 500,000 people via both on-grid and off-grid systems [13].

Electricity access is also a fundamental part of the Ugandan Government's Poverty Eradication Action Plan (PEAP) which strongly aligns energy access and poverty alleviation. The document specified that 10% of rural populations should have electricity access by 2012, and this figure is further repeated in the Rural Electrification Strategy and Plan.

In April 2010 the Ugandan Government released the National Development Plan which further specifies specific objectives for increasing electricity access and consumption so as to incentivize growth, employment, and socioeconomic conditions. The plan specifies that impacts on forestry should be marginalised in the process and highlights key challenges for the power sector.

In 2012 the government set aside \$160M so as to subsidise electricity tariffs. By mid-2012 the government reported a significant imbalance between the UMEME grid operator's cost-reflective tariff (\$0.37 / kWh) and the cost of the customer (\$0.15 / kWh) by allowing tariffs to be increased by 55%. To date the grid tariff based on customer category, is around \$0.26 for domestic customers, and is no longer cross-subsidized. Besides UMEME distribution there are 5 additional distribution firms in the country that use tariffs adapted for off-grid applications. Special tariffs granted for mini grid range from around \$0.18 - \$0.26.

8.1.2 Buvu Island Profile

8.1.2.1 Island Communities

Kachanga is the largest community on the island and has a population of around 3,000 people and 122 registered fishing boats. The Kachanga landing site is shown in Figure 25:



Figure 25 Kachanga Landing Site

The second largest community is Luwungulu which has around 100 households. It was reported that the community's population is currently in decline due to the government's new regulations on fish stock controls. A photo of the Luwungulu landing and its houses site are shown in Figure 26 and Figure 27:



Figure 26 Luwungulu Landing Site



Figure 27 Luwungulu Houses

The final community, Kamese, is the smallest on the island with 450 reported male residents and 82 registered fishing boats. A trip to the mainland from any of Buvu's communities costs 200 UGX (0.54 USD). Fishing is the island's largest industry. The fishing boats use petrol powered motors and across the island petrol costs between 4,600 and 4,800 UGX (1.25 – 1.3 USD).

8.1.2.2 Island Facilities

There are two savings cooperatives on the island that operate from 07:30 – 17:00 every day and are both located in Kachanga. Both institutes are powered by solar home systems (SHS) and it was reported that when there are days without sunshine it becomes difficult to work due to lack of light. The savings cooperatives reported that if reliable power was attained they would procure a computer and regularly use printing services.

There is a club / lodge located in Kachanga that is powered by a diesel generator and solar system. It was reported that the generator is the main energy source and is primarily used to power sound equipment and a refrigerator. The generator consumes 7-8 litres of diesel on a Sunday and a Wednesday from 19:00 until 06:00. The major challenge faced by the business was reported to be the difficulty in making it profitable due to the high fuel cost of the generator. The generator was reported to often require spare parts and servicing from a mechanic located on the main land. Additionally, when there is a fuel shortage the diesel price increases to unobtainable levels. It was reported that if reliable electricity was obtained then the business would purchase a photocopier as well as look for expansion opportunities.

Kamese has only one bar/club/library and it is powered by a 1kVA 4 stroke generator that consumes 5 litres of diesel per day, and a small SHS. The bar has a fridge which is sometimes powered by the generator but more commonly filled with purchased ice. The generator is only used at weekends as it is expensive to use whereas the SHS is used during the week.

The island has one primary and one secondary school. The primary school has 360 pupils with many other children on the island commuting daily by boat to neighbouring islands. The school currently has a SHS used for lighting and that in the evening it becomes difficult for student to study. The school does not have a diesel generator as they were reported to be too expensive to operate. It was also reported that if reliable and affordable electricity was provided then the head teacher would mobilise parents to contribute towards the cost. It was reported that the school currently spends 300,000 UGX (82 USD) per month on fuel wood for cooking. A photo of the primary school is shown in Figure 28:



Figure 28 Kachanga Primary School

The island currently has two health facilities both of which are powered by SHSs. In both cases the SHS is used to power lighting, vaccine freezers, a TV, and a microscope. It was reported at one health facility that the SHS is faulty and insufficient. Both health facilities reported that they used charcoal for cooking and that annual expenditure on charcoal amounts to 70,000 UGX (\$20 as of June 2019).

Fresh water on the island is sourced from a bore hole on the island which is currently powered by a solar system. On days of sufficient solar irradiance, water flows freely from designated taps on the island. On days with insufficient solar irradiance, water needs to be taken from the lake.

8.1.2.3 Island Energy Supply

SHS are common and are the village's main source of energy. The most commonly found electrical appliances among households, businesses, and community institutes are lights, phone chargers, and entertainment equipment such as TVs, radio's, speakers, and PA systems. Most households with SHS use them to power a lamp and DC solar television. A commonly held complaint among community members was that replacement batteries had to be regularly purchased and that the SHS were not fit for purpose.

Besides SHS and a selection of 1kVA diesel generators, the only other sources of electrical power on the island are two privately owned 4 kVA diesel generators. These generators are owned by one individual who sells electrical power to the local populace. The owner sells electrical power by power/appliance/day. The generators were found to be overloaded and it was reported that they suffered from frequent breakdowns with downtimes as long as a week. The reasoning for the long downtimes was that spare parts often had to be sought from the mainland. Customers of the

generator had to pay upfront for future electrical use and it was reported that during downtimes customers received no compensation on their down payment. At the time of the study the owner was offering one bulb light for 1,000 UGX / per day and 2,000 UGX / per television / per day which equates to significantly higher than that of grid electrical supply. It was reported that kerosene is not used on the island for lighting due to fire hazard; it is however used by the fisherman for catching silver fish.

The local fishing industry relies heavily on the use of flake ice for preserving the fish. Ice is currently transported from the Kigungu landing site on the nearby Buggala Island and stored in containers near the port at each of the island's three communities. Ice is purchased in 83 kg bags and this does not differ by season. It was reported that a large quantity of this melts enroute and alternative preservation methods such as salting or smoking is not allowed. It was further reported that Buvu's fisherman face strong competition from fisherman on other islands.

The major agricultural industries on the island are that of rice, maize, and cassava and products are mainly used for self-consumption. There is no official figure for livestock numbers on Buvu however one leader reported that there were approximately 400 cows. The cows are sometimes used for their milk which is consumed by the local populace however much of this goes to waste. The cow dung is also gathered by local Kraal owners and sold for fuel.

It was reported during focus group discussions that electricity is highly sought after within all communities so that local businesses can improve. The local generator is used for welding, lighting, and using mechanical appliances that are needed for repairs. As specified above, this electricity is unreliable and highly overpriced and thus more traditional practices are used for overcoming this challenge, for example, ironing is mostly conducted through a method that employs charcoal.

A major barrier to the uptake electricity is the high investment cost required for solar home systems which are currently the most financially viable electricity source available.

8.1.3 Solar Photovoltaics

Solar PV systems utilise the photovoltaic effect to convert solar irradiance into electrical energy through the creation of an electric current due to exposure of electrons to light. This phenomenon takes place within solar cells, typically made up of semiconductor materials such as Silicium. Solar cells consist of a p-n junction which is combination of p-type semiconductors, which have a lack of electrons, layered with n-type semiconductors that have a surplus of electrons. When exposed to the photons of light, electrons are stimulated and move from the n-type to the p-type semiconductor and an electric current is generated. The relationship between current and voltage within a solar cell is characterised by its I-V curve, conveyed in Figure 29. The maximum achievable current is denoted by the short circuit current I_{SC} and the maximum achievable voltage is denoted by the open circuit voltage, V_{OC} . The maximum power achievable from the solar panel is denoted by the maximum power point, MPP.

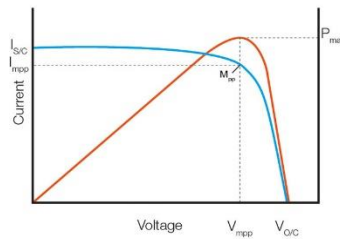


Figure 29 Solar Cell I-V Curve

The area occupied by a PV array is determined by the efficiency of the panels selected and the inclination angle of the PV system. It is assumed that the panels will be in a fixed position throughout the day and this the ideal angle of inclination specific to Buvu is determined to be 0.18° . The spacing required between panels is calculated using Equation 15 and Equation 16 – characterised by Figure 30. The angles α and β are determined by the latitude of the location. For Buvu Island, α is 0.8° and thus β is 89.2° [19].

Equation 15 Spacing length required between panels

$$L = \frac{l * \sin \alpha}{\tan \beta}$$

Equation 16 Area required between panels

$$\text{Area of two rows} = L * l * \left(\cos \alpha + \frac{\sin \alpha}{\tan \beta} \right)$$

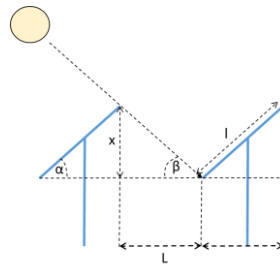


Figure 30 PV Array Area Calculation Diagram (author's own diagram)

8.1.4 WTP and ATP

Willingness to pay (WTP) is defined as the maximum amount an individual indicates that they are willing to pay for a good or service. WTP surveys are fundamental in assessing the financial feasibility of the mini grid as the results will have significant influence on the final tariffs set, and thus, the amount of revenue that can be generated to cover costs of operations, maintenance, and repairs [60].

WTP is largely determined through data acquisition and analysis of potential customers of the mini grid. Using interviews and focus group discussion conducted in field, analysts can assess the actual expenditure of current energy sources and other aspects of daily life that suggest availability of economic resources for paying for electricity [19].

The first step in the WTP is for the demand assessment team to conduct interviews and ask people what they currently pay for their current energy sources such as kerosene, candle, batteries, etc., and this is known as the 'ability to pay' or 'revealed WTP' [60].

Additionally, the maximum amount the customer would be willing to pay in the hypothetical scenario of a mini grid being in place is then assessed. This is known as 'contingent valuation' and reveals the 'expressed WTP'. This process is reported in literature to be potentially unreliable and inaccurate as the customer may not be fully aware of the system described and can thus not provide an accurate response. To conduct this assessment to the highest level of accuracy, it is fundamental that the community be already well informed on what a mini grid can do [60].

Both the 'revealed WTP' and 'expressed WTP' should be gathered during the prefeasibility stage of a mini grid project. For inputting information into the financial analysis and tariff design, the 'revealed WTP' is preferable as it is not speculative, however, ascertaining both values are useful for future comparative analyses.

Additionally, whilst conducting interviews it is reported as useful to ask about what else people spend or invest their money on i.e., housing, food, transportation, education, health, and communication. This information can also serve as an indicator as to how much they are willing to pay for the new electric service as their income can be assessed [60].

Guidelines for ascertaining necessary information are highlighted in multiple online resources [19] [60] [29] [49].

8.1.5 HOMER Calculation Methodology

8.1.5.1.1 HOMER Technical Analysis

The first step in the HOMER analysis is inputting the load demand data, which is typically available as hourly power demand, as well as local energy resource data such as hourly wind speed (m/s) and solar irradiance (kWh/m²). Subsequently, specific technical components are chosen by the user and associated costs allocated. This includes all power generating technologies and storage. Upon initiating a simulation HOMER calculates the energy balance for the system and assesses the optimal means by which the power generators and storage can be used to meet demand.

Following on from assessing a selection of viable configurations HOMER conducts an economic assessment of the each possible solution.

HOMER assesses the output of the solar PV array using Equation 17. The terms referenced are noted in the List of Equation Terms:

Equation 17 PV Array Output Power

$$P_{PV} = P_{PV,STC} f_{pv} \left(\frac{G_T}{G_{STC}} \right) (1 + \alpha_T (T_T - T_{STC}))$$

Where;

P_{PV} = Power Output of PV array (kW)

$P_{PV,STC}$ = Rated capacity of PV array (kW)

f_{pv} = The derating factor of the solar array

G_T = Solar irradiation on the PV array (kW/m²)

G_{STC} = Solar irradiation on PV array at standard test conditions (kW/m²)

α_T = Temperature coefficient of power (%/°C)

T_T = Temperature of the PV array (°C)

T_{STC} = Temperature of the PV array (°C)

Homer assesses the maximum amount of power that can be allocated to the battery storage system at each time step taking into account the kinetic storage model, maximum charge rate, and maximum charge current. The takes into account the discharge losses that arise from the batteries discharging power to serve the load - this is assessed from Equation 18:

Equation 18 Battery discharge losses

$$P_{D,max} = P_{D,max,ksm} \eta_D$$

Where;

$P_{D,max}$ = Maximum discharge power of the battery (kW)

η_D = Discharge efficiency of battery (%)

$P_{D,max,ksm}$ = Maximum discharge power of the battery calculated from kinetic model

8.1.5.1.2 HOMER Economic Analysis

Following on from sizing the best suited energy system HOMER conducts an economic analysis which considers the LCOE and NPC for each of its proposed systems. The system costs included in the HOMER analysis include capital costs for investments, component replacement costs, O&M costs, grid purchases (if applicable), and fuel costs. The revenues considered in the analysis come from the salvage cost of components over the projects lifetime. HOMER does not consider the tariff cost to the end users. The annualised cost of the system is assessed using Equation 19:

Equation 19 Annualised Cost of Mini Grid

$$C_{ann,tot} = NPC \text{ CRF } (iL_T)$$

Where;

$C_{ann,tot}$ = Total annualised cost of mini grid (USD)

NPC = Net Present Cost (USD)

CPF = Capital recovery factor

i = Interest rate

L_T = Project lifetime (years)

The Capital Recovery Factor (CRF) is assessed using Equation 20, where 'n' is the number of years by which the investment capital must be recovered:

Equation 20 Capital Recovery Factor

$$CRF = \frac{i (i + 1)^n}{(i + 1)^n - 1}$$

HOMER subsequently calculates the systems Levelized Cost of Electricity (LCOE) using the Equation 21:

Equation 21 HOMER LCOE Calculation

$$LCOE = \frac{C_{ann,tot}}{E_{supplied}}$$

Where;

$E_{supplied}$ = The sum of the energy served to the load (J)

Having deduced an LCOE a decision can be made on the financial feasibility of the system. It should be noted that within this work HOMER is used as means of sizing the mini grid and doing a preliminary financial assessment. As noted in section 3.5, a full financial analysis is conducted using practices used by MEI, and multiple developers associated with the Green Minigrid Helpdesk Facility [19].

8.1.6 Mini Grid Operative Models

8.1.6.1 Utility Operated

Within this model the regional or national utility both owns and operates the mini grid. Collection of payments is also the responsibility of the utility staff. If there are wider social objectives of the mini grid then a tariff is usually set in and around that of the local grid cost for electricity whereby any disparities in production cost and the tariff are equalised by government subsidies. Initial funding in this model is usually provided by the utility, government, or third party donors [24].

8.1.6.2 Private Sector Operated

Within this operative mode a private sector actor is in charge of building, managing, and operating the system. Funding will typically come from a variety of sources including equity and commercial loans. It is also common for such projects to garner support from public sector bodies who grant funding. To date, there are very limited examples of mini grids that are entirely funded from private sources. Private sector projects are often difficult to make financially feasible when only relying on the electricity demand of domestic customers. It is therefore common for business models to be built around anchor loads. An anchor load is a commercial customer of high demand that has, steady, and regular electricity demand. Telecommunications towers are the most commonly found anchor loads with privately owned mini grid projects [24].

8.1.6.3 Community Operated

In this operative mode the mini grid is managed by the end-users. The local community thus has to organise a cooperative that is tasked with running the mini grid in accordance with government regulations. Much evidence does suggest that community involvement supports the success of the mini grid as it incentivizes people to use it appropriately. Financing of such projects usually comes from grants and the local community provide services such as labour and land. In theory these projects are useful for developing local economies and improving the livelihoods of the local populace [24].

A major drawback of such projects is that there is typically a lack of technical expertise to ensure the mini grid is suitably maintained. It is therefore advisable for tariffs to be set by the cooperative that ensure that appropriate skills are on hand and that O&M is appropriately valued within the project [24].

8.1.6.4 Hybrid Models

Hybrid operative models combine different elements of models previous discussed. This implies that different bodies could be involved in the investment, ownership, and operation of the mini grid. This can also include the division of labour between privately hired employees and community members. The generation and distribution of electricity can also be divided between government companies, private firms, and local communities [24].

8.1.7 Mini Grid Business Models

8.1.7.1 The Franchise Approach

Within this model the management costs are the responsibility of the franchiser with marginal costs to the franchisee. If the franchiser has many franchisees then the costs of management are reduced due to economies of scale. This model is hugely popular in India and it was estimated that by 2012, 37,000 franchises were operational. Most of these projects are associated with the local electricity board however mini grid projects are using this model are becoming of increasing popularity [24].

8.1.7.2 Anchor, business, and Consumers “ABC” Approach

Within the ABC model the operator builds the mini grid centred on a so called ‘anchor’ customer. This is most commonly a load such as a telecoms tower, an agri-processing plant, or a petrol station. Local businesses are typically the second largest revenue to the operator, followed by domestic customers [24].

This model has been successfully used across areas of Northern India by the operator OMC Power and recently GIZ highlighted there are 150,000 telecoms towers across SSA that within close proximity of un-electrified communities – this poses as a promising opportunity for ABC model mini grids. GIZ trialled such a model in Uganda and it was reported that the local telecoms company saved 40% on their energy bills through being supplied by the PV mini grid [24].

A major drawback to this business model is that domestic customers are prioritised the lowest in electricity provision and thus during times of low energy resource, domestic customers may not receive power of suitable quality [24].

8.1.7.3 Clustering Approach

The business model is when multiple un-electrified villages are categorised into clusters whereby one mini grid supplies electricity to one cluster. The major benefit of this is that in theory the capital and O&M costs are vastly reduced. The feasibility of this business model is highly dependent on the local context and policy framework [24].

A major drawback to this approach is that it requires a high level of technical and management skills on hand.

8.1.8 Mini Grid Distribution System

The distribution of lines and anchor loads between the powerhouse and end user are characterised in Figure 32:

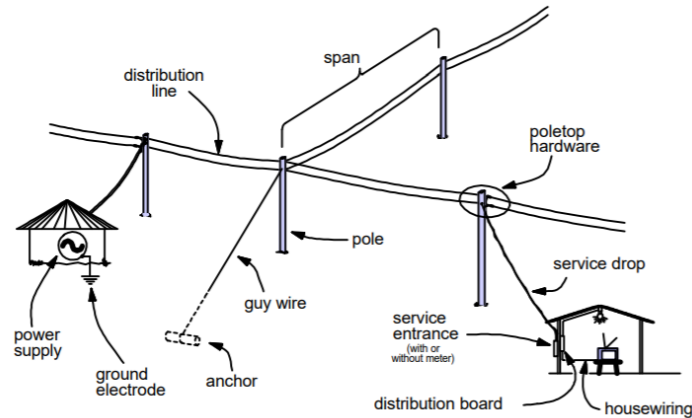


Figure 31 Mini Grid Cabling Diagram – Distribution Lines and Anchor Loads [25]

8.1.8.1 Powerhouse Location

The powerhouse location is affected by several factors however in most projects there are only very limited locations where such a system can be placed. The main factors affected by the powerhouse location are the voltage drop; the energy resource availability; the impact the system has on the local community; and the size of the system; and land ownership considerations [25].

8.1.8.1.1 Location of Lines

Having defined a location for PV system the layout of the distribution lines needs to be determined. The optimal layout is one that meets the criteria for voltage drops whilst minimizing cost and meeting safety and reliability requirements.

In general, when less cabling is used the system is cheaper. This is largely due to the lesser requirement for poles which are typically the most costly component of the distribution system. The following subsections detail the main factors considered in line layout process.

8.1.8.1.1.1 Location Of Transport Arteries

It is preferential to install cables along roads, trails, and paths. This is firstly due to the simplicity of the installation process and associated reduced costs. It is also due to the fact that potential new customers to the mini grid are likely to build houses/businesses that are also located on main transport routes and thus it is easy to connect them to the system. Having distribution infrastructure located along transport routes makes it easy to connect to already existent street lighting, or indeed install new street lighting [25]. Additionally, in many countries, crossing property lines causes major problems to mini grid developers due to ownership and land disputes. Following established paths and trails that are regarded as open amongst the community minimizes this potentially major project risk [25]. Despite this, through communication with GRS Commodities it is confirmed that cables are allowed to be planned to go over private property. As shown in the methodology, proposed lines mainly follow roads but also are planned across low lying household buildings.

8.1.8.1.1.2 Presence of Trees

Tress in most cases will need to clear so as to make way for distribution infrastructure. Despite this, trees often represent a source of income for the local community (from the sale of produce such as

nuts and fruit). Trees often need to be cleared to reduce the risks of interference with lines and to prevent branches falling and breaking conductors or from shorting circuiting lines. Conversely, older, more robust trees, are sometimes used as poles themselves. These natural poles have the advantage of not needing regular treatment to prevent decay, particularly at the foundations [25].

Religious Buildings – Buildings of religious significance need me identified and considered within infrastructure planning. Religious buildings may put constraints on line routing options [25].

8.1.8.1.1.3 Topography

Certain aspects of the landscape should be avoided for construction works due to the long-term O&M costs they could incur. Examples of this include slopes of high gradient, areas susceptible to erosion, swamp area, and flood risk areas, amongst other

8.1.8.1.1.4 Line length

Poles and conductors are typically the most costly component of the mini grid project and thus line length should be minimised wherever possible to reduce costs [25]. Additionally, the allowable line length is determined by the country specific grid codes and the selection of pole to be used to uphold the lines. In accordance with Ugandan grid code and through discussion with field workers, a maximum length of 40m is allowable for low and medium voltage cables for mini grids on Ugandan islands [25] [54].

8.1.8.1.1.5 Minimizing alignment changes

When there is a change in direction of a line, the conductor under tensile stress imposes a force on the pole which supports it. Oftentimes there will be multiple cables connected to one pole and thus the different tensile stress vectors need to be balanced. This can either be achieved by the balancing the direction of the cables or by adding an additional guy and anchor which is course comes at an additional cost and efforts needed for installation. Additionally, the guy and anchor can pose as a safety hazard as they are not readily visible, see Figure 31. Accordingly, when conductor tension is large, all efforts should be made to marginalising deviations in direction over as great a distance as possible [25].

8.1.8.1.1.6 Loading

In instances there are multiple lines coming from the powerhouse the aim should always be to equalize the kW*km loading for each line during peak demand ours. Doing so allows for the use of the same size conductor which thus allows for quantity discounts in purchasing. In a similar regard, another important consideration is that within three phase power is imperative that single phase loads be as balanced as possible. This consideration becomes of increasing importance as the size of the powerhouse increases [25].

8.1.8.1.1.7 Planning horizon

When planning the distribution system the project lifetime needs to be taken into considerations do the likely ways in which the community will develop over that same period. Considerations to be taken in account are the location of potential future customers, the impact that emerging industries will have, and area for expansion.

The above considerations were taken into account in this work with the specific steps and results being detailed in section 3.3.

8.1.8.1.1.8 Location of Poles

Having decided upon a layout for the lines, poles need to be placed along such lines so as to support conductors with adequate clearance thus minimizing safety risks to people, animals, or vehicle. The major factors considered in this work are detailed in the following sub-section.

Line bending – As noted in section 8.1.8.1.1, guying may be required for balancing the tensile forces imposed on individual poles. So as to minimize the need for guys and anchors and subsequent costs and safety risks, bends in lines need to be kept to a minimum and poles need to be located at all bends that are present [25].

Location of load clusters – It is recommended by multiple sources as well as from industry that all drop cables that connect the customer to the grid come from a pole and not mid cable. Accordingly, poles will need to be strategically positioned amongst clusters of customers. It is therefore paramount that the location of all customers be assessed prior to pole location planning [25] [59].

Ground clearance – The height at which the cabling is above the ground is highly project specific. Based on communications with GRS Commodities and INENSUS, a pole height of 10m is recommended for the local context.

Pole strength – In addition to tensile stress incurred from the connected cabling, the poles need to be strong enough to withstand climatic conditions, primarily wind. Accordingly, communication channels with industry will be exploited to assess the best option for Buvu Island.

The above considerations were taken into account in this work with the specific steps and results being detailed in section 3.3

8.1.8.1.2 Line Configurations

The configuration of the lines has implications on the cost of the distribution system and again, the optimal configuration is highly project specific. The line configuration also has impact on the conductor sizing and the associated allowable voltage drop. Within mini grids there are four commonly used line configurations which are detailed in this subsection. A diagram of such configurations is conveyed in Figure 32. The methodology employed for selecting and thus designing lines is detailed in section 3.3.

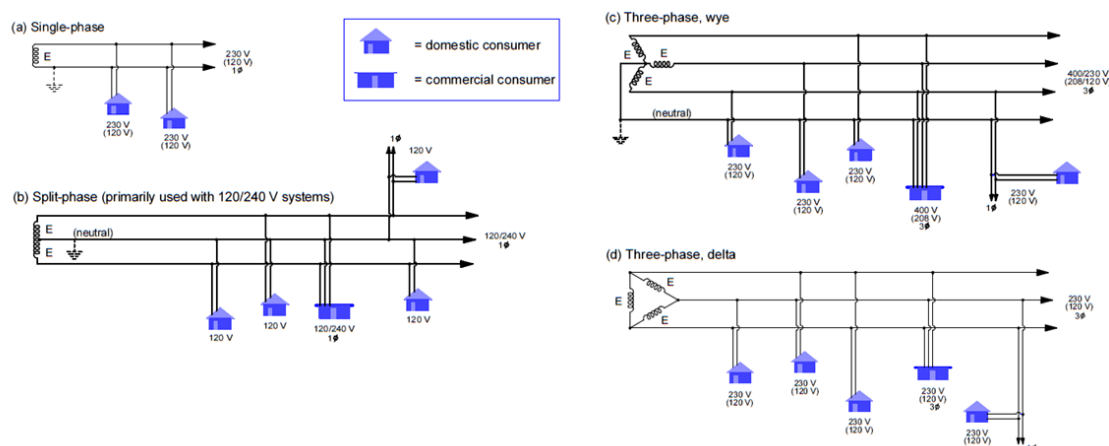


Figure 32 Four basic distribution line configurations for village scale mini grids [25]

8.1.8.1.2.1 Single Phase Supply

Within the Single Phase Two Wire configuration two conductors from the powerhouse serve all loads at a voltage of typically either 120 or 230 V - within the Ugandan context the voltage would be 230 V.

Customer connections in this configuration are relatively straightforward in that for each customer a service drop line simply taps both conductor lines. This is simplest configuration for mini grid and unsurprisingly is the most commonly found however not the most efficient. This configuration is conveyed in Figure 32 (a) [25].

Within the single-phase three wire split phase configuration three conductors are required and this arrangement is primarily used in systems that have a nominal consumer voltage of 120 V. In this case the generator is connected so as to generate twice the nominal voltage (250 V as opposed to 120 V). The two phase conductors are connected at the end of the power generators windings and the neutral conductor, which could be grounded, is connected to the centre tap of that windings. As shown in Figure 32 (b) one of the conductors on the customer drop line always taps into the neutral conductor's main line. The other conductor of the drop line always taps into one of the two main phase cables. When selecting which phase to tap into an important consideration is to ensure that there remains to be balanced load at the mains [25].

The split phase configuration has two major advantages compared to the previous case. Firstly, it has a reduced cost because a smaller conductor can serve the same load. This is because twice the voltage and half the current are being used in this results in a fourfold reduction in voltage drop and power loss. Likewise, if the load is completely balanced then a conductor of cross section four times smaller can be used [25].

8.1.8.1.2.2 Three Phase Supply

Three phase brings the benefit of being able to serve single phase users as well as larger consumers who require three phase supply. It is common to find systems that only serve single phase users however still have a main three phase line. The community layout, powerhouse location, and potential governing requirements for a three-phase power cable along a main road for commercial purposes usually dictate that a three phase system be employed [25] [54] [12]. There are multiple disadvantages to using three phase over single phase configurations:

- Using three conductors can necessitate the need for more poles so as to maintain the same ground clearance. Additionally, more poletop hardware may be required unless a bundled, insulated cable is used
- Particular attention, and thus resources, needs to be paid to balancing peak-time loads for the difference phase conductors.
- If the load demand of the mini grid exceeds that of its design capacity, then conductors will have to be replaced with larger ones which is a time and cost intensive process.
- For systems of low demand, mechanical strength is the main deciding factor in the minimum size of the required conductor. This can result in conductors being sized that are unnecessarily large for the load and indeed a single phase line would suffice.

Three phase systems can be connected in two different ways – four-wire delta, or three-wire wye.

Three-phase, four-wire (wye) is the most commonly used configuration for low voltage networks of three phases that are used by electrical utilities. As shown in Figure 32 (c), this configuration can serve both single and three phase customers. This is the most commonly used three phase configuration due to an increased efficiency which comes a lesser requirement for current which reduces the percentage voltage drop and power loss.

The three phase, three-wire, delta configuration is far less frequently used in electricity distribution due to its increased cost. It does however have the advantage that the voltage drops associated with unbalanced loads are far less than that of the wye configuration. Indeed, a delta configuration can

handle double the level imbalance of the wye configuration and this is advantageous in projects where load balancing is not possible due to the layout of the village loads.

8.1.8.1.2.3 Conductor Costs

Table 34 represents a cost comparison of the four conductor types discussed in the chapter and taken from the *Mini Grid Design Manual* [25]. Within this cost comparison it is assumed that:

- The same line performance is same for each case
- Conductor cost is proportional to its cross-sectional area
- Conductors are available in any size.

Table 34 Conductor Costs for Distribution Systems

Relative Conductor Cost		
Line Configuration	Balanced Loads	50% unbalance
Single-phase, two wire	4	4
Single-phase, three wire	1.5	2
Three-phase , four wire	1.3	2
Three-phase, three-wire	3	3.2

8.1.8.1.2.4 Conductor Sizing

The conductor is one of the more costly aspects of the mini grid system and there is often high incentive to make the system more affordable through using cheaper conductor material. Unfortunately, in the process of electrical transmission, resistive loads generated by the conductor result in voltage drops along the line and thus power loss. This implies that reducing the conductor size can result in poor power supply quality to the consumer and/or more power needs to be generated to meet demand.

On the assumption that the mini grid power generating technology is fully operational, the size and type of the conductor as well as the power factor are the main variables that determine if suitable voltage is being applied to the system. It is therefore paramount when designing the distribution network to calculate the voltage drop of the system. An example distribution line that would be subject to a voltage drop assessment is conveyed in Figure 33:

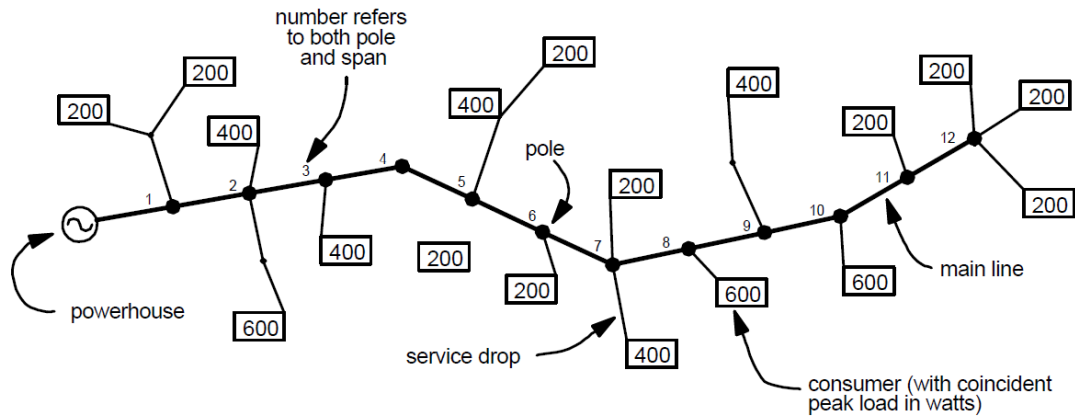


Figure 33 Sample of Village Mini Grid subject to a voltage drop assessment

The means by which the voltage drop is assessed within this work is discussed in the Methodology Section 3.3. The results of the voltage loss assessment are cross-referenced with the Quality Assurance Framework, discussed in Section 1.5.7.

8.1.9 Quality Assurance Framework

The National Renewable Energy Laboratory framework has the overall goal of filling in the gaps that prevent wide scale deployment of mini grids. Frameworks are categorised as follows:

- Level of Service Framework
 - Power Quality
 - Power Availability
 - Power Reliability
- Accountability and Performance Reporting Framework
 - Customer Accountability
 - Utility Account Ability

The availability, reliability, and quality of power are fundamental to the mini grids design and business mode. The availability and reliability refer to the quantity and security of power provided.

The quality of the power refers to the how efficiently and safely electrical energy is transported around the systems grid infrastructure. *Chattopadhyay et al, 2011* defined power quality as “maintaining the near sinusoidal waveform of power distribution bus voltages and currents at rated magnitude and frequency” [61]. The power quality of a mini grid is assessed according to the following variables, as specified in the NREAL Quality Assurance Framework:

- Voltage Imbalance
- Transients often Caused by Lightning
- Short Duration Variations
- Long Duration Variations
- Frequency Variations
- Resistive Voltage Drops
- DC Ripple (DC)
- Switching Noise (DC)

The QAF specifies three tiers for the quality of electricity supplied: base, standard, and high. Base service ensures minimum levels of safety, with high level being similar to that of centralised grid infrastructure. Table 35 provides a summary of how the levels of service a mini grid can provide, as defined by QAF framework:

Table 35 Levels of Service of Power Quality [26]

Issue	Base Level of Service	Standard Level of Service	High Level of Service
AC Power Quality Phenomena			
Voltage Drop	<10%	<5%	<2%
Transients	No protection	Surge protection	Surge protection
Short voltage-duration variations	<5/day	<1/day	<1/week
Long voltage-duration variations	<10/day	<5/day	<1/day
Frequency variations	48 Hz < f < 52 Hz	49 Hz < f < 51 Hz	49.5 Hz < f < 50.5 Hz
DC Power Quality Phenomena			
Resistive voltage drop	<10%	<5%	<2%
Percent ripple¹⁴	50% peak to peak (pk-pk)	20% pk-pk	10% pk-pk
DC ripple & switching noise	Unfiltered	Transient noise minimized	Ripple noise also minimized
Transients	No protection	Surge protection	Surge protection
Faults allowed per day	<5 per day	<2 per day	<1/day
Power Reliability			
Unplanned-interruptions	<52 per year	<12 per year	<2 per year

Within this study of Buvu Island electrification, the proposed system will be assessed according to the above framework. A voltage drop assessment will be conducted of the proposed grid infrastructure and compared against the quality recommendation suggested in Table 35.

8.2 Appendix B – Multi Tier Framework

Table 36 Multi-tier framework for household electricity access [62]

	Tier 0	Tier 1	Tier 2	Tier 3	Tier 4	Tier 5
Power		Very low power, minimum 3 watts	Low power, minimum 50 watts	Medium power, minimum 200 watts	High power, minimum 800 watts	Very high power, minimum 2 kW
Daily capacity		Minimum 12 watt-hrs	Minimum 200 watt-hrs	Minimum 1.0 kW-hrs	Minimum 3.4 kW-hrs	Minimum 8.2 kW-hrs
Hours per day		Minimum 4 hours	Minimum 4 hours	Minimum 8 hours	Minimum 16 hours	Minimum 23 hours
Hours per evening		Minimum 1 hour	Minimum 2 hours	Minimum 3 hours	Minimum 4 hours	Minimum 4 hours
Affordability				Cost of a standard consumption package of 365 kilowatt-hours per annum is less than 5% of household income		
Reliability					Maximum 14 disruptions per week	Maximum 3 disruptions per week of total duration less than 2 hours
Quality					Voltage problems do not affect use of desired appliances	

8.3 Appendix C – Interview Findings

Table 37 Interview Findings Summary

Developer	Country	Operations	Interview Findings
GRS Commodities	Uganda	Scaled one mini grid and plan to develop one on Buvu Island	First mini grid over-sized by a factor of 4. Mini grid operator not willing to share data
Absolute Energy	Uganda	230kW PV mini grid at Kitobo Island. 600 households and businesses	Unwilling to share data regarding mini grid performance as already collaborating with academic partners
E-Sam Energy	Nigeria	100 kW mini grid planned but not financed	Load demand assessment conducted by connecting diesel generator to half of community. Strategy in place for implementing productive end uses
OnePower	Lesotho	One pilot system and 24 mini grids planned	Interested in strategy for productive use energy strategy due to low load demand. Data obtained for one pilot site
Rift Valley Energy	Tanzania	3 MW Hydro plant. 80% of production if for industrial scale uses. 20% is for rural communities.	Plant has been operational for three years but funded by grants so little commercial interest in increasing demand. More interest in socio-economic impact of electrification
JUMEME	Tanzania	One operational mini grid	Not willing to share mini grid production data however interested in understanding how productive end uses could be introduced to incentive demand

8.4 Appendix D – Photos of Landing Sites



Figure 34 GIS Map of Kachanga and its key features

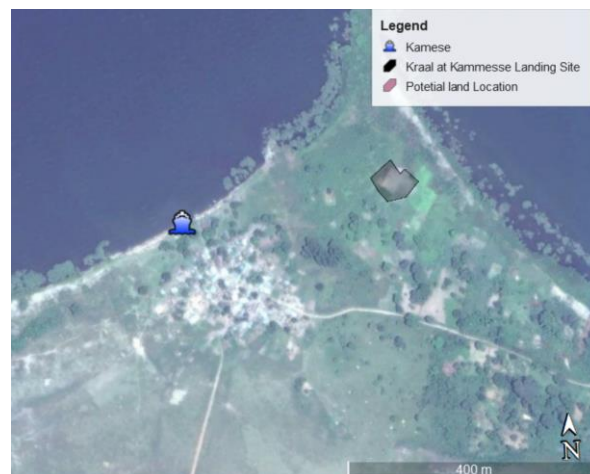


Figure 35 GIS Map of Kamese and its key features

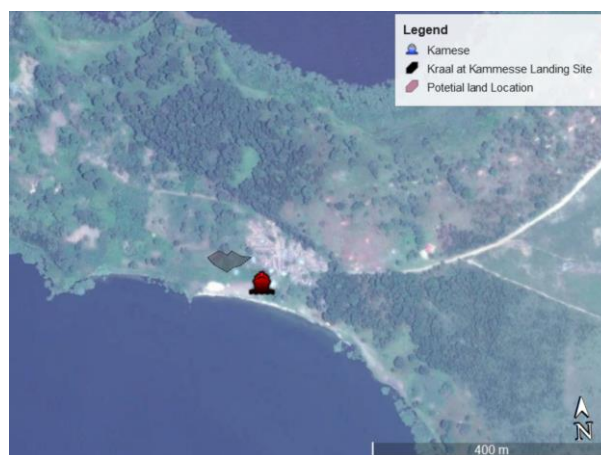


Figure 36 GIS Map of Luwungulu and its key features

8.5 Appendix E – Electricity Survey

Correspondent information																	
Name:																	
Mob:					GPS:												
Type of client:		Household ☐ Business ☐ Anchor load ☐			Description:												
No. of inhabitants:					Age of correspondent												
Current energy expenditure					Highest education level												
Type of energy		Expense per day		capacity		Who pays for the energy?											
Diesel/petrol ☐						Are/is he resident on the island?											
Candle ☐						Type of structure		Permanent ☐ semi-permanent ☐									
kerosene ☐																	
Firewood ☐																	
charcoal ☐																	
Batteries ☐																	
Mobile phone charging ☐																	
Battery charging ☐																	
Prepaid solar system ☐																	
Source of income					Electricity needs of the consumer												
Income source		Ugx/ day (low season)		Ugx/day (high season)		Appliances		Power (in W)		No of appliances		daily use (in hours)		Range used: (time, from, to)			
						light bulbs (8-20)											
						TV (45-120)											
						Radio/CD (20-60)											
						Phone (2-5)											
Stated ability to pay					Fridge(45-60)												
Amount per day		In low season		In high season		Computer (45-200)											
						Welding											
						Others											
						other											
Planned load and demand forecast						Which appliances would you buy if we were to provide them to you on credit if you were using electricity? How much are u willing to spend on them?											
TV ☐						☐											
Fridge ☐						☐											

Radio ☐		☐	
Fan ☐		☐	
Computer ☐		☐	
Light bulbs ☐		☐	
iron ☐		☐	
Popcorn machine ☐		☐	
others ☐			
Willingness to pay			
Do you already have an individual solar system or generator or battery?	Yes ☐	No ☐	Under consideration ☐
How satisfied are you with your current electricity supply	Very satisfied ☐	Not satisfied ☐	can live with it ☐
Does your current solar system supply all your energy needs?	Yes ☐	No ☐	Up to what time in the night ☐

8.6 Appendix F – Customer List

Table 38 Buvu Island Customer Segments

Buvu Island Customer Segments				
Segment	Kachanga	Luwungulu	Kamese	Total
Low-Income Household	81	86	23	190
Medium-Income Household	401	0	79	480
High Income Household	63	3	15	81
Retail stores	41	0	6	49
Bar	30	2	9	39
Entertainment Centre	1	0	1	2
Night Club	1	0	0	1
Barbers	3	0	3	6
Hairdressers	10	0	3	13
Lodges	21	0	5	21
Restaurant	8	0	0	13
Savings Cooperative 1	1	0	0	1
Savings Cooperative 2	1	0	0	1
Primary School	1	0	0	1
Secondary School	1	0	0	1
Health Centre 1	1	0	0	1
Health Centre 2	1	0	0	1
Clinic	1	0	0	1
Mosque	1	0	1	2
Church	5	0	1	6
Ice Plant	1	0	0	1
Total	674	91	146	911

8.7 Appendix G – Appliance List

Table 39 Appliance List

Appliance	Rated Power (W)
Indoor lighting points	5
Outdoor lighting points	5
Television and decoder	36
Phone Charger	5
DVD Player	12
Fan	85
Projector	550
Decoder	18
Mixer + amplifier	100
PA speakers	50
Community Radio (Mega phone)	25
Fridge	93.125
Laptop	60
Public Address System	30
Medical freezer 100 l	93.125
Computer	120
Computer	120
Microscope	12
Printer (HP Laserjet Pro MFP M426)	600
Hair driers	1400
Shaving machine	35

8.8 Appendix H - Sample Inventory List for Secondary School

Table 40 Secondary School Energy Assessment

Secondary School					
Types of Appliances per customer	Quantity	Power (W)	Hours/day	Total Power of Appliances (W)	Daily Energy Demand (kWh/day)
Indoor Lighting points	8	5	1	40	0.04
Outdoor lighting points	4	5	1	20	0.02
Phone Charging	4	5	1	20	0.02
Lighting points in school	7	5	1	35	0.035
Television + decoder	1	36	1	36	0.036
Total				151	0.151

8.9 Appendix – I – Voltage Drop Analysis Table

Table 41 Line 42-43 Voltage Drop Table

End Segment		Phase	1		Type of Conductor	Copper	Size of Conductor	50	mm2	
Line Segment 43-42	Customer Segment	Pole	Demand (KW)		Voltage (V)	Current (A)		Length (m)	Voltage drop	
Connection			Spurs	Total		Spurs	Total		(V)	%
1	MHH	44	0.015	0.015	230.000	0.082	0.082	43	0.022	0.01
2	MHH		0.015	0.030	230.000	0.082	0.163			
3	MHH		0.015	0.045	230.000	0.082	0.245			
4	MHH		0.015	0.060	230.000	0.082	0.326			
5	MHH		0.015	0.075	230.000	0.082	0.408			
6	MHH		0.015	0.090	230.000	0.082	0.489			
7	MHH		0.015	0.105	230.000	0.082	0.571			
8	MHH		0.015	0.120	230.000	0.082	0.652			
9	MHH		0.015	0.135	230.000	0.082	0.734			
10	MHH		0.015	0.150	230.000	0.082	0.815			
11	CHU		0.015	0.165	230.000	0.082	0.897			

8.10 Appendix J – Sizing Study with no Ice Plant

An optimised system was sized for the load demand of households, commercial facilities, and community facilities, excluding the ice plant – this load profile is shown in Figure 9. Within this model the peak load demand is 42.74 kW and thus the inverter size will have to be modified. 9 Sunny Island 8.0Hs are used which have a combined total capacity of 54kW. The BAE 26 PVS 3610 was the chosen battery. The results of the sizing study are conveyed in Table 42:

Table 42 Optimised Size for System without Ice Plant

Ice Plant (Y/N)	Renewable Penetration (%)	Solar PV (kW)	Diesel Generator (kW)	Battery (kWh)	N° Batteries	Converter (kW)	Dispatch	NPV (USD)	LCOE (USD/KWH)
No	50.8	110	48	163	24	54	CC	950342	0.425
Yes	55.44	184	71	163	24	72	CC	1266596	0.39

As show in Table 42, the model without the ice plant has a noticeably smaller PV array. The model without the ice plant has a larger LCOE which at this stage in the work indicates that the project may be less profitable - as the tariff is fixed to grid cost. The state of charge of the battery, and output power of the generator over the course of a typical day are conveyed in Figure 37:

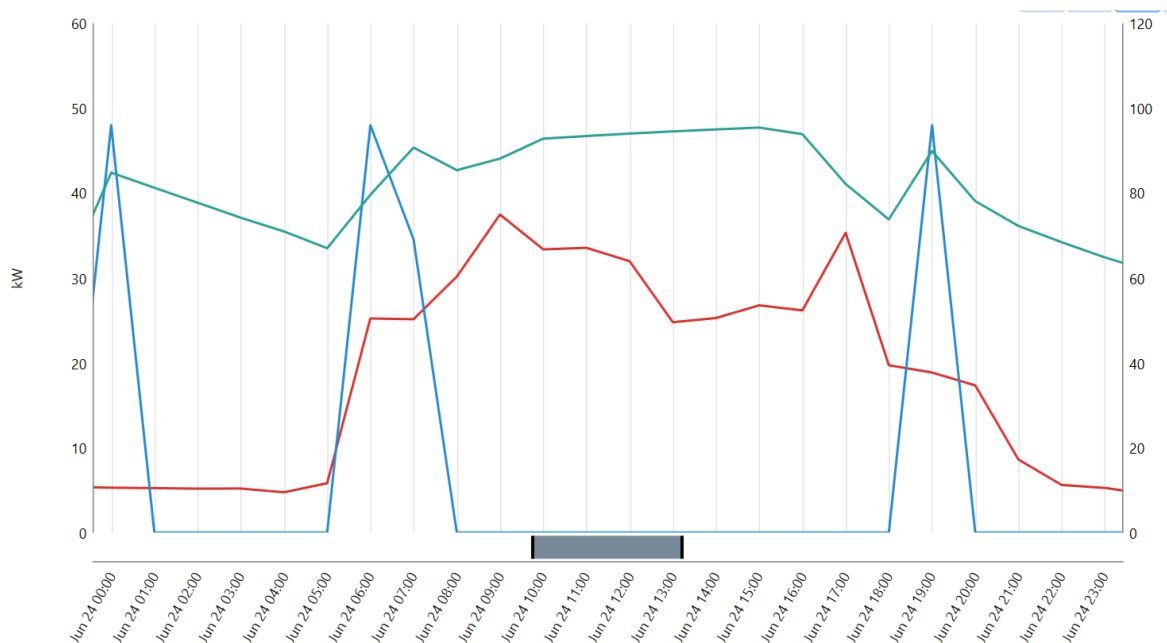


Figure 37 Battery state of charge (green), generator output power (blue), and power demand (red) for Battery Homer Simulation

As shown Figure 37, the generator is used multiple times throughout the day and the battery has a minimum state of charge of 60%.

8.11 Appendix K – Financial Results for Grid Infrastructure Included

An analysis was conducted of the project feasibility if the predicted grid infrastructure costs needed to be paid for by GRS Commodities. Reference case conditions have been used in this analysis and it is assumed that the grid infrastructure will need to be replaced after 20 years.

Table 43 Grid Infrastructure Included in Costs

KPI		
	Grid CAPEX Included	Ref
NPV	1,482	114,651
Project IRR	15.3%	23.5%
Equity IRR	15.4%	24.3%
Lowest DSCR	244.0%	529.0%
LCOE in [USD/kWh]	0.17	0.15
Revenues in [USD/kWh]	0.16	0.16
Avg. Operating Profit Margin	0.31	0.35
Lowest Operating Profit Margin	0.19	0.26

As highlighted in Table 43, the project still returns infeasible results when the costs of grid infrastructure are included in projections. The project however costs approximately \$100,000 more over the lifetime of the project in real terms. The project becomes in infeasible if one of the following conditions occur:

- Grant for investment is below 47%
- Debt from investment is above 19%
- Daily ice sales are below 2200 kg per day

Although the impact of GRS Commodities paying for grid infrastructure is severe, the probability of this occurring is incredibly low as it is local policy that the grid operator pay for the distribution network - mitigation action therefore does not need to be taken. This is further discussed in the Risk Analysis, see Section 4.4.

8.12 Appendix L – Financial results for no Ice Plant

The following sub-section highlights the results of the financial analysis of the mini grid without the ice plant and used the optimised sizing configuration highlighted in Table 42. The KPIs for the project are listed in Table 44:

Table 44 KPIs for Mini Grid without Ice Machine

KPIs	
NPV	-100,481
Project IRR	4.2%
Equity IRR	3.6%
Lowest DSCR	-1284.2%
LCOE in [USD/kWh]	0.97
Revenues in [USD/kWh]	0.60
Avg. Operating Profit Margin	-0.12
Lowest Operating Profit Margin	-0.49

As shown in Table 44, if the project is scaled to not encompass an ice plant and only generates revenues from single phase electricity sales then the project is financially unviable due to inability to repay debt loans taken at a suitable rate, as well as the NPV being negative implying an overall net loss project. The project costs are highlighted in Figure 38 :

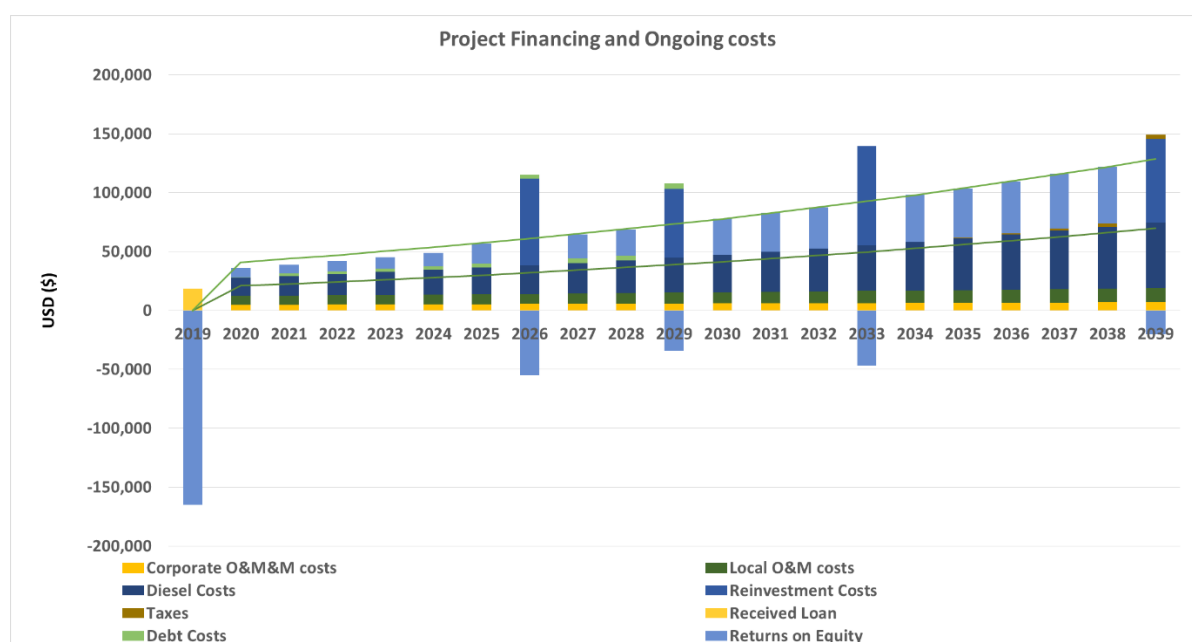


Figure 38 Project Financing and Ongoing Costs

As shown in Figure 38, by year 20 the revenues remain below the total costs.

The main variable that can make the system financially feasible is the Tariff cost. However, as this is set by the grid operator in reality it is fixed. A study was conducted to assess the tariff that would need be in place for the reference scenario to become economically viable. It was found that for the project to become economically viable to tariff would have to be above 2160 UGX / kWh (0.6 USD/kWh) which

is not possible for the island as few members of the community could be able or willing to pay this rate.

This study largely confirms the necessity for the ice plant to be in place on the island for making the solar mini grid a reality.

8.13 Appendix M – Comparison of IRR and DSCR

Table 45 Comparison of IRR and DSCR for Sensitivity Study

Variable	Condition	Number	IRR	DSCR
Reference Scenario			26%	880.1%
Investment Costs	High	1.1	13.60%	216.00%
	Low	1.2	19.10%	551.10%
Connection Rate (%)	0-100	2.1	16.00%	331.20%
	25-100	2.2	16.00%	346.00%
	75-100	2.3	16.00%	359.00%
	50-75	2.4	15.40%	315.10%
	50	2.5	14.70%	271.00%
Connection Fee (USD)	0	3.1	11.90%	237.30%
	50	3.2	12.90%	265.10%
	100	3.3	14.00%	296.10%
	150	3.4	15.20%	330.90%
	200	3.5	16.50%	370.50%
Ice Sales (tonnes)	0	4.1	0.00%	-1332.00%
	0.5	4.2	-1.30%	-911.00%
	1	4.3	4.80%	-489.90%
	2	4.4	12.90%	179.20%
	1.25-2.5	4.5	10.60%	-279.40%
	0	5.1	8.30%	195.90%
Grants	20	5.2	10.70%	240.70%
	40	5.3	13.90%	308.50%
	60	5.4	18.90%	423.00%
	80	5.5	28.30%	657.70%
	0	6.1	15.80%	N/A
Debt (%)	20	6.2	16.40%	166.60%
	40	6.3	17.00%	71.20%
	60	6.4	17.50%	39.40%
	80	6.5	17.80%	23.50%
	100	6.6	18.00%	14.00%
	10	7.1	16.90%	411.80%
Load Per Capita (%)	25	7.2	16.70%	402.80%
	50	7.3	16.50%	387.60%
	75	7.4	16.30%	372.50%
	High	1.1	13.60%	216.00%

8.14 Appendix N – Data Sheets

Ice Plant Data Sheet

Subject		Unit	Technical parameter
Power supply		Voltage/Phase/Frequency	380V/3P/50Hz
Refrigerant			R22 or R404A
Capacity		Kg/day	5000
Cooling way			Water cooling
Temperature	Ambient temperature	℃	35
	water inlet temperature	℃	20
Compressor	Manufacturer		Bitzer
	Refrigeration capacity	Kw	30.3
	Total running power	Kw	15.18
	Compressor COP/EER	Kw/kw	2
	Horse power	Hp	25
Operating power	Total running power	Kw	17.72
	Installment power	Kw	22
	Operating power of reducer	Kw	0.37
	Operating power of cold water pump	Kw	0.12
	Cooling water pump	Kw	1.5
	Fan of cooling tower	Kw	0.55
Dimension	Size of ice maker	(L×W×H)(mm)	2200×1050×1550
	Size of cooling tower	(O.D.×H)(mm)	φ1380×2170
	Size of water feeding pipes of cooling tower		1/2"
	Size of cooling water pipes		2"
	Size of water feeding pipe		3/4"
	Size of ice outlet	mm	Φ712
Weight of unit	Operating weight of flake ice maker	Kg	1250
	Operating weight of cooling tower	Kg	460
Water supply pressure requirement		Bar	1.5

Table 46 BAE Lead Acid Battery Range

BAE SECURA PVS SOLAR

Technical Specification for Vented Lead-Acid Batteries (VLA)

1. Application

BAE SECURA PVS SOLAR batteries are the optimal solution for a reliable and robust storage of regenerative energy under extreme conditions in the industrial sector.

The special electrode design with tubular electrodes distinguishes the BAE SECURA PVS SOLAR batteries leading to high security and reliability as well as high cycle life time.



Similar to the illustration

2. Technical data (Reference temperature 20 °C)

Type	C_{1h} Ah	C_{10h} Ah	C_{20h} Ah	C_{72h} Ah	C_{100h} Ah	C_{120h} Ah	C_{240h} Ah	R_i 1) mΩ	I_k 2) kA	Length (L) mm	Width (W) mm	Height (H) mm	Weight dry kg	Weight filled kg
U_e V/cell	1.67	1.80	1.80	1.80	1.80	1.80	1.80							
2 PVS 140	63	111	127	141	143	144	148	1.52	1.37	105	208	420	9.1	14.5
3 PVS 210	95	167	191	211	215	217	222	1.06	1.96	105	208	420	11.2	16.4
4 PVS 280	127	223	254	282	287	289	295	0.84	2.46	105	208	420	12.8	18.0
5 PVS 350	159	279	318	352	359	361	369	0.70	2.98	126	208	420	15.3	21.7
6 PVS 420	191	334	382	424	431	434	444	0.60	3.47	147	208	420	18.1	25.7
5 PVS 550	223	389	432	486	496	500	513	0.57	3.61	126	208	535	20.0	28.8
6 PVS 660	267	467	518	583	595	601	616	0.49	4.18	147	208	535	23.5	34.0
7 PVS 770	310	544	604	681	694	700	720	0.44	4.69	168	208	535	26.8	39.1
6 PVS 900	352	665	748	856	877	888	916	0.47	4.41	147	208	710	33.0	47.4
7 PVS 1050	415	777	872	993	1,020	1,033	1,065	0.36	5.66	215	193	710	42.1	61.5
8 PVS 1200	473	886	996	1,137	1,160	1,178	1,216	0.32	6.36	215	193	710	46.6	65.4
9 PVS 1350	522	992	1,116	1,274	1,300	1,320	1,365	0.33	6.20	215	235	710	51.4	75.4
10 PVS 1500	585	1,100	1,240	1,418	1,450	1,464	1,516	0.28	7.25	215	235	710	56.0	79.4
11 PVS 1650	635	1,210	1,362	1,555	1,590	1,608	1,665	0.28	7.36	215	277	710	61.0	89.6
12 PVS 1800	698	1,320	1,486	1,699	1,740	1,752	1,816	0.24	8.41	215	277	710	65.4	93.4
11 PVS 2090	790	1,470	1,636	1,836	1,870	1,884	1,941	0.24	8.38	215	277	855	72.7	105.9
12 PVS 2280	869	1,600	1,784	2,001	2,040	2,052	2,116	0.22	9.48	215	277	855	77.4	110.4
13 PVS 2470	978	1,740	1,938	2,174	2,210	2,232	2,292	0.16	13.03	215	400	815	90.8	137.8
14 PVS 2660	1,051	1,880	2,080	2,332	2,380	2,400	2,448	0.15	13.82	215	400	815	95.3	142.4
15 PVS 2850	1,123	2,010	2,220	2,498	2,550	2,568	2,640	0.14	14.43	215	400	815	100.2	146.9
16 PVS 3040	1,195	2,140	2,380	2,664	2,710	2,736	2,808	0.13	15.20	215	400	815	105.4	151.6
17 PVS 3230	1,280	2,290	2,540	2,858	2,910	2,940	3,000	0.12	16.91	215	490	815	117.7	175.1
18 PVS 3420	1,352	2,420	2,680	3,024	3,080	3,108	3,192	0.11	17.55	215	490	815	121.9	179.1
19 PVS 3610	1,425	2,560	2,840	3,189	3,250	3,276	3,360	0.11	18.36	215	490	815	126.8	183.6
20 PVS 3800	1,496	2,690	2,980	3,355	3,420	3,444	3,528	0.11	18.92	215	490	815	132.0	188.3
22 PVS 4180	1,635	2,950	3,280	3,686	3,750	3,780	3,888	0.10	19.92	215	580	815	145.4	213.9
24 PVS 4560	1,777	3,220	3,560	4,010	4,090	4,128	4,224	0.09	21.26	215	580	815	155.2	223.0
26 PVS 4940	1,917	3,480	3,860	4,341	4,420	4,464	4,584	0.09	22.49	215	580	815	165.0	232.0

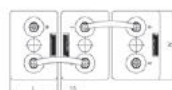
1, 2) Internal resistance R_i and short circuit current I_k according to IEC 60896-11

Height (H) is the maximum height between container bottom and top of the bolts in assembled condition.

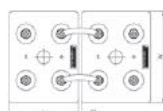
BAE SECURA PVS SOLAR batteries are also available as dry pre-charged version. They are titled with additional "TG", e.g. 4 PVS 280 TG.

All values published in the table correspond to 100 % discharge of current depending capacity without voltage drop of connectors. Please consider item 7.

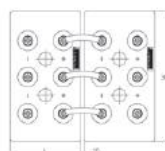
3. Terminal positions



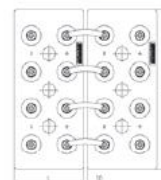
2 PVS 140 to 6 PVS 900



7 PVS 1050 to 12 PVS 2280



13 PVS 2470 to 16 PVS 3040



17 PVS 3230 to 26 PVS 4940

Terminals are designed as female poles with brass inlay M10 for flexible insulated copper cables with cross-section 25, 35, 50, 70, 95 or 120 mm² or insulated solid copper connectors with cross-section 90, 150 or 300 mm².

Table 47 Sunny Island 8.0H data sheet

SUNNY ISLAND 6.0H / 8.0H

Technical Data	Sunny Island 6.0H	Sunny Island 8.0H
Operation on the utility grid or generator		
Rated grid voltage / AC voltage range	230 V / 172.5 V to 264.5 V	230 V / 172.5 V to 264.5 V
Rated grid frequency / permitted frequency range	50 Hz / 40 Hz to 70 Hz	50 Hz / 40 Hz to 70 Hz
Maximum AC current for increased self-consumption (grid operation)	20 A	26 A
Maximum AC power for increased self-consumption (grid operation)	4,6 kVA	6 kVA
Maximum AC input current	50 A	50 A
Maximum AC input power	11,500 W	11,500 W
Stand-alone or emergency power operation		
Rated grid voltage / AC voltage range	230 V / 202 V to 253 V	230 V / 202 V to 253 V
Rated frequency / frequency range (adjustable)	50 Hz / 45 Hz to 65 Hz	50 Hz / 45 Hz to 65 Hz
Rated power (at Unom, from / 25°C / cos φ = 1)	4,600 W	6,000 W
AC power at 25°C for 30 min / 5 min / 3 sec	6,000 W / 6,800 W / 11,000 W	8,000 W / 9,100 W / 11,000 W
AC power at 45°C permanently	3,700 W	5,430 W
Rated current / maximum output current (peak)	20 A / 120 A	26 A / 120 A
Total harmonic distortion output voltage / power factor at rated power	< 4 % / -1 to +1	< 4 % / -1 to +1
Battery DC input		
Rated input voltage / DC voltage range	48 V / 41 V to 63 V	48 V / 41 V to 63 V
Maximum battery charging current / rated DC charging current / DC discharging current	110 A / 90 A / 103 A	140 A / 115 A / 130 A
Battery type / battery capacity (range)	Li-Ion*, FLA, VRLA / 100 Ah to 10,000 Ah (lead-acid) 50 Ah to 10,000 Ah (Li-Ion)	Li-Ion*, FLA, VRLA / 100 Ah to 10,000 Ah (lead-acid) 50 Ah to 10,000 Ah (Li-Ion)
Charge control	IUoU charge procedure with automatic full charge and equalization charge	IUoU charge procedure with automatic full charge and equalization charge
Efficiency / self-consumption of the device		
Maximum efficiency	96 %	96 %
No-load consumption / standby	26 W / 7 W	26 W / 7 W
Protective devices (inverter)		
AC short-circuit / AC overload	● / ●	● / ●
DC reverse polarity protection / DC fuse	- / -	- / -
Overtemperature / battery deep discharge	● / ●	● / ●
Overvoltage category as per IEC 60664-1	III	III
General data		
Dimensions (W / H / D)	467 mm / 612 mm / 242 mm (18.4 inch / 21.1 inch / 9.5 inch)	467 mm / 612 mm / 242 mm (18.4 inch / 21.1 inch / 9.5 inch)
Weight	63 kg (138.9 lb)	63 kg (138.9 lb)
Operating temperature range	-25°C to +60°C (-13°F to +140°F)	-25°C to +60°C (-13°F to +140°F)
Protection class as per IEC 62103	I	I
Climatic category as per IEC 60721	3K6	3K6
Degree of protection as per IEC 60529	IP54	IP54
Features / function		
Operation and display / multifunction relay	External via SRC-20 / 2	External via SRC-20 / 2
Three-phase systems / battery backup function	● / ●	● / ●
State of charge calculation / full charge / equalization charge	● / ● / ●	● / ● / ●
Battery temperature sensor / data cables	● / ●	● / ●
Certificates and approvals	www.SMA-Solar.com	www.SMA-Solar.com
Warranty	5 years	5 years
For off-grid applications		
Automatic rotating magnetic field detection / generator support	● / ●	● / ●
Parallel connection / Multicluster	● / ●	● / ●
Integrated soft start / generator support	● / ●	● / ●
Accessory		
For off-grid applications		
Battery cable / battery fuse	○ / ○	○ / ○
Interface SI-COMSMA (RS485) / SISYSCAN (Multicluster)	○ / ○	○ / ○
Load-shedding contactor / external battery current measurement	○ / ○	○ / ○
For on-grid applications		
Interface SI-COMSMA (RS485) / Interface SWDMSI-10 (Speedwire)	○	○
Sunny Home Manager / SMA Energy Meter / automatic transfer switch for battery backup (procurement via external supplier)	○ / ○ / ○	○ / ○ / ○
Type designation	SI6.0H-11	SI8.0H-11
● Standard features ○ Optional features – Not available		
* from Akasol, Leclanché, LG-Chem, SAFT, Samsung, Sony, Dispatch Energy, Hoppecke		

Appendix O - Risk Assessment

SWOT Analysis

The following table details the results of the SWOT analysis

Table 48 SWOT Analysis

Strengths Prefeasibility findings • Prefeasibility review demonstrates project feasibility • The ice sales are able to support the electrification for the entire island • The grid operator will cover the cost of grid infrastructure • Electricity will be offered at grid price which is cheaper than what islanders currently pay for electricity from SHS/generator • Cheaper flake ice for fish preservation • The community already have experience using electricity • The community have expressed clear interest in adopting a system • GRS Commodities have experience facilitating a mini grid installation • GRS Commodities have local contacts to partner with • GRS Commodities have experience operating mini grids • GRS Commodities have experience operating an ice plant • GRS Commodities have experience engaging local communities • No threat of grid extension due to being on an island	Weaknesses • Load demand is highly uncertain and could impact on project feasibility • Project costs are highly uncertain which could impact project feasibility • The conditions of loans are highly uncertain • Ice sales are highly uncertain which could impact project feasibility • The tariff is not controlled by GRS Commodities and thus cannot be adapted if project runs over budget • The project is very isolated and this increases O&M costs • Project highly sensitive to local fishing industry • Project highly sensitive to daily ice production • Large distribution network that needs managed
Opportunities • Expansion to neighbouring island • Support from the government in the form of funding • Improved energy supply encourages the uptake of new productive uses • Improved ice storage capabilities allow for higher fish yields • The system can be have its capacity expanded • Project highly dependent on ice sales • Jobs can be created through employing people to work on the system	Threats • Grants may not awarded for asset which will have negative impact on the long term financial benefit • The local community does not accept the mini grid and continue to take electricity from alternate expensive sources • Seasonality of demand impacts finances • Fishing quotas are tightened impacts finances • Negative impact on local businesses • Resistance to paying tariff • The project has high reliance on diesel and is thus sensitive to price increases • Difficulty collecting payment

Risk Impact Assessment

Projects risks have been categorised along with the specific impacts they could have on the mini grid itself. The risks are characterised by likelihood of occurrence and severity of occurrence. Both of these variables been ranked on a scale of 1-5 with 5 being highly probable/severe, and 1 being highly unlikely/inconsequential. These are listed in the Table 49:

Table 49 Risk Impact Assessment

Risk Category	Risks	Potential Impacts	Severity	Likelihood
Inaccurate load demand assessment	- Errors in assessing current electricity in the field - Inaccuracies in estimating future demand - Incorrect modelling of load daily dispersion - Inaccurate demand for ice	- Oversized system - The system does not generate enough revenue to make project feasible - Undersized system: - Reduced service quality and technology is mistreated and this leads to high O&M costs - Ice plant over sized	4	4
Social Acceptance	- Lack of cultural awareness from project team - Lack of understanding of supposed benefits of mini grid - Lack of WTP or ATP - Suspicion of unfamiliar developer	- Low connection rates and/or electricity use impede on project's financial feasibility - Fishermen do not purchase ice - Project assets mistreated	2	2
Fee Collection	- Problems with metering system or payment collection method - Load fluctuations throughout year prevent payments - Local industries suffer and this effects payments	- Electricity use is below predicted levels - Increased O&M costs - Breakdown of communication between operator and community and lack of trust - Ice sales reduce	4	3
Cost of Resources	- Inaccuracy in CAPEX projections for assets - Inaccuracy in logistical costs	- Higher operation costs - Higher investment costs - Potential need to buy cheaper equipment with shorter lifetimes	2	3

	• Inaccuracy in operative cost projections • Diesel price fluctuations	• Reduced service quality		
Engineering, Procurement, Construction	• Difficulty in tendering • Difficulty finding partners • Lack of skills for construction • Logistical problems • Quality of technology issues • Technology compatibility issues • Damage to local land/property	• Increased CAPEX and OPEX costs • Elongated timeframes which impact upon social acceptance and costs • Breakdown of relations with community • Damage to components	4	2
Environmental Risks	• Adverse weather damage equipment • Adverse weather prevents construction works	• Delays to project delivery • Increased OPEX and CAPEX costs • Reduced service quality • Lack of manpower	2	2
Currency & Loan Conditions	• Inflation is different than expected • Interest rates change for foreign currency • Difficulty acquiring loan	• Reduced returns • Reduced returns to external investors • Problems meeting loan conditions	4	2
Theft and Vandalism	• Assets damaged or stolen • Electricity supply tapped into unofficially	• Increased operative costs due to repairs, and additional security • Power outage reduces service quality	1	1